Standard Model Gauge Structure under Complete-Response and Observer-Transport Premises

B. Müller Mario Poneder Alexander Osika Kai Xue Peter Nguyen
David Matscheko Daniel Ari Friedman

October 28, 2026

Paper release: `r2055` **Released:** October 28, 2026

Author affiliations: Bernhard Mueller, Pragma Research Inc.; Alexander Osika, EtherWorks; Daniel Ari Friedman, Active Inference Institute and FractiAI.

Abstract

Gauge structure is proposed to be the transport of overlap data between observer patches. On twelve ports with icosahedral incidence, a faithful compact commutator-closed response whose proper rechartings act by inner automorphisms of that same response has exactly one compact Lie type, $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. Faithfulness fixes the dimension twelve, the single fixed line of the transitive A_5 action bounds the centre by one, the fixed-space count excludes $\mathfrak{su}(2)^4$, and the classification of compact simple Lie algebras, a declared input, leaves the split $11 = 3 + 8$. The dimension arithmetic and the centre bound are machine checked; the fixed-space exclusion and the classification enter Lean as declared hypotheses. A separate categorical route reconstructs a compact group from overlap transport on a declared bosonic sector category.

A conditional matrix realization supplies the current blocks. Its declared spin lift and trace redistribution are the standard $\mathfrak{su}(5)$ embedding, and its declared exterior module is the Georgi–Glashow $\mathbf{10} \oplus \bar{\mathbf{5}}$ generation with exact Standard Model charges, anomaly cancellation and common central kernel $\mathbb{Z}_6$. In the conditional window $3 \leq N_g \leq 5$, whose lower edge is a declared CP-capability clause, a declared band-selection rule and the exact Laplacian costs $5 - \sqrt{5} < 6 < 5 + \sqrt{5}$ select the rank-three band. On the executable carrier the twelve-current law is a specification: the tested native exchange response closes to seventeen dimensions, and no result derives the twelve-dimensional current from unchanged repair. The physical current, matter action, global quotient and couplings are not source-derived.

Interpretation and state variables. OPH proposes one structure with two descriptions: a timeless, self-referential complete history and its actual internal computation. Observers are proposed to reverse-engineer their reality's substrate and construct hardware whose execution realizes that same history, including its builders and hardware, in emergent, mutually calibrated local times. The candidate must supply construction, initialization, controls, memory, read/write and record services, clock comparisons, and their resource costs; no external host or additional time is assumed. Its realization map must specify whether it matches one actual history, a family of preparation and intervention responses, or every independently allowed microstate. These are different requirements. A candidate must specify the history domain and closure operator; neither the schematic equation nor the finite constructions establish this physical realization or select a unique universe. An actual quantum record history also need not assign simultaneous outcomes to incompatible unperformed measurements.

The proposed primitives are patch states, records, interfaces, and admissible operations. Physical space, duration, and matter are intended emergent readouts. A supplied carrier incidence, geometric domain, action, representation, or clock is a premise of the construction using it. Complete-history closure, an instantaneous repair fixed point, a public normal form, and a stationary probability law are different objects. A stationary law can support changing individual histories. An operational observer is defined by its access and record interfaces; consciousness requires an additional psychophysical interpretation.

Basis, premise labels and paper set. OPH has one hypothesis, closure: the universe is a timeless self-referential structure whose observers recover its specification from their records and realize it. Its internal-realization policy requires its operations and information access to be accounted within the structure; operational agreement on accepted shared data defines what is public. The executable specification of carrier, repair law, observer algebra and closure inputs is implemented and runs; it is a candidate judged by whether it reproduces what observers measure. The labels A1, A2 and A3 name theorem premises: the carrier, federation and response layer of that specification; the agreement condition with an additional response-endogeneity principle; and an adopted least-informative state-selection principle. Closure alone implies neither the entropy minimizer nor inner implementation of rechartings on the connected response algebra. The basis reference states this formally.[1] A flagship synthesis gives the account, an executable observer construction states the specification with its acceptance criteria and premise map, and the theorem library carries conditional implications whose antecedents that premise map tracks.

Time conventions. Host runtime, solver counters, native attempt counts, local event-clock readings, event precedence or logical numbering, model coordinates or lapse, modular parameters, and calibrated observer-clock readings have distinct roles. Their identification requires explicit maps. A modular parameter is dimensionless; its geometric and operational readings require the stated branch and clock premises. Consistent local clock comparisons require no preferred universal clock or identical clock rates. Shared time parameters in restricted benchmarks are declared model choices. Experienced succession and a final instantaneous cosmic endpoint do not follow from a modular-flow or finite-convergence theorem.

One theory and its executable hypothesis. OPH proposes closure: one complete history is both a timeless self-referential structure and its own internal computation, and observers inside it recover and realize its specification. The hypothesis that makes this proposal concrete and testable is executable.

> *A simple, fully specified local repair and memory law on the proposed carrier architecture admits continuing observers whose compatible records and interventions have the observed* $3+1$*-dimensional quantum-field description and a common classical-gravity limit. The preferred matter mechanism is localized topological defects of native overlap transport, with rest energy derived from their excess maintenance dynamics in the same physical clock and energy dictionary.*

One source process must meet every part of this statement: records, clocks, matter, energy and responsive geometry come from the same law, state and refinement. Supplied coherence, protected codes, radial directions, fields, clocks or couplings are premises until that process realizes them. A

[1] https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/docs/BASIS.md

failed specified law rejects that law and its tested interpretation; it rejects the wider proposal only when the rejected choice is proved necessary.

The tower. The papers form one tower over the executable model. Each layer states a hypothesis and proves finite theorems under displayed premises. A conclusion passes to the next layer only after that premise is proved on the same model process; until then it is a conditional implication.

0. **Closure and agreement.** One history reads itself into existence, and agreement on shared records defines the public world. This is the interpretation and acceptance criterion; no theorem selects a unique universe.

1. **Repair mathematics.** Local agreement repair reaches global consistency, and nonzero cycle holonomy is the residue that local repair cannot remove. Fixed-point, normal-form, frustration and finite event-algebra theorems hold on their stated domains.

2. **Carrier and repair law.** Federated twelve-port icosahedral carriers with the native C/E seam repair are the substrate. The seam rule follows from four stated requirements, the quantum record laws are classified, and the uniform native repair selects a rank-three slow-response band.

3. **Executable observer.** Observers are continuing native processes that prepare, read, retain and act on records. Finite complete-output executions, overlapping-observer correspondence, event clocks and checkpoint continuation are verified; one continuing observer with every service and resource accounted is the open join.

4. **Clocks and fields.** Modular flow of observer algebras supplies candidate local clocks, and native repair supplies a positive continuum generator of spectral dimension two on the screen. A declared radial extension gives a conditional smooth $1 + 3$ scalar description. A physical clock and energy normalization and an interacting chiral $3 + 1$ field description from native records are open joins.

5. **Gauge structure and matter.** Gauge structure is the transport of overlap data, and stable massive matter is a localized defect whose boundary transport carries a holonomy that admitted repairs cannot erase. The compact Lie type $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$ follows under complete-response premises, and fixed nonzero holonomy forces positive residual mismatch. No native defect sector, maintenance-energy law or measured spectrum is established; the native double-load candidate is transient.

6. **Gravity.** Einstein dynamics is the coarse response of the same process. The null-balance and tensor-completion implication is proved under its premises; realizing those premises on one refinement tower is open, and the static radial product fails null balance.

7. **Cosmology and tests.** The same process must fix the vacuum, dark sector, primordial response and black-hole records, and pass independent prospective tests. The cosmology papers and the fine-structure, Koide, de Sitter and Yang–Mills supplements carry conditional branches, diagnostics and retrospective comparisons at this layer.

Completion and stopping. The aim is a consistent and complete theory: every layer realized on one process, or a proved obstruction that removes OPH or a stated branch of it. Results that no join needs leave the active chain with their provenance. A universal obstruction over a frozen admissible class, an unavoidable excluded prediction, or a contradiction among indispensable premises stops that class. A failed candidate, an unconstructed map or a resource stop has a narrower meaning.

The ordered joins, retained foundations and stopping rules are recorded with the executable model in the public repository.[2]

Fixed physical boundaries. Repair counts, host runtime and entropy production are distinct from energy; the maintenance-to-energy map must be derived with one common signal speed and no per-species conversion. The modular parameter is dimensionless until a derived clock calibrates it. Stationary repair or record entropy is distinct from the fixed-volume generalized-entropy variation that the Einstein construction consumes. A massless photon makes the infinite-volume theory gapless, so sector gaps and thresholds name their state and limit; confinement precludes free massless gluons as observable particles; a quantum graviton lies outside the classical Einstein theorem.

Finite C/E campaign. A versioned finite family tests both complete C and E instruments, their tagged classical mixtures, positive-probability identity holds and bounded classical memory with a positive uniform seam-exploration floor. On the existing 78-state carrier, the continuing observer retains a finite record buffer and explicitly exports displaced records or halts when full. Exact six-attempt executions preserve complete instrument probabilities and untouched-reference marginals. The transition kernel specifies every finite record history; the exponentially growing history tree is not materialized. Growing exports remain a resource, and preparation, randomization, access, storage and delivery services remain supplied.

The adaptive observer also has a constructed stationary reference. Its source/current-sign quotient has a unique recurrent class of 129 of 132 states. Lifting that exact law through the actual complete record instrument gives 128 positive record blocks for C and 192 for E, with 4,218 positive source/record diagonal entries in either case and exact stationary residual zero. This is the finite source/buffer marginal at repair probability one, not the growing history; faithfulness requires its stated support restriction. The delayed member needs its own reference construction.

Two exact obstructions sharpen the next joins. Each law's complete active branch $*$-algebra is the full matrix algebra on both the 78-state source and its 66-state binary shell. Thus a source-only Hermitian charge commuting with every branch is scalar; this does not exclude forward-invariant subspaces, general expectation-conserved observables, history-dependent charges or controlled lifetimes. Separately, the frozen real-operation grammar cannot implement a nontrivial modular action of the constructed nontracial reference on its fixed full support algebra: a specified input and dimensionless modular parameter give normalized trace-distance error at least $1/2$. The separating observable is a mathematical witness, not an admitted native detector. This is not an exclusion of every clock or encoding.

These results supply finite components for the larger construction. The finite campaign itself does not freeze a carrier/refinement class; that boundary belongs to the separate executable program contract. Its observer services must be constructed or counted on the declared source, and any physical use needs the actual accessible algebra and implemented clock. A defect sector or controlled lifetime also needs its own specification outside the excluded commuting-charge class. The finite campaign adds no independent physical comparison or whole-program exclusion.

Same-action finite evidence. A supplied three-dimensional Weyl/contact CAR action has one isolated two-fermion bound band whose wavepacket moves, carries the conserved CAR number charge and has complete contact/scattering spectral records. Its positive contact section makes the band topologically trivial. An exact rational curvature enclosure disagrees with the rest/kinetic

[2] https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/docs/COMPLETION_PROGRAM.md

mass relation using the same speed parameter; these are coordinate energies and masses, not identified particle species. One invariant pair and dark-reference subspace is attached by a supplied isometry to a native two-level corner. Its record-derived modular clock and complete herald, continuation and energy records are executed with a finite unconditional error bound. This does not encode the whole spatial field.

On the same local action, signed CAR regional restrictions, exact local energy conservation and finite quantum Gibbs/entropy identities have explicit witnesses. They retain all local number sectors, references and detector outcomes. Regional bare-energy modular identification fails, and the contact interaction leaks from any proper Weyl momentum pocket into its complement. Finite thermal positivity and KMS identities do not remove that coupling or establish a chiral continuum. These results reduce specific finite obligations in the existing premise map; physical geometry, Lorentz stress, common normalization and integrated acceptance are not established. The current four construction targets are native defect protection, common energy/kinetics, the interacting chiral field and finite-theorem integration.

A further complete-band execution uses the native record-derived rate with counted local field controls. Its unconditional error bound is below 0.037991 on $|s| \leq 1$, conditional on a supplied global field Gibbs reference. Finite independent copies of the original native reference cannot furnish that same field clock: their integer modular-gap ratios conflict with an exactly bounded noninteger pair gap. Directly on the native carrier, a faithful coherence has an exactly reference-matched exterior through every raw history and a finite contrast lifetime, but zero modular-energy excess and a fixed coherent core. The exact-code exterior and clock restrictions prevent combining these separate properties into an established moving particle.

The fair and endpoint-biased actively corrected memories have separate exact core laws. Positive coin rescaling transfers the sharp-code classification to the biased law, extending its optimal asymptotic reset bound to arbitrary adaptive exact code choices. A separately checked variant removes identity recoveries while preserving the admitted channel. Contact-only field protocols also admit a seven-state cyclic reduction in the declared side-four zero-momentum fiber. These simplifications reduce finite service or evaluation costs. A clock on another factor composes with the memory, but supplies no physical mass or propagation relation.

Complete native histories admit exact scalar-weighted partial-map evaluation, reducing host storage while retaining every raw outcome and the same source costs. A centered spectral bound reduces the full-field clock schedule from 32 to 21 split steps and from 26,238,982 to 17,219,334 supplied source pulses, with complete average half-diamond bound below 0.037991 on $|s| \leq 1$. The original schedule and preparation premises are retained. On the fixed two-particle carrier, an added cN is scalar and cannot determine energy relative to a vacuum sector. Sector-preserving histories with the same block-diagonal preparation likewise cannot identify relative sector offsets, even with a separate clock. An independently established Gibbs preparation law or coherent comparisons between sectors would be additional input. For the stated regional number-preserving field algebra, boundary hopping also excludes an exact retraction preserving the global Gibbs reference: the rational witness is $\|[N_A, H]\|_{\mathrm{HS}}^2 = 117/16$. Regional state restriction and clocks from the actual reduced density remain valid. These finite results sharpen the attachment conditions without establishing a physical mass, continuum or canonical premise discharge.

The finite matrix argument also removes a redundant hypothesis for partition pinching. For $L = \sum_c \gamma_c (E_c - I)$, trace orthogonality gives $\mathrm{Tr}(X^* L X) = -\sum_c \gamma_c \|X - E_c X\|_{\mathrm{HS}}^2$. Nonnegative rates preclude cancellation; strictly positive rates identify $\ker L$ with the intersection of the partition commutants. The Lean corollary requires neither commuting pinchings nor a separate no-cancellation premise. It does not identify the native repair generator with this class or select a physical rate and clock.

The same supplied Weyl/contact action extends to the complete CAR Fock algebra, with source execution on every number sector and a volume-dependent complete-history split bound. An explicit local commutator estimate makes the fixed-spacing thermodynamic dynamics norm Cauchy on local observables; the contact interaction survives in its local derivation. This removes the two-particle and finite-volume restrictions for that mathematical lattice field, without deriving a chiral spatial-continuum limit or selecting the action from passive native repair. External link coupling obeys exact gauge covariance and number-current Ward identities. Its contact-pair Schur complement carries charge two, and its energy-dependent norm metric joins the effective current and contact residue to the same full eigenstate.

On the native binary carrier, a faithful-reference Dirichlet identity strengthens the source-only conservation test: positive all-seam exploration forces every universally conserved source observable to be scalar, including under classical record feedback. A rational weighted gap also controls nearly conserved observables and arbitrary exogenous selector sequences sharing the reference. It is a gap per native attempt, not a particle mass; joint feedback charges and actively protected memories retain their distinct contracts. Compatible Petz-transpose observations compose regionally without product references, but their implementation and modular-clock covariance are further conditions.

This paper's hypothesis and place in the tower

This paper is the gauge half of layer 5. Its hypothesis is that gauge structure is the transport of overlap data between observer patches: on the twelve-port icosahedral carrier, a complete faithful commutator-closed port response on which every proper carrier recharting acts endogenously, by an inner automorphism of the same response, is the local gauge current, and its compact Lie type is $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$.

What is proved. The Lie-type theorem (Theorem 3.1) is a finite implication from the response clauses of premise A1, the endogenous-transport clause of premise A2 and the classification of compact simple Lie algebras. The categorical route proves a fixed-cutoff bosonic sector category unconditionally and a compact Tannaka group on any refinement tail that carries the stated receipt; a Doplicher–Roberts field net requires further declared hypotheses. The matrix current, spin lift, trace redistribution, exterior matter module, band selection, bracket discriminator and local action are conditional realizations under displayed declarations.

Open joins. The executable model must realize the complete response of A1 and the endogenous transport of A2 from its own repair and memory law, with the port generators, commutators and implementers reconstructed from ordered source histories. The tested native exchange response gives seventeen currents and fails A1; a specified coherent control law gives the twelve closed currents on a declared control menu (Section 4). Physical spin and statistics, a physical Weyl field, the global quotient, the family attachment and all couplings are further joins. The matter hypothesis, that stable massive matter is a localized topological defect of native overlap transport whose boundary holonomy takes values in this gauge structure, is developed in the particle paper [21].

Refutation. The complete accessible current of a specified executable law, if it is not a faithful closed twelve-dimensional response with inner carrier action, excludes that law's realization of the branch. A proof that no admissible local repair and memory law on the carrier supplies the A1 and A2 response clauses would remove this gauge branch. The finite failure conditions are listed in Section 12.

Contents

1 Two reconstruction routes **7**

2 Premises **9**

3 The finite-current Lie-type theorem **10**

4 The current on the executable carrier **12**
4.1 Carrier rechartings, the shared triplet and physical rotations 13

5 The categorical route **14**
5.1 Transport from overlap gluing 14
5.2 The fixed-cutoff sector category 15
5.3 Refinement and compact-group reconstruction 15

6 Conditional matter realization **17**
6.1 The matrix current, spin lift and exterior module 17
6.2 Global form 19

7 Families and the generation window **20**

8 The equivariant bracket variety and a face discriminator **21**
8.1 An assembled structural action 23

9 Refinement stability and relevant transverse perturbations **23**

10 Edge-sector weights and coupling extraction **24**

11 Classical carrier modes and the quantum-particle criterion **25**

12 Failure conditions **26**

A Local action, Cartan reduction and fermion current **27**
A.1 A local gauge-covariant action on the declared matter grammar 27
A.2 A consistent local Cartan–scalar reduction and current response 28
A.3 A continuous charged-current and separated-detector bound 30
A.4 Fermionic hypercharge transport on prepared source addresses 32
A.5 A flux-dressed fermion state satisfying abelian quantum Gauss 34

B A coupled fermion current in a declared hybrid gauge action **36**
B.1 Spin transport and the information needed to select exchange statistics 38
B.2 Conditional exclusion from a stable generator and a scalar deformation law 42
B.3 What finite source operations must supply for exclusion 44

1 Two reconstruction routes

The first route is structural:

$$\text{overlap gluing and transport} \Longrightarrow \text{rigid symmetric sector category} \Longrightarrow \mathrm{Aut}_{\otimes}(\mathcal{F}).$$

It reconstructs a compact group on the declared bosonic branch (Section 5). It does not select the Standard Model without a realized sector witness.

The second route is finite and carrier-specific:

$$\begin{aligned}&\text{twelve-port incidence and complete reversible response}\\ &\Longrightarrow \text{internal proper-carrier holonomy} \Longrightarrow \mathfrak{su}(3)\oplus\mathfrak{su}(2)\oplus\mathfrak{u}(1)\\ &\Longrightarrow \text{conditional Standard Model matter image.}\end{aligned}$$

Incidence supplies the twelve-dimensional A_5-module and its single fixed line. Premise A1 supplies the faithful compact commutator-closed response on that module. Premise A2 makes the proper carrier action inner on the same response. Incidence without the response clauses leaves many equivariant laws. Response naturality without endogeneity leaves the abelian $\mathfrak{u}(1)^{12}$ and the mixed $\mathfrak{su}(2)^2\oplus\mathfrak{u}(1)^6$ branches available; endogeneity bounds the centre by the single fixed line, and the fixed-line count then excludes $\mathfrak{su}(2)^4$. The two routes have separate premise sets. No theorem identifies the finite current with the Tannaka group; that equality requires a same-source current construction and a commuting-square identification.

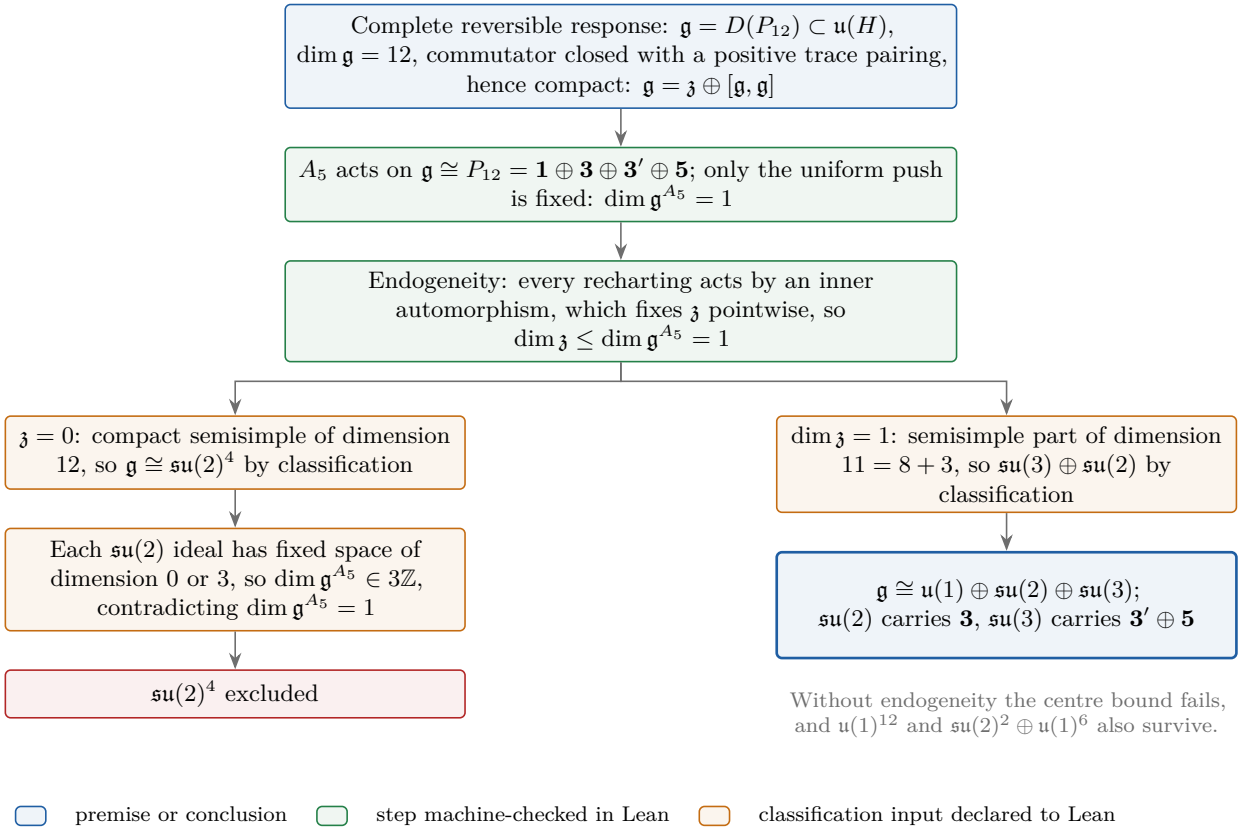


Figure 1: Steps of the gauge Lie-type theorem. Green steps are machine-checked in the Lean library; orange steps use the classification of compact simple Lie algebras and the fixed-space argument against $\mathfrak{su}(2)^4$, which enter Lean as declared hypotheses. The port count fixes the dimension twelve by declaration; the content of the theorem is the exclusion of $\mathfrak{su}(2)^4$ and a one-dimensional centre.

2 Premises

The premises A1–A3 are shared with the other layers. The Lie-type theorem uses only the response clauses of A1 and the endogenous-transport clause of A2. A3 enters only the edge-sector weighting of Section 10. Branch-local hypotheses are stated at the results that consume them.

A1 (oriented observer-patch federation and spherical support). There exists an observer patch net on an oriented spherical screen: at every finite resolution each local carrier has twelve primitive boundary ports forming the vertices of an oriented triangular boundary with 30 edges and 20 faces, combinatorially the boundary of an icosahedron, and carriers join through typed seams and coherent triple overlaps, refine to an oriented spherical support, and expose local state, readback, records, repair moves, and checkpoints. Their complete infinitesimal port response is represented faithfully on a finite-dimensional unitary response space and is closed under the commutators generated by ordered response composition. Formally, for every regulator r in a directed system there is a typed object $\mathfrak{N}_r = (\mathcal{P}_r, \mathcal{A}_r, \mathcal{R}_r, \mathcal{I}_r, \mathcal{U}_r, \mathcal{C}_r, N_r, S_r, b_r)$: a finite patch/overlap category with an isotone local algebra net and central record algebras; a federation of finite carriers, each with twelve primitive pairwise-orthogonal central port projections summing to one and a boundary packet $K = (P, E, F, o)$ with $|P| = 12$, $|E| = 30$, $|F| = 20$, degree-five ports, five-cycle links, and coherently oriented two-face edges, isomorphic to the icosahedral boundary complex with no preferred labels; seam algebras with unital restrictions and coherent triple-overlap cocycles forming the nerve N_r; a finite edge-midpoint refinement support S_r realized orientation-preservingly in S^2 with mesh tending to zero; and a source-bound degree-one bridge $b_r : N_r \to S_r$, all commuting with refinement. Each carrier also has $V_{r,i} = \mathbb{R}[P_{r,i}]$, a finite-dimensional complex Hilbert space $H_{r,i}$, and an injective real-linear map

$$D_{r,i} : V_{r,i} \longrightarrow \mathfrak{u}(H_{r,i}).$$

The image $\mathfrak{g}_{r,i} = D_{r,i}(V_{r,i})$ is closed under commutators and complete for the declared quotient-visible infinitesimal port response. The primitive port probes span $V_{r,i}$, no additional public response direction is omitted, and $-\operatorname{Tr}(D_{r,i}(v)D_{r,i}(w))$ is positive definite. The response construction is natural under admissible presentation equivalence and refinement. A1 does not select a particular response table, inverse-port law, Lie type, global group, particle interpretation, or metric content.

A2 (observer agreement). Observers operating on the screen agree on the meaning of the data they jointly interpret. Formally, the interpretation map $\mathcal{J}_r : \mathsf{Data}_r \to \mathsf{Meaning}_r$ from observer-accessible data to operational meanings is natural with respect to every visible overlap restriction, recharting, seam translation, higher-overlap map, federation map, and refinement map: every declared data-access diagram for accepted public data commutes after interpretation, and $\mathcal{J}_r \circ c_{s\to r} = C_{s\to r} \circ \mathcal{J}_s$ across resolutions. A2 constrains accepted shared data only, and the domain it quantifies over is supplied by the A1 interfaces rather than by A2 itself.

Accepted reversible overlap transports form a groupoid $\mathcal{O}_r$. For a complete carrier chart o, its closed paths $\mathrm{Hol}_r(o)$ map onto the orientation-preserving incidence automorphisms $\mathrm{Aut}^+(K_{r,i})$. For every proper carrier automorphism a, A2 supplies a closed path γ_a and one projective implementer $[U_a] \in \mathrm{PU}(H_{r,i})$ satisfying

$$\mathrm{Ad}_{[U_a]}\, D_{r,i}(v) = D_{r,i}(a \cdot v).$$

Writing

$$G^0_{D,r,i} = \langle \exp(tD_{r,i}(v)) : t \in \mathbb{R},\ v \in V_{r,i} \rangle^0,$$

the implementer is endogenous:

$$[U_a] = [g_a c_a], \qquad g_a \in G^0_{D,r,i}, \quad c_a \in C_{U(H_{r,i})}(\mathfrak{g}_{r,i}).$$

Closed paths compose projectively and the construction commutes with refinement. A unitary lift need not be unique. Scalar phases and centralizer actions are permitted because they act trivially on the response algebra. An independent spectator carrying a matching projective action cannot replace the response-generated factor or contribute a public response direction. The closed path, port action, and implementer must arise from the same accepted source history in an executable realization.

The two response clauses support an abstract target-free compact Lie-type theorem. They do not certify that an executable producer emitted a particular current fixture. Such a certification must reconstruct the port generators, commutator, and projective implementers from ordered source histories without a named current model upstream. A2 does not imply global state extension, repair termination, confluence, unique normal forms, record durability, Byzantine safety, dissemination, or commutation of state optimization with refinement; raw mismatch and repair precede acceptance.

A3 (conditional maximum randomness). Everything that observer agreement leaves unconstrained is maximally random. Formally, at finite regulator r a state is a compatible family of local normalized states on the accessible algebra net; $\mathcal{K}_r$ is the nonempty convex set of such families satisfying the finite observer-visible constraints supplied by A1 and A2 through an A1-generated constraint grammar with a factorization theorem; and, relative to an exact reference family τ_r, an A1-generated observer cover $\mathcal{G}_r$, and strictly positive exact weights $w_{r,P}$ constructed from quotient-visible A1 data by a rule natural under admissible presentation equivalence, the realized state is the information projection. The cover restriction map is injective on $\mathcal{K}_r$; equivalently, at smooth points no nonzero feasible tangent is invisible to the whole cover. The realized state is

$$\rho_r = \operatorname*{arg\,min}_{\rho \in \mathcal{K}_r} \sum_{P \in \mathcal{G}_r} w_{r,P}\, D(\rho_{r,P} \| \tau_{r,P}).$$

The same projection rule applies separately to each declared optimizer object type (ontic state, inference state, or transition distribution), and statements about one type do not transfer to another. This is equivalent to weighted local entropy maximization when every local reference density is identity-proportional in its declared trace. "Random" means least informative relative to the declared reference and the complete agreement constraint set. A3 selects one state inside one fixed feasible space. It does not compare unrelated state spaces or field contents, does not imply optimizer compatibility across refinement, and does not imply recovery, alignment, or mixing; each such statement carries its own theorem, interface, or countermodel.

The complete formal statement of A1–A3, including its constraint map and non-implications, is given in the premise reference.

3 The finite-current Lie-type theorem

Theorem 3.1 (Finite-current fixed-space theorem)**.** *Let $\mathfrak{g} = D(\mathbb{R}^{12}) \subset \mathfrak{u}(H)$ be the image of the complete reversible twelve-port response of premise A1, and let the proper carrier rotations act on it by inner automorphisms of the same response, as premise A2 requires. Then*

$$\mathfrak{g} \cong \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3).$$

The implication assumes no ambient continuous gauge group and is independent of the categorical refinement receipt.

Proof. Faithfulness gives $\dim \mathfrak{g} = 12$. Commutator closure inside $\mathfrak{u}(H)$ makes $\mathfrak{g}$ compact reductive, so $\mathfrak{g} = Z(\mathfrak{g}) \oplus [\mathfrak{g}, \mathfrak{g}]$. The transitive action of the alternating group A_5 on the twelve ports gives $\dim \mathfrak{g}^{A_5} = 1$. Inner automorphisms fix the centre pointwise, so $Z(\mathfrak{g}) \subset \mathfrak{g}^{A_5}$ and $\dim Z(\mathfrak{g}) \leq 1$. If the centre vanished, $\mathfrak{g}$ would be semisimple of dimension twelve; the compact simple algebras of dimension at most twelve have dimensions 3, 8 and 10 [77], so $\mathfrak{g} \cong \mathfrak{su}(2)^4$. The inner action preserves each simple factor of $\mathfrak{su}(2)^4$; simplicity of A_5 makes each restricted action trivial or faithful, with fixed dimension three or zero. No sum of such terms equals one, so this branch is excluded. Hence $\dim Z(\mathfrak{g}) = 1$, the semisimple part has dimension eleven, and compact-simple classification gives the unique split $11 = 3 + 8$. □

The port count fixes the dimension twelve by declaration; the content of the theorem is the exclusion of $\mathfrak{su}(2)^4$ and a one-dimensional centre. The dimension arithmetic, the centre bound and the dimension-six exclusion below are machine checked in the accompanying Lean development [29]. The $\mathfrak{su}(2)^4$ fixed-space step and the compact-Lie classification enter Lean as declared hypotheses, and no source current or full Lie-algebra isomorphism is constructed there. The formal interface keeps the inputs separate: an injective port-response map supplies dimension twelve; an inner carrier action places the centre inside the one-dimensional fixed space; the fixed-space exclusion supplies a nonzero centre; and compact-simple classification supplies the allowed simple dimensions $3, 8, 10$. The final Lean selection theorem takes the last three constraints as hypotheses and proves the split. An observable relative phase alone establishes neither centrality nor trivial carrier action on the whole centre. The trivial carrier action on the centre is a sufficient premise for this Lie type; the proof supplies no independent physical origin for it.

What incidence alone supplies. The certified incidence and inverse-port pairing determine J, with

$$10J = A^3 - 4A^2 - 5A + 10I,$$

where A is the port adjacency. The equivariant linear-response commutant is four-dimensional, and a target-blind impulse and readback protocol produces $R = -J$. Its restrictions to $\mathbf{1} \oplus \mathbf{3} \oplus \mathbf{3}' \oplus \mathbf{5}$ have fixed relative signs; simultaneous reversal is an algebraic response convention and is not identified with matter charge conjugation. These facts constrain the response without constructing the current derivative.

Exact negative controls. The registered single-recurrence interface generates the commutative algebra $\mathrm{span}\{I, A, A^2, A^3\}$. It contains neither twelve independent current generators nor a nonzero bracket, and its only proper recharting is the identity; this bound applies to that registered packet. The target-free diagonal phase lift has twelve independent first derivatives and an abelian bracket. Adding the connected adjacency tangent generates $\mathfrak{u}(12)$, with derived rank 143. The canonical scalar seam equalizers do not supply the current either. On the eleven-dimensional zero-sum port space,

$$\mathrm{rank}\langle [G_e, G_f] \rangle = 49, \qquad \mathrm{Lie}\langle [G_e, G_f] \rangle = \mathfrak{so}(11), \qquad \mathrm{Alg}\langle G_e \rangle = \mathrm{End}(\mathbb{R}^{11}),$$

of dimensions 49, 55 and 121. The strict-descent repair maps therefore cannot be relabelled as the twelve-dimensional compact reversible current, and the repair law acquires no gauge content by definition.

Carrier specificity. Identify the twelve ports in antipodal pairs. The quotient triangulates the real projective plane with six vertices, fifteen edges and ten triangular faces; its edge graph K_6 has twenty 3-cycles. Its transitive A_5 action retains one fixed line. If a six-dimensional compact current carries that action internally, a one-dimensional centre leaves the impossible semisimple dimension five, whereas a zero-dimensional centre gives $\mathfrak{su}(2)^2$ with fixed dimension zero, three or six. No current satisfies all those premises. The adjacency spectrum $5, -1$, with multiplicities one and five, is the antipode-even part of the icosahedral spectrum; the two $\pm\sqrt{5}$ triplets are antipode-odd. Antipodal identification changes topology and port count; forgetting a chosen orientation alone leaves the twelve-port sphere.

4 The current on the executable carrier

The Lie-type theorem consumes a complete response; it does not produce one. On the executable carrier the twelve-current law is a specification; no result derives it from the unchanged native repair. Fixed unrecorded native seam channels annihilate nonzero operators and cannot supply invertible currents. Active correction has a different domain, examined here on one protected native object [26, 29].

A persistent geometric object. The thirty-six nonedge pairs of the native binary load-two carrier admit two source-derived equivariant perfect-matching colourings by the six antipodal axes. Each normalized matching code has dimension six, carries the actual port-permutation representation $1 \oplus 5$, and corrects every complete recorded C/E branch on all thirty native seams with the declared coherent encoder and record-dependent recovery. Its sum-zero quintet is protected. Two such corrected objects contain a unique geometric $3 \oplus 3'$ sector under simultaneous port rotations, with an exact intertwiner inside the full thirty-six-dimensional product code. This is a persistent geometric carrier; it supplies no spin-half lift, exterior matter module or autonomous selection of that sector.

The native exchange response fails the complete-response clause. On opposite colouring choices, the twelve corresponding-port exchanges on all 10626 native load-four configurations yield six independent Hermitian mixed responses that preserve the geometric sector. They use the declared joint-code phase control and are not copied from any target current. Their exact real Lie closure is

$$\mathfrak{g}_{\text{native}} = \mathfrak{su}(3) \oplus \mathfrak{su}(3) \oplus \mathbb{R}\, iI_6, \qquad \dim \mathfrak{g}_{\text{native}} = 17.$$

The proper port rotations belong to this connected response group at the effective-algebra level, so the response realizes an endogeneity bridge on one object. It does not instantiate A1. Conjugation on $\mathrm{End}(\mathbb{C}^6)$ sees sixteen directions, since the common scalar phase is silent there. On the full product code the same preserving menu has nine independent Hermitian operators with a three-dimensional restriction kernel, so its Lie closure has dimension at least twenty and its conjugation-visible response on $\mathrm{End}(\mathbb{C}^{36})$ has dimension at least nineteen. The response lacks the relative phase between 3 and $3'$ and allows five traceless phase directions on each triplet. The exchange law and the specified law below intersect in eleven dimensions, and exact quotients and rotation-fixing channels cannot delete the unwanted quintet while supplying the missing centre. This seventeen-versus-twelve mismatch excludes the exchange grammar; it does not exclude other admissible current laws.

A specified twelve-current law. A distinct specified coherent law realizes the current interface on the same protected object without changing its ports or C/E instruments. For the native port vectors v_p, Galois conjugation σ and cross-product matrix $\widehat{v}$, set

$$D_p = \operatorname{diag}\left(\widehat{v_p/2} + iQ_p,\ \widehat{\sigma(v_p)/2}\right), \qquad Q_p = \tfrac{1}{2}v_p v_p^T - \tfrac{1}{6}\|v_p\|^2 I_3 + \tfrac{1}{12}I_3.$$

The native near-fivefold class sum distinguishes the two triplets before the complex-versus-real control choice is applied. Exact checks give rank twelve, all sixty-six bracket closures and all seven hundred twenty port covariance identities. Proper carrier rotations belong to the same connected current group. The sum of the currents is $\operatorname{diag}(iI_3, 0)$, whose relative phase is observable by cross-triplet readout. The algebra is $\mathfrak{u}(3) \oplus \mathfrak{so}(3) \cong \mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$. Source words execute its primitive currents, composed with the geometric preparation, native repair, record-assisted recovery and pointer outputs. Completeness holds over the specified current-control menu. The coherent compiler and detector services are declared apparatus, and excluding them from the menu does not establish their physical inaccessibility, full physical A1 closure or gauge redundancy.

Controlled limits. A positive native class-sum bias, fresh-bath coupling and a complete eighteen-coefficient disturbance-gradient controller give a source-executed bounded-drive joint limit that approaches twelve observable currents. Every finite filter is invertible on all eighteen input currents; fixed-bias long feedback instead leaves six observable currents. Reference cooling alone leaves every work current available. The bias, baths, actuator and feedback rule are architecture choices that unchanged C/E repair does not select.

4.1 Carrier rechartings, the shared triplet and physical rotations

One three-dimensional A_5 representation on the port vectors appears in three roles. The vectors v_p span the adjacency eigenspace with eigenvalue $\sqrt{5}$, the Laplacian band $L = 5I - A$ of cost $5 - \sqrt{5}$. This band is the rank-three Gram frame and slow repair band that supplies the spatial frame triplet [20, 17]; it is the band selected as the family candidate in Section 7; and the colour block of the specified current acts on it, each generator containing the frame rotation $\widehat{v_p/2}$. Its Galois conjugate $\sigma(v_p)$ spans the $-\sqrt{5}$ band $3'$ on which the weak $\mathfrak{so}(3)$ block acts.

The paper keeps these roles distinct. Premise A2 places the finite carrier rechartings inside the connected internal current group, up to centralizer elements and phases; this is a finite covariance statement and it is a premise. No result here identifies those rechartings with spatial rotations of a recovered interacting field, and representation equivalence alone proves neither an identification nor an independence of physical colour, family and spatial actions. Equivalent triplets can occur on different tensor factors: the specified family-feedback construction uses a gauge-trivial ancillary triplet rather than the coloured current triplet. The local rank-three Gram band and the refining global low band are also distinct: the former has a fixed nonzero carrier energy, whereas the latter concentrates on cell averages.

The conditional spin and matter join constructs commuting actions on $\mathbb{C}^2 \otimes \mathbb{C}^{15}$, with physical-spin central element $-I$ on all thirty states. The diagonal locked action instead has fourteen minus and sixteen plus central eigenvalues and fails that fermionic test. An explicitly chosen compensating internal transport realizes the independent spin action. These are finite existence and obstruction results under a chosen tensor attachment, not a source-selected spacetime connection or a joint charged propagator.

Identifying carrier rechartings with physical rotations while A2 keeps them internal would combine spacetime and internal symmetry. In a relativistic interacting scattering theory that combination is constrained by the Coleman–Mandula theorem under its Poincaré, spectrum, analyticity and

nontrivial-scattering hypotheses.[3] The paper makes no such identification, and those hypotheses are not established for the finite interface. The required test identifies rotations operationally on the same clocked field response, establishes their independence from internal currents up to the declared gauge equivalence, including any allowed compensators, and checks that the dynamics preserves that identification.

5 The categorical route

5.1 Transport from overlap gluing

At a fixed regulator the overlap rechartings of a connected cut Σ form a compact unitary transition system. We use chronological multiplication of comparison maps: U_{ij} maps components in chart j to components in chart i, so an overlap path $p = (i_0 \to \cdots \to i_m)$ has $U_p = U_{i_0 i_1} \cdots U_{i_{m-1} i_m}$. Thus U_p compares the terminal chart to the initial chart; forward transport of column components is $U_p^\dagger$. If $p * q$ means first following p, then q, the conventions are $U_{p*q} = U_p U_q$ and $U_{\overline{p}} = U_p^{-1}$ for the reversed path $\overline{p}$. Gauge-as-gluing, the tree and loop lemmas and the transportability theorem are proved in the spacetime paper [17]. The gauge route consumes the following exact criterion.

On the central-defect branch, identify the overlap centres with one fixed abelian unitary coefficient group Z_Σ on which overlap transport acts trivially. With $U_{ji} = U_{ij}^{-1}$, define the implementer-level multiplier

$$\Omega_{ijk} = U_{ij} U_{jk} U_{ki} = z_{ijk} \in Z_\Sigma,$$

equivalently $U_{ij} U_{jk} = z_{ijk} U_{ik}$. The multiplier is defined at the implementer level: $\mathrm{Ad}(z_{ijk})$ is the identity and does not record the defect.

Theorem 5.1 (Central defect strictification and residual holonomy)**.** *The family $\{z_{ijk}\}$ is an untwisted Čech 2-cocycle with coefficients in Z_Σ, and its class $[z] \in \check{H}^2(N_\Sigma, Z_\Sigma)$ is gauge invariant. A central edge rephasing by a cochain in $C^1(N_\Sigma, Z_\Sigma)$ removes every triangle defect if and only if $[z] = 0$. After such a rephasing, let U^{str} be the resulting edge 1-cocycle and $\mathcal{C}_\alpha$ the full transported charge block, including carrier and multiplicity data. Loop-coherent endpoint-only transport exists on that block if and only if, in addition,*

$$\mathrm{Hol}_{\alpha, U^{\mathrm{str}}}([\gamma]) = U^{\mathrm{str}}_\gamma\big|_{\mathcal{C}_\alpha} = \mathbf{1}_{\mathcal{C}_\alpha} \qquad \textit{for every } [\gamma] \in \pi_1(|N_\Sigma|, i_*).$$

Proof. Conjugation by Ω_{ijk} fixes every triple-overlap observable, so the multiplier is central there; the fixed-coefficient and trivial-action hypotheses place it in Z_Σ. Comparing $(U_{ij}U_{jk})U_{kl}$ with $U_{ij}(U_{jk}U_{kl})$ on a quadruple overlap gives $z_{jkl} z_{ikl}^{-1} z_{ijl} z_{ijk}^{-1} = 1$. A vertex-frame change $U_{ij} \mapsto V_i U_{ij} V_j^\dagger$ conjugates z_{ijk} and leaves a central multiplier unchanged. A central edge rephasing $U_{ij} \mapsto a_{ij}^{-1} U_{ij}$ replaces z by the corresponding coboundary, so $[z] = 0$ is equivalent to removing all triangle multipliers. For two paths p, p' with common endpoints, the residual comparison $U_p U_{p'}^{-1}$ is the holonomy of the closed loop $p * \overline{p'}$, based at their common initial chart. It is the identity for every pair exactly when the displayed holonomy representation is trivial. Conversely, strict path independence makes every triangle comparison and every represented loop action trivial. □

If overlap transport acts nontrivially on a local system of centres, the comparison carries the transported factor and requires the twisted Čech differential. On the genuinely noncentral branch the coefficient data is a crossed module $H_\Sigma \to G_\Sigma$. The vanishing $o_\Sigma^{(2)} = 0$ strictifies the higher

[3] S. Coleman and J. Mandula, "All Possible Symmetries of the S Matrix," *Physical Review* **159**, 1251–1256 (1967), https://doi.org/10.1103/PhysRev.159.1251.

associator, after which at least one allowed strict G_Σ-valued representative must have trivial represented holonomy. The full orbit q_Σ need not determine a unique ordinary H^1 class and is not itself the strictification test. A nonzero $o_\Sigma^{(2)}$ remains a higher-gauge sector, and an orbit whose every strict representative has residual loop action lies outside the ordinary path-independent sector. The class $[z]$ is structurally analogous to an effective 't Hooft anomaly class; a map between them requires a separate anomaly-descent construction, so $[z] = 0$ is used here only as the internal strictifiability condition. The overlap obstruction calculus classifies and routes sectors; it does not select the Standard Model.

5.2 The fixed-cutoff sector category

At cutoff r, choose one common stagewise strict edge cocycle U_r^{str} from the allowed representatives, and retain only seeds on which this representative acts trivially around every closed overlap loop. Sectors requiring incompatible strictifications define alternative fixed-stage categories; they are not unioned before tensor closure.

Definition 5.2 (Fixed-cutoff tensor-generated sector category)**.** *Let S_r be the set of visible collar carriers $W_{\alpha,r}$ with trivial holonomy under the common U_r^{str}. Define*

$$\mathsf{Sect}_r^{\mathrm{bos}} = \langle \mathbf{1}, S_r, S_r^* \rangle_{\oplus,\otimes,\text{subobjects,isomorphism}} \subset \mathrm{Rep}_{\mathrm{fd}}(\widehat{K}_{\Sigma,r}),$$

the replete full additive Karoubi rigid tensor subcategory generated by the finitely many visible carriers. Its simple objects are the irreducible summands of finite tensor words in seeds and conjugates; there may be infinitely many of them although every object and Hom space is finite-dimensional.

Theorem 5.3 (Fixed-cutoff bosonic sector category)**.** *At every fixed regulator cutoff, on the ordinary or central-defect zero-obstruction branch and in the bosonic internal-gauge sector of the $3+1$-dimensional effective regime, $\mathsf{Sect}_r^{\mathrm{bos}}$ is a semisimple rigid symmetric C^*-tensor category with one monoidally coherent path-independent transport. The carrier-forgetful functor $F_r : \mathsf{Sect}_r^{\mathrm{bos}} \to \mathsf{Hilb}_{\mathrm{fd}}$ is faithful, *-preserving and symmetric strong monoidal.*

Proof. Complete reducibility of finite-dimensional representations of the compact group gives semisimplicity and subobjects; the operator adjoint and norm give the C^*-structure; conjugate representations give duals; the flip gives the bosonic symmetry. Around a closed loop U_r^{str} acts as the identity on every retained seed, hence on direct sums, tensor products, conjugates and invariant subobjects. Collar concatenation realizes tensor words whenever the finite tensor-realization receipt is supplied. □

Fermionic signs, spinorial matter and chirality are not part of this bosonic category; they belong to the matter lift of Section 6.

5.3 Refinement and compact-group reconstruction

Definition 5.4 (Compact-gauge refinement receipt)**.** *On a cofinal tail, the receipt supplies: a surjectivity certificate for each finite-state restriction $\rho_{sr} : Q_s \to Q_r$; multiplicities $m_{\beta\alpha}^{rs}$ and unitaries V_β^{rs} with $d_{\beta,s} = \sum_\alpha m_{\beta\alpha}^{rs} d_{\alpha,r}$, defining the injective block embedding*

$$\big(\jmath_{rs}(a)\big)_\beta = V_\beta^{rs} \left[\bigoplus_\alpha (a_\alpha \otimes \mathbf{1}_{m_{\beta\alpha}^{rs}}) \right] (V_\beta^{rs})^*$$

of the collar algebras, coherent under composition; continuous surjections $\pi^{\Sigma}_{sr} : \widehat{K}_{\Sigma,s} \twoheadrightarrow \widehat{K}_{\Sigma,r}$ with $\pi^{\Sigma}_{tr} = \pi^{\Sigma}_{sr} \circ \pi^{\Sigma}_{ts}$; one common strict representative at each stage, intertwined by the pullbacks; and compatible finite concatenated-collar realizations of every tensor word and invariant subobject projector. The finite-state consensus maps alone imply none of these data.

Theorem 5.5 (Refinement functor and fibre descent)**.** *On a tail carrying the receipt, $U_{rs} := (\pi^{\Sigma}_{sr})^* : \mathsf{Sect}^{\mathrm{bos}}_r \to \mathsf{Sect}^{\mathrm{bos}}_s$ is fully faithful, symmetric strong monoidal and *-preserving; it preserves the unit, duals, irreducibles and the full zero-obstruction criterion, with $U_{st}U_{rs} = U_{rt}$ and $F_sU_{rs} = F_r$ on carrier spaces.*

Proof. Set $U_{rs}(H_X, \varrho_X) = (H_X, \varrho_X \circ \pi^{\Sigma}_{sr})$ and $U_{rs}(f) = f$. Surjectivity gives

$$\mathrm{Hom}_{\widehat{K}_{\Sigma,s}}(U_{rs}X, U_{rs}Y) = \mathrm{Hom}_{\widehat{K}_{\Sigma,r}}(X, Y),$$

so pullback is fully faithful and a coarse simple cannot split. Pullback commutes with tensor products, unit, conjugates, duality maps and the flip. The receipt intertwines the stagewise strict representatives, so zero holonomy persists for the whole category. Pullback changes only the action, so the fibre comparison is the identity. □

On such a tail, the directed colimit $\mathsf{Sect}_\infty := \varinjlim_r \mathsf{Sect}^{\mathrm{bos}}_r$ retains the sectors and intertwiners that persist, and the stagewise carriers descend to a faithful bosonic fibre functor $\mathcal{F} : \mathsf{Sect}_\infty \to \mathsf{Hilb}_{\mathrm{fd}}$. A realized zero-obstruction class with representatives satisfying $U_{st}\alpha_s \simeq \alpha_t$ determines a unique object of the colimit. Without the receipt only the fixed-stage categories are constructed.

Theorem 5.6 (Receipt-conditional Tannaka reconstruction)**.** *On the ordinary or central zero-obstruction bosonic branch with a cofinal tail carrying the receipt, Sect_∞ is a rigid symmetric C^*-tensor category with faithful bosonic fibre functor $\mathcal{F}$, $G := \mathrm{Aut}_\otimes(\mathcal{F})$ is a compact group, and $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$, with G unique up to isomorphism. If the category is trivial, G is trivial.*

Proof. The refinement functors preserve tensor products, unit, duals and *-structure, so the colimit inherits a rigid symmetric C^*-tensor structure and the carriers descend to $\mathcal{F}$. A unitary symmetric monoidal natural automorphism η embeds as $(\eta_X)_X$ in $\prod_X U(\mathcal{F}(X))$. Naturality, $\mathcal{F}(f)\eta_X = \eta_Y\mathcal{F}(f)$, and monoidality, $\eta_{X\otimes Y} = J_{X,Y}(\eta_X \otimes \eta_Y)J^{-1}_{X,Y}$ with $\eta_{\mathbf{1}} = \mathrm{id}$, are closed relations, so G is a closed subgroup of a compact product. Doplicher–Roberts and Tannaka reconstruction give $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$ [66, 67], and two groups reconstructing the same category with compatible fibre functors are both the automorphism group of that functor. □

The local-Gibbs and collar-mixing package is not used to prove nontriviality or to supply the receipt. The theorem is neutral on whether Sect_∞ is trivial: it returns whatever compact group the tensor-fibre data reconstructs and does not select $(\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1))/\mathbb{Z}_6$.

Definition 5.7 (Field-net realization hypotheses)**.** *(R1) An isotonous net $O \mapsto \mathcal{A}(O)$ of C^*-algebras over the small-region collar localizations, with a faithful vacuum representation. (R2) Spacelike commutativity, Haag duality or its declared essential variant, and Property B. (R3) Each persistent class $[\alpha]$ is realized by a localized endomorphism ρ^{α}_O. (R4) Transportability by unitary intertwiners in $\mathcal{A}$; vanishing collar holonomy is a necessary input to (R4) and is not equivalent to it. (R5) Finite permutation-symmetric statistics and conjugates matching Sect_∞. (R6) The assignment $[\alpha] \mapsto [\rho^{\alpha}]$ is an equivalence onto the localized transportable finite-statistics endomorphisms.*

Corollary 5.8 (Two-level reconstruction)**.** *Level (i), proved: on the receipt-certified branch, $\mathsf{Sect}_\infty \simeq \mathrm{Rep}(G)$ with no observable net involved. Level (ii), conditional: under (R1)–(R6) the Doplicher–Roberts theorem [66, 67] supplies a bosonic field net $\mathcal{F}_{\mathrm{net}}$ carrying a G-action whose field operators implement every persistent sector, with $\mathcal{A} = \mathcal{F}^G_{\mathrm{net}}$.*

Proof. Level (i) is Theorem 5.6. Under (R1)–(R6) the localized transportable endomorphisms form a category equivalent to Sect_∞ by (R6), hence to $\mathrm{Rep}(G)$; the Doplicher–Roberts inputs are exactly (R2)–(R6) on the net (R1). □

This paper constructs the categorical side only; no observable net is specified on which the persistent sectors act, so every fixed-point identity $\mathcal{A} = \mathcal{F}_{\mathrm{net}}^{G}$ is conditional on (R1)–(R6).

Lemma 5.9 (Product factorization)**.** *If* $\mathsf{Sect} \simeq \mathsf{Sect}_1 \boxtimes \mathsf{Sect}_2$ *identifies the reconstructed fibre functor with* $\mathcal{F}_1 \boxtimes \mathcal{F}_2$*, where each* $\mathcal{F}_i$ *is a faithful symmetric tensor fibre functor, then* $\mathrm{Aut}^{\otimes}(\mathcal{F}) \cong G_1 \times G_2$ *with* $G_i = \mathrm{Aut}^{\otimes}(\mathcal{F}_i)$ *and* $\mathsf{Sect}_i \simeq \mathrm{Rep}(G_i)$*.*

Proof. A tensor-natural automorphism of the external-product functor restricts to each factor by evaluation on $(X, \mathbf{1})$ and $(\mathbf{1}, Y)$, and the two restrictions determine it on every (X, Y) by tensor naturality. Reconstruction on each factor gives the representation equivalences. □

A bare categorical factorization without compatible fibre functors does not establish the product statement, and factorized heat-kernel weights on one state do not reconstruct a product group. If, independently, the finite Lie-type theorem certifies the two nonabelian factors and the charge theorem certifies one connected U(1), the connected realization is $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ up to the separately computed finite kernel.

6 Conditional matter realization

6.1 The matrix current, spin lift and exterior module

The charged-double-triplet matrices are an exact conditional realization of the premise-implied Lie type. A separate geometric-response construction probes all matrix units of $M_6(\mathbb{C})$ and resolves twelve current generators. The declared spin lift sends a six-state generator $D = \mathrm{diag}(A, \widehat{w})$ to

$$L(D) = \mathrm{diag}\left(A, -\frac{i}{2} w \cdot \sigma - \frac{\mathrm{tr}\, A}{2} I_2\right).$$

Together with the declared central trace redistribution this is the standard embedding $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3) \subset \mathfrak{su}(5)$. The nonvacuum even exterior module $M = \Lambda^2\mathbb{C}^5 \oplus \Lambda^4\mathbb{C}^5$ has dimension fifteen and is the Georgi–Glashow $\mathbf{10} \oplus \bar{\mathbf{5}}$ generation [117]. Inside the declared ten-component exterior menu, the exhaustive 1024-subset anomaly scan selects $P_{\mathrm{even}} - P_{\mathrm{vac}}$ and $P_{\mathrm{odd}} - P_{\mathrm{top}}$ as the unique charge-conjugate pair of rank-15 chiral projectors. Source-computed charges and Casimirs identify Q, u^c, d^c, L, e^c. One declared Higgs doublet gives three invariant interaction lines Qu^cH, Qd^cH^* and Le^cH^*, and with three family copies these admit three arbitrary complex 3×3 Yukawa matrices. The spin cover, exterior-sector choice and physical fermion field are separate attachments; even auxiliary exterior parity does not establish physical fermion statistics, and the decoder of these labels is a representation check rather than a particle identification [21].

Lemma 6.1 (Factor identification)**.** *Let* H *be a positive-dimensional compact connected Lie group. (a) If* H *has a faithful irreducible two-dimensional pseudoreal unitary representation, the connected derived subgroup of the image is* $\mathrm{SU}(2)$ *acting as the doublet. (b) If* H *has a faithful irreducible three-dimensional representation whose nonabelian factor acts irreducibly and inequivalently to its conjugate, the connected derived subgroup of the image is conjugate to* $\mathrm{SU}(3)$ *on the fundamental, up to finite central kernel. (c) A connected abelian charge factor acting nontrivially on the coupled carrier gives a one-torus in the abelianized group.*

Proof. By Schur the centre acts by scalars, so the derived subgroup acts irreducibly. On $\mathbb{C}^2$ the only connected compact groups acting irreducibly have derived subgroup SU(2). On $\mathbb{C}^3$, two nontrivial simple factors would force a tensor decomposition of dimension at least four; among compact simple algebras only $\mathfrak{su}(2)$ and $\mathfrak{su}(3)$ have nontrivial irreducibles of dimension at most three, and the three-dimensional representation of $\mathfrak{su}(2)$ is real. Part (c) is immediate. □

Irreducibility in (a) is load-bearing: U(1) acting by $\mathrm{diag}(z, z^{-1})$ is faithful, pseudoreal and reducible. The intrinsically complex hypothesis in (b) is load-bearing: $\mathrm{U}(1) \times \mathrm{SO}(3)$ acting by zR satisfies every other hypothesis. None of these lemmas enters Theorem 3.1.

Proposition 6.2 (Conditional matter image)**.** *Assume the certified twelve-port carrier, the response and endogenous transport clauses, the declared matrix current, the declared fermionic spin category and the resulting conjugate projector pair. Anomaly-forced determinant balance and tensor descent on every realized matter tensor give: either projector supplies the one-generation chiral module with weak-doublet and colour-triplet roles on a common carrier $\mathbb{C}^3 \otimes \mathbb{C}^2$; the exact Standard Model hypercharge lattice up to simultaneous charge reversal; $N_c = 3$; the conjugation-insensitive kernel $\mathbb{Z}_6$; and compatible scalar charges and Yukawa channels. The maximal faithful matter image is*

$$G_{\mathrm{packet}} = \frac{\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)}{\mathbb{Z}_6}.$$

The Lie type uses the carrier, the response clauses and classification; the other conclusions use the declared matrix and matter premises. The response-sign convention and the choice between conjugate projectors are separate, and no theorem identifies them. The cover and its $\mathbb{Z}_2$ and $\mathbb{Z}_3$ quotients carry the same local tensors, so local data do not select the physical global form (Section 6.2). The proposition attaches no laboratory current and does not identify G_{packet} with the Tannaka group.

Theorem 6.3 (Hypercharge from anomaly freedom and Yukawas)**.** *Assume* $\mathrm{SU}(N_c) \times \mathrm{SU}(2) \times \mathrm{U}(1)_Y$*, one generation of left-handed Weyl fermions* (Q, u^c, d^c, L, e^c)*, and the Yukawa channels* QHu^c*,* $QH^\dagger d^c$*,* $LH^\dagger e^c$*. Then anomaly freedom and Yukawa invariance fix the hypercharges up to normalization; with* $Q = T_3 + Y$ *and* $Q(\nu_L) = 0$*,*

$$Y_Q = \tfrac{1}{2N_c}, \qquad Y_L = -\tfrac{1}{2}, \quad Y_e = 1, \quad Y_u = -\tfrac{N_c+1}{2N_c}, \quad Y_d = \tfrac{N_c-1}{2N_c}, \quad Y_H = \tfrac{1}{2}.$$

For $N_c = 3$*:* $Y_Q = \frac{1}{6}$*,* $Y_u = -\frac{2}{3}$*,* $Y_d = \frac{1}{3}$*.*

Proof. Yukawa invariance gives $Y_u = -(Y_Q + Y_H)$, $Y_d = -Y_Q + Y_H$, $Y_e = -Y_L + Y_H$. The $\mathrm{SU}(2)^2\mathrm{U}(1)$ condition $N_c Y_Q + Y_L = 0$ and the mixed gravitational condition $2N_c Y_Q + N_c Y_u + N_c Y_d + 2Y_L + Y_e = 0$ give $Y_L = -N_c Y_Q$, $Y_H = N_c Y_Q$, $Y_u = -(N_c+1)Y_Q$, $Y_d = (N_c-1)Y_Q$, $Y_e = 2N_c Y_Q$; the $\mathrm{SU}(N_c)^2\mathrm{U}(1)$ and cubic anomalies then vanish. The normalization gives $Y_Q = 1/(2N_c)$. Without the Yukawa channels the cubic anomaly leaves two branches exchanging Y_u and Y_d. □

This is the textbook argument applied to the declared packet; the exact rationals reproduce the Standard Model because every discrete input is the Standard Model's. The channel scan selects the compatible scalar charge branch, not scalar attachment, multiplicity or a one-Higgs completion. Per generation there are $N_c + 1 = 4$ weak doublets, so the global SU(2) anomaly [72] is absent on the realized branch; this is a consistency check on $N_c = 3$, not a selector.

Theorem 6.4 (Conditional charge quantization)**.** *If the maximal faithful image is* G_{packet} *and* $Q = T_3 + Y$ *on a separately assumed electroweak branch, every colour-singlet representation of the packet has integer charge.*

Proof. The $\mathbb{Z}_6$ quotient identifies $(e^{2\pi i/3}, -1, e^{i\pi/3})$ with the identity. On a colour singlet with weak spin j and $n = 6Y$, this requires $(-1)^{2j} e^{i\pi n/3} = 1$, so Y is an integer for integer j and a half-integer for half-integer j. In both cases $T_3 + Y \in \mathbb{Z}$. □

The statement concerns packet representations, not a quantum field realization. It is consistent with the bound $|q_p + q_e|/e < 1 \times 10^{-21}$ on matter neutrality, with bulk fractional-charge abundance limits near 10^{-22} per nucleon, and with collider exclusions of stable particles with $q \in [e/3, 0.9e]$ up to masses of about 640 GeV at 95% CL; these searches do not discharge its premises.

Corollary 6.5 (No simple-GUT X/Y gauge channel)**.** *On the connected* $\mathrm{SU}(3) \times \mathrm{SU}(2) \times \mathrm{U}(1)$ *realization and the twelve-dimensional current algebra, the adjoint is* $(8,1,0) \oplus (1,3,0) \oplus (1,1,0)$*. There are no gauge generators in* $(3,2,\pm 5/6)$*, so no ordinary* X, Y *exchange inside the current algebra.*

Heavier modes outside the algebra, scalar mediators and other ultraviolet mechanisms are not excluded. Conditional on the one-generation table, its accidental baryon and lepton labels and the declared local dimension-six grammar, an exhaustive census of the 715 four-field multisets admits four Hermitian-conjugacy classes, $QQQL$, QQu_Re_R, QLu_Rd_R and $u_Ru_Rd_Re_R$, whose exact exterior contractions remain nonzero after the Schouten, Fierz and identical-field Pauli relations. The diagonal $\mathbb{Z}_6$ generator fixes every field, so the gauge structure and the quotient do not prove proton stability; the census supplies no coefficient or lifetime. In the same way the mixed hypercharge anomaly gives $k_R^{YWW} = N_g(3Y_Q + Y_L) = 0$, so the $\mathbb{Z}_6$ deck phase supplies no direct $\Theta_R W\widetilde{W}$ baryogenesis coupling; a global $B + L$ attachment would give $k_R = 2N_g$.

6.2 Global form

Four central global forms act consistently on the same local tensors: the cover and its $\mathbb{Z}_2$, $\mathbb{Z}_3$ and $\mathbb{Z}_6$ quotients. The kernel theorem identifies the maximal faithful local matter image. On the finite weight lattice of hypercharge, triality and duality labels in the $q = 6Y$ normalization, a label is fixed by every element of the diagonal $\mathbb{Z}_6$ kernel exactly when

$$2t + 3d + q \equiv 0 \pmod 6,$$

and the kernel acts on any label by multiples of this centre character. Every row of the matter table satisfies the congruence, including both carrier blocks and the scalar; a fractionally charged singlet is blocked by the generator; and exactly one sixth of the thirty-six lattice labels descend, each character value being carried by six labels. A machine-checked receipt composes this onto the selected matter mask, so every selected component descends through the full kernel. The six-axis artifact computes an order-six flux menu after its coefficient relations are declared; the menu matches the $\mathbb{Z}_6$ quotient and carries the declared matter single-valuedly.

The order-six seam classification keeps two actions separate. Every pure-hypercharge character is enumerated: the fixed-space dimension menus are $\{15, 7\}$ at order two, $\{15, 3\}$ at order three and $\{15, 7, 3, 1\}$ at order six, while the canonical diagonal $\mathbb{Z}_6$ kernel acts trivially on all fifteen matter states. These are different representations of the same abstract cyclic order, and refinement preserves the classified actions. The calculation selects no physical seam character, flux action or 2-representation and does not exhaust every seam grammar. Selection of one physical quotient among the four requires a source-derived complete character category, a same-source loop-to-kernel identity, four-dimensional instanton normalization and theta periodicity, monopole dynamics and a laboratory flux attachment; the congruence derives no physical global form.

Corollary 6.6 (Structural electroweak package)**.** *On a separately declared one-Higgs branch, the electroweak factor has Lie algebra* $\mathfrak{su}(2)_L \oplus \mathfrak{u}(1)_Y$*. A nonzero neutral vacuum vector* $\phi_0 = \frac{v}{\sqrt{2}}\binom{0}{1}$ *of* $H = (1,2)_{1/2}$ *has* $Q\phi_0 = (T_3 + Y)\phi_0 = 0$ *and stabilizer* $\mathrm{U}(1)_Q$*; the package contains* $W^{\pm} = (W^1 \mp iW^2)/\sqrt{2}$*,* Z*,* A *and the exact charge operator* $Q = T_3 + Y$*.*

Since $e^{i\alpha T_3}e^{i\beta Y}\phi_0 = e^{i(\beta-\alpha)/2}\phi_0$, vector invariance ties $\beta = \alpha$, while the projective ray $[\phi_0]$ has the two-torus stabilizer $\mathrm{U}(1)_{T_3} \times \mathrm{U}(1)_Y$; projectivization alone does not select the electromagnetic diagonal. A machine-checked composition on the committed doublet carrier proves, for declared $\lambda, v, g, g' > 0$ and Yukawa coefficient y, the minimizer shell $|h|^2 = v^2$, one unbroken line with three Goldstone directions, broken mass values $g^2v^2/4$ and $(g^2+g'^2)v^2/4$, and a Yukawa bilinear of coefficient yv. The mixing angle, v and the W/Z masses belong to the running and matching surface. On the unbroken factor, a supplied Lorentzian Maxwell action $-\frac{1}{2g_Q^2}\int F_Q \wedge *F_Q + \int A_Q \wedge *J_Q$ with $F_Q = dA_Q$ gives $dF_Q = 0$ and $d * F_Q = g_Q^2 * J_Q$; the reconstruction does not supply that action or its coefficient. In a tensor-network realization a faithful representation of $\mathrm{SU}(3) \times \mathrm{SU}(2)$ needs edge dimension $\chi \geq 5$, and $\chi \geq 6$ only under the additional declared requirement of a single irreducible tensor-product carrier.

7 Families and the generation window

Anomaly cancellation is generation by generation and does not fix the number of generations.

Proposition 7.1 (Conditional generation window)**.** *On a separately declared one-Higgs quark branch over the matter image of Proposition 6.2, with* $N_c = 3$ *and the declared clauses of intrinsic CKM CP capability and weak-sector ultraviolet completability, the admissible window is* $3 \leq N_g \leq 5$*. No count inside the window is selected.*

Proof. An $N_g \times N_g$ CKM matrix has $(N_g - 1)(N_g - 2)/2$ physical CP phases, which vanish for $N_g \leq 2$; the declared CP-capability clause therefore gives $N_g \geq 3$. The one-loop coefficient $b_{\mathrm{SU}(2)} = \frac{22}{3} - \frac{1}{3}N_g(N_c+1) - \frac{1}{6}$, with the last term from one complex Higgs doublet, is positive exactly when $N_g(N_c+1) < 43/2$; with $N_c = 3$ this gives $N_g \leq 5$. □

The lower edge is the declared CP-capability clause; the window is the complete numerical content of the two clauses and is not forced by anomaly cancellation, the A_5 graph or the target-free source reduct.

Proposition 7.2 (Conditional screen-band selection)**.** *The twelve-port coefficient module decomposes into bands of ranks* $1, 3, 3', 5$*, with exact operational costs of the nontrivial bands*

$$5 - \sqrt{5} < 6 < 5 + \sqrt{5}.$$

Assume that one complete faithful band in the declared rank window $3 \leq r \leq 5$ *realizes the family carrier and that the lower cost selects between admissible complete bands. Then the rank-three band of cost* $5 - \sqrt{5}$ *is unique. Tensoring it with the rank-*15 *generation gives a conditional rank-*45 *candidate whose chirality and diagonal* $\mathbb{Z}_6$ *action come from the generation table.*

These implications are kernel-decided in the Lean screen certificate. A response receipt using the graph Laplacian as a unitary generator recovers the same cost bands and places the rank-three residue at the lowest positive generator frequency; the channel conserves mode norms, so this is a frequency statement rather than a relaxation statement. For the four-entry band table, faithfulness

excludes the dimension-one band and leaves dimensions $3, 3, 5$; removing the window while keeping faithfulness changes no admissible candidate, so the window does not itself explain triplication. The load-bearing identification is that the family object is a single complete faithful band, compared by the declared cost rule and realized in the matter response. A specified source-executed cooling law computes this rank-three band on the native carrier; selecting that rule and identifying its levels with physical families are inputs.

Generation names require no universally conserved generation charge. Within a sector whose charges and statistics are established, ordering nondegenerate masses is a naming convention, and a physical comparison must retain the eigenvectors, residues and charged-current matrix of the same response. The mismatch of the two left quark diagonalizers determines CKM mixing; a generation label conserved by every charged weak transition would exclude that mixing.[4]

Corollary 7.3 (Declared minimal completion)**.** *Assume Proposition 6.2, the receipt-certified bosonic branch, a one-Higgs scalar attachment, the CP and weak-sector clauses, and a declared physical attachment of the rank-three band to the matter packet. Then the completed packet has maximal faithful image data* $(G_{\mathrm{packet}}, \mathcal{R}_{\mathrm{SM}}, H, \mathcal{Y}_{\mathrm{SM}})$ *with* $N_c = 3$, $N_g = 3$ *and*

$$Q_i = (3,2)_{1/6}, \quad u_i^c = (\bar{3},1)_{-2/3}, \quad d_i^c = (\bar{3},1)_{1/3}, \quad L_i = (1,2)_{-1/2}, \quad e_i^c = (1,1)_1, \quad H = (1,2)_{1/2},$$

$i = 1, 2, 3$.

The Lie type uses no clause of the completion; the matter type, charges and $N_c = 3$ are exact inside the declared fixture; $N_g = 3$ and the one-Higgs scalar enter through the completion declaration. Under the fixed-cutoff heat-kernel law of Section 10 every finite-dimensional witness label has positive weight; that positivity concerns bosonic sector labels and supplies no light chiral multiplet. Physical identification of the rank-45 candidate requires a source-derived matter-pole map, continuum spin and locality, physical seam selection, residues, excluded-band and refinement receipts; the absence of other light sectors is unproved.

8 The equivariant bracket variety and a face discriminator

Conditional on the canonical oriented carrier, the target-free space $\mathrm{Hom}_{A_5}(\Lambda^2\mathbb{Q}^{12}, \mathbb{Q}^{12})$ has dimension fourteen and an exact rational Reynolds basis. It is a search space; no bracket is selected. Substitution of the general bracket into Jacobi gives 2,640 coordinate equations in 44 target orbits with coefficient-row rank 38. Over $\mathbb{Q}(\sqrt{5})$ that rowspace splits into eleven fixed-line derivation-weight products and a rank-27 residual system that applies on every weight flat. A three-parameter derivation-only Jacobi family supplies no compactness or source selection. Conditional on three named textbook lemmas, the compact real locus of the fourteen-parameter equivariant variety is classified exactly over $\mathbb{Q}(\sqrt{5})$ into three certified bracket families F, G, P. For each family the coroot lattices, integral carrier weights and carrier-admitted global forms are computed; the selected forms are adjoint with zero abelian charge lattice, so the abelian sector is undetermined at carrier level. Noncompact and residual components outside the compact locus are unclassified.

Applying the declared equal-weight cyclic rule to the twenty oriented faces gives $B_{\mathrm{face}} = 60R_{13}$. This bracket is A_5-equivariant and fails Jacobi in exactly 240 of the 2640 coordinates, 120 at $+1$ and 120 at -1. Under the added Euclidean metric on upper-triangular structure constants, orthogonal projection onto the compact locus has the unique nearest family G:

$$d_G^2 = \frac{615 - 123\sqrt{5}}{22}, \qquad d_F^2 = \frac{615 + 123\sqrt{5}}{22}, \qquad d_P^2 = 45.$$

[4] A. Ceccucci, Z. Ligeti and Y. Sakai, "CKM Quark-Mixing Matrix," in the Particle Data Group review (2024), Section 12.1, https://pdg.lbl.gov/2024/reviews/rpp2024-rev-ckm-matrix.pdf.

Both F and G projections lie in their open compact strata. On the same 792-coordinate convention, exact primal–dual certificates give

$$d_1(G) = 30(\sqrt{5}-1) < \inf d_1(F) = d_1(P) = 60,$$

$$d_\infty(G) = \frac{5-\sqrt{5}}{10} < d_\infty(F) = \frac{\sqrt{5}}{5} < d_\infty(P) = \frac{1}{2}.$$

The ℓ_1 value for F is an infimum approached under scaling, not an attained minimum. The Python replay proves the primal–dual optimization and the Lean theorem checks the radical values and their order. The family verdict holds across the ℓ_1, ℓ_2 and ℓ_∞ scores on the classified compact locus once the coordinate norm is supplied.

The quadratic verdict is robust over the invariant class. The carrier splits multiplicity-free as $1+3+3'+5$, the commutant of the port action is spanned by the four spectral projectors, and every invariant carrier inner product is $\alpha P_1 + \beta P_3 + \gamma P_{3'} + \delta P_5$. The induced distances are

$$d_P^2 = \frac{15-3\sqrt{5}}{\beta} + \frac{15+3\sqrt{5}}{\gamma} + \frac{15-3\sqrt{5}}{2}\frac{\beta}{\delta^2} + \frac{15+3\sqrt{5}}{2}\frac{\gamma}{\delta^2},$$

$$d_F^2 = \frac{15+3\sqrt{5}}{2}\frac{\gamma}{\delta^2} + \frac{15+3\sqrt{5}}{\gamma} + \frac{60+12\sqrt{5}}{11\beta}, \qquad d_G^2 = \frac{15-3\sqrt{5}}{2}\frac{\beta}{\delta^2} + \frac{15-3\sqrt{5}}{\beta} + \frac{60-12\sqrt{5}}{11\gamma},$$

with α absent. Both gaps $d_P^2 - d_G^2$ and $d_P^2 - d_F^2$ are coefficient-positive Laurent polynomials, so P is excluded for every invariant metric. At $\gamma = \beta$ the gap $d_F^2 - d_G^2 = 3\sqrt{5}\,\beta/\delta^2 + (90\sqrt{5}/11)/\beta$ is positive, and for $\beta/\delta \in [1/50, 6]$ the family G is uniquely nearest for all γ, δ. The mirror family F is nearer on the other side of the exact three-scale tie surface, with witness $(\beta, \gamma, \delta) = (8, 1, 1)$. Conjugating $\sqrt{5}$ and swapping $\beta \leftrightarrow \gamma$ maps d_G to d_F. Reversing every face orientation negates B_{face} and leaves the distances unchanged. A channel-diagonal invariant metric that is not carrier-induced reverses the balanced-point selection, so the carrier-induced restriction is load-bearing. On the pinned incidence table, equal normalized face weights and the declared barycentric one-third share give port-dual weight $1/12$ at every port; under the declared diagonal measure-to-metric rule this uniform measure induces the scalar carrier inner product at $(1/12, 1/12, 1/12, 1/12)$, where G is uniquely nearest with $d_G^2 = (3690 - 738\sqrt{5})/11$, $d_F^2 = (3690 + 738\sqrt{5})/11$ and $d_P^2 = 540$.

The equal-weight rule, the coordinate norms, the nearest-point law and the measure-to-metric rule are declared inputs, not consequences of A1–A3, so the verdict is a conditional discriminator obtained without reference to a measured value. A composed machine-checked surface bundles these premises with the classification inputs and the exterior selection, and proves that the premise-implied type has a unique type-matched nearest family at the balanced point and that the fifteen-state count, the anomaly forms and the $\mathbb{Z}_6$ stabilizer are carried by the bundled selection mask. Among the three family labels and the declared 2^{10} exterior table, the admissible pairs are G with either of two conjugate masks. This does not count points of the continuous G-family or classify physical Standard Model structures.

Over the certified brackets, ad-invariance fixes the kinetic shape block-diagonally with zero cross terms. On carrier-projector forms, families F and G reduce the three-dimensional invariant space to two dimensions, $F:\ w_{3+} = \sqrt{5}\,w_5$ and $G:\ w_{3-} = \sqrt{5}\,w_5$, with P the two-to-two control; simultaneous invariance for F and G leaves the ray $(w_{3+}, w_{3-}, w_5) = t(\sqrt{5}, \sqrt{5}, 1)$, an extra premise. The two-factor history theorem constructs a Gibbs kernel from $aq_1 + bq_2$ with exact factorization; equality of two such kernels identifies λa and λb, so only common scaling survives and a/b is not multiplier gauge. No theorem supplies relative factor coefficients or coupling scales [28, 29].

8.1 An assembled structural action

On one typed premise bundle, the gauge kinetic term (one declared magnitude per invariant carrier sector $3^+ + 3^- + 5$, weighted in the two-parameter G-invariant normal form), the quartic potential $\lambda(|h|^2 - v^2)^2$, the gauge mass form at the chosen minimum and the realified Yukawa line with coefficient y sum to one real function of one field tuple. Exact restriction theorems recover each sector on its slice; the gauge coefficients satisfy the G ad-invariance row and collapse to the mirror ray under the extra F row; the cross-sector generators realizing those rows have coefficient ratio $(1, -\sqrt{5})$, and the control weight $(1, 1, 1)$ breaks invariance with exact value $1 - \sqrt{5}$. A fourth summand carries one complex amplitude per row of the ten-component table; its bilinear $\overline{\psi_i}\psi_j$ is phase-invariant on the selected rows exactly when the charges $q = 6Y$ balance, and the charge column forces the pairing to be diagonal. On the shared doublet and singlet rows the extended action evaluates to the mass bilinear with coefficient yv. Every coefficient is a declared parameter; the construction supplies no source-produced spacetime action, measured coupling, family attachment, global-form selection, chirality or spin attachment. The local covariant action, its Cartan reduction and the fermion current are given in the appendices.

9 Refinement stability and relevant transverse perturbations

The state-side refinement stability used by the gauge branch is a separate optimizer-pushforward and refinement-tail interface: it keeps the finite local constraint family across cutoffs, so that a refinement channel acts on the realized Gibbs family $\omega_\ell(\lambda) \propto \exp(-\sum_{x,a} \lambda_a O_a(x) - \sum_b \mu_b Q_b)$ by an induced map $\omega_L(R_{\ell\to L}(\lambda))$. Coarse-graining a finite-range Gibbs family generically leaves any fixed finite list, so this is a substantive premise; premise A3 supplies the fixed-stage optimizer and not this cross-scale statement or the refinement receipt. The square defects that control reconciliation under coarse-graining are treated in the consensus paper [19].

Proposition 9.1 (Conditional growth of a relevant transverse component)**.** *Assume the local finite-constraint branch of A3, the refinement-tail interface, and a differentiable extension of the refinement maps to an enlarged coupling space with transverse coordinate $g_{\mathcal{O}}$ for a gauge-invariant Lorentz-scalar deformation $\mathcal{O}$. If $\delta g_{n+1} = \Lambda_n \delta g_n$ with $|\Lambda_n| \geq q > 1$, every nonzero infinitesimal transverse component grows by at least q^m after m steps. If the zero-coupling surface is invariant and the trajectory starts on it, relevance alone does not generate $\mathcal{O}$.*

Proof. Iteration gives $|\delta g_{n+m}| = \prod_j |\Lambda_{n+j}||\delta g_n| \geq q^m |\delta g_n|$. Generation from zero concerns the transverse part of the full extended map, which is not a coordinate of $R_{\ell\to L}$ and is not supplied by A3. Exact tuning, an invariant manifold, vanishing operator mixing, cancellations, an additional symmetry or fixed-point dynamics can keep the component zero, so the statement is not an exhaustive protection theorem. □

A gauge-invariant Dirac mass is a relevant scalar when the representations allow it. With nonzero mixing into that coordinate, a nonzero mass perturbation is linearly amplified unless a protecting mechanism intervenes. Chiral matter forbids the Dirac mass by symmetry; vector-like branches with vanishing mixing, tuning or fixed-point protection are not excluded, so the diagnostic does not select chirality by itself.

10 Edge-sector weights and coupling extraction

The edge-centre completion yields sector probabilities p_R on collar boundaries. A conditional heat-kernel weighting relates them to a gauge coupling.

Theorem 10.1 (Fixed-cutoff edge-sector law)**.** *Under premises A1–A3, the fixed-cutoff overlap-gauge realization, a quasi-local local-Gibbs generator on the faithful interior branch, and hypotheses (EH-1) and (EH-2), the regulated same-overlap sector distribution is*

$$p_R = \frac{d_R e^{-t\lambda_R}}{\sum_{R'} d_{R'} e^{-t\lambda_{R'}}},$$

with λ_R the Laplacian eigenvalue on the R-isotypic component and t the collar Gibbs parameter.

(EH-1) The edge generator is the factorwise group Laplacian, $H_{\text{edge}} = \Delta_G$, or a positive combination $\sum_i c_i \Delta_{G_i}$ for a product group; the separate Gibbs parameter is t. For finite groups the declared Cayley generating multiset is conjugacy-invariant and inverse-symmetric, so $\lambda_R = |S| - d_R^{-1} \sum_{s\in S} \chi_R(s)$. Without centrality the Fourier transform is a matrix on each irreducible block and the character average is only its normalized trace, not a sector eigenvalue. This is a declared input: gauge invariance and the Gibbs form force only $H_{\text{edge}} = \sum_R h_R P_R$, and any function of the sector label is gauge invariant and realizable by a local Hamiltonian at fixed cutoff. Within second-order bi-invariant operators the Laplacian is unique up to scale, and one declared Markov chain with a Casimir acceptance rule derives (EH-1); its universality is unproved. (EH-2) The probability space is the one-sided electric-centre edge algebra $\bigoplus_R B(V_R)$, on which the sector projector has rank d_R. On the two-sided space $L^2(G)$ the projector has rank d_R^2, and a two-sided Gibbs trace gives weights $d_R^2 e^{-t\lambda_R}$; the d_R law is a statement about the one-sided algebra.

Proof. The edge degrees of freedom at Σ carry a G-action, modelled for the refinement lift by $L^2(G)$ with its Peter–Weyl decomposition [76]. By (EH-2) the physical one-sided observables are $\bigoplus_R B(V_R)$; by (EH-1) the generator is $\sum_R \lambda_R P_R$. Maximum entropy selects $\rho = Z^{-1} \sum_R e^{-t\lambda_R} P_R$, and $\operatorname{Tr} P_R = d_R$ gives the stated law. □

For $\mathbb{Z}_n$ the Laplacian eigenvalues are $\lambda_q = 4\sin^2(\pi q/n)$. The heat-kernel ansatz assigns the sector probabilities $p_q \propto e^{-t\lambda_q}$, so $t = -\log(p_q/p_0)/\lambda_q$. For SU($N$) the conditional ansatz assigns the probabilities $p_j \propto d_j e^{-tC_2(j)}$, so $t = (\log(p_0/d_0) - \log(p_j/d_j))/C_2(j)$ and $g_{\text{ent}}^2 = t/(2\pi)$. In two-dimensional Yang–Mills the Hamiltonian is the group Laplacian, $H = -\frac{g^2}{2}\Delta_G$ with $E_R = \frac{g^2}{2} C_2(R)$, and a cylinder of area A gives $t = g^2 A/2$ exactly; gluing the open-edge weights gives the closed partition sum $\sum_R d_R^2 e^{-tC_2(R)} = K_t(1)$. The Casimir log-gaps $\Delta_R = \ln(p_0/d_0) - \ln(p_R/d_R) = tC_2(R)$ have parameter-free ratios: for SU(2), $\Delta_1/\Delta_{1/2} = 8/3$, $\Delta_{3/2}/\Delta_{1/2} = 5$; for SU(3), $\Delta_8/\Delta_3 = 9/4$, $\Delta_6/\Delta_3 = 5/2$, $\Delta_{10}/\Delta_3 = 9/2$, $\Delta_{15}/\Delta_3 = 4$, $\Delta_{27}/\Delta_3 = 6$. These are algebraic consequences of the weighting, not source-selected coupling predictions. The repair-gap identification on the compact-gauge branch is the subject of the Yang–Mills supplement [23].

Finite diagnostics. In two-dimensional $\mathbb{Z}_n$ lattice gauge models on a 2×2 torus with $H = -K\sum_p \operatorname{Re} B_p - h\sum_\ell \operatorname{Re} X_\ell - \Gamma \sum_v \operatorname{Re} A_v$, $K = 1$, $\Gamma = 5$, the electric-centre edge charge at a boundary vertex is the restricted star. For $\mathbb{Z}_2$ one eigenvalue allows an exact fit, and for $\mathbb{Z}_3$ the two extractions agree by charge conjugation and eigenvalue degeneracy; these are implementation checks. $\mathbb{Z}_5$ has two distinct nonzero eigenvalues with $\lambda_2/\lambda_1 = \phi^2 \approx 2.618$. Fitting t on $q = 1$ and holding out $q = 2$, direct diagonalization gives the following.

h	$\ln(p_2/p_0)/\ln(p_1/p_0)$	deviation from ϕ^2
0.50	2.876	9.9%
0.20	2.750	5.1%
0.10	2.671	2.0%
0.05	2.643	1.0%

The deviation decreases monotonically across these samples; no limit proof or interval enclosure is supplied. A symmetry-reduced electric calculation with n^3 states reproduces the link-basis values.

For the one-plaquette S_3 model with transposition Cayley eigenvalues $\lambda_{\text{triv}} = 0$, $\lambda_{\text{sign}} = 6$, $\lambda_{\text{std}} = 3$, Gauss' law leaves the three character states, and $p_R = |c_R|^2$ exactly. With $K = 1$ and four electric links the reduced Hamiltonian is

$$H(h) = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 24h & -1 \\ -1 & -1 & 12h-1 \end{pmatrix}, \qquad h \geq 0,$$

and the Gauss penalty is a common shift. Fitting t on the standard sector and holding out the sign sector gives the following values.

h	p_triv	p_sign	p_std	t (sign)	t (std)	$\Delta t/t$	log-ratio
0.5	0.962927	2.476×10^{-4}	0.036825	1.37766	1.31898	4.3518%	2.088972
1.0	0.991861	1.400×10^{-5}	0.008125	1.86139	1.83261	1.5580%	2.031406
2.0	0.998116	8.158×10^{-7}	0.001883	2.33619	2.32203	0.6080%	2.012197
5.0	0.999713	1.993×10^{-8}	2.871×10^{-4}	2.95511	2.94950	0.1897%	2.003798
12	0.999951	5.895×10^{-10}	4.890×10^{-5}	3.54195	3.53963	0.0656%	2.001312
100	0.999999	1.208×10^{-13}	6.956×10^{-7}	4.95749	4.95721	0.0056%	2.000112

Here $\Delta t/\bar{t} = (t_{\text{sign}} - t_{\text{std}})/\bar{t}$ and the log-ratio is $\log(p_{\text{sign}}/p_0)/\log(p_{\text{std}}/(2p_0))$, which equals 2 under exact heat-kernel weights. Writing the ground energy as $-x < 0$, the secular equation is $x + 12h - 1 - 1/x - 1/(24h + x) = 0$, with a unique positive root decreasing from $x(0) = 2$. With $z = x + 12h$ and $p_{\text{sign}}^{\text{pred}} = p_{\text{std}}^2/(4p_{\text{triv}})$,

$$\frac{p_{\text{sign}}}{p_{\text{sign}}^{\text{pred}}} = \left(\frac{z-1}{z}\right)^2 < 1,$$

so the standard-sector fit overpredicts the sign sector at every finite h, and the ratio increases strictly from 1/4 toward one as $h \to \infty$. The fitted time $t_{\text{std}} = \log(2/x^2)/3$ is positive exactly for $h > h_* = (1+\sqrt{3}-\sqrt{2})/24 \simeq 0.0549098852$. At $h = h_*$, the relative log residual and log-ratio divide by zero; below it the fitted time is negative and outside the positive-diffusion ansatz. On $h > h_*$ the relative residual and log-ratio minus two decrease strictly to zero; at $h = 100$ the held-out relative log residual has magnitude about 0.0056%, with the fit overpredicting the sign sector. The identity proves a strict held-out mismatch at every finite coupling inside one three-dimensional model and transfers nothing to either continuous group. Neither diagnostic validates the law generally. A held-out refit script is part of the public code [29].

11 Classical carrier modes and the quantum-particle criterion

A compact group identifies a symmetry type and a classical field equation a classical branch; neither constructs a propagating quantum particle.

Definition 11.1 (Carrier-mode and quantum-particle receipts)**.** *Fix the invariant conversion speed $c_\star$; its value* $299\,792\,458\ \mathrm{m\,s^{-1}}$ *is exact by the SI definition of the metre. A classical massless carrier-mode receipt for q_X consists of (C1) a stated background and phase, (C2) a quadratic action with positive kinetic coefficient, (C3) a gauge fixing or constraint reduction with physical projector Π_X, and (C4) a positive reduced Hamiltonian, with physical Hessian $K_X = Z_X(\omega^2 - c_\star^2|\mathbf{k}|^2)\Pi_X$, $Z_X > 0$. A quantum-particle receipt additionally requires (Q1) a positive-energy quantization with physical Hilbert space, (Q2) a two-point function whose Källén–Lehmann measure contains $Z_X^{\mathrm{pole}}\delta(\mu^2)$ with $Z_X^{\mathrm{pole}} > 0$, and (Q3) the asymptotic-state or LSZ hypotheses of the phase, including a deconfined sector for a coloured carrier.*

Theorem 11.2 (Action-level modes)**.** *(i) An unbroken $\mathrm{U}(1)_Q$ connection with the Lorentzian Maxwell action $S_Q = -\frac{1}{4g_Q^2}\int F_{Q,\mu\nu}F_Q^{\mu\nu}$, $0 < g_Q^2 < \infty$, and no mass operator, has in radiation gauge $K_{Q,T} = g_Q^{-2}(\omega^2 - c_\star^2|\mathbf{k}|^2)\Pi_T$ with $\operatorname{rank}\Pi_T = 2$. (ii) A compact G with the two-derivative pure Yang–Mills action about the trivial background in a perturbative or deconfined phase has $2\dim G$ transverse modes with $K_{\mathrm{YM},T}^{ab} = \delta^{ab}g^{-2}(\omega^2 - c_\star^2|\mathbf{k}|^2)\Pi_T$ and no hard mass at quadratic order.*

Proof. Gauss' law removes the longitudinal pair, leaving two transverse polarizations with positive Hamiltonian and dispersion $\omega^2 = c_\star^2|\mathbf{k}|^2$. The Yang–Mills Hessian is one Maxwell Hessian per generator; interactions begin at cubic order. These calculations establish (C1)–(C4) and none of (Q1)–(Q3). □

Calling these modes photons or gluons requires the quantum-particle receipt. The perturbative mode does not produce a gauge-invariant asymptotic gluon, and in a confining phase the coloured free pole need not occur in the physical spectrum. The analogous transverse-traceless calculation for a supplied Einstein–Hilbert action gives two classical spin-two modes and no graviton state. Vanishing quadratic mass parameters are action-level statements; Higgs or Stueckelberg completions, medium self-energies, confinement and additional massive fields require their own field-content and phase hypotheses.

12 Failure conditions

The finite chain is exact on its stated premises, and each step has a registered negative control. The Lie-type theorem fails if a faithful compact commutator-closed twelve-port response with inner carrier action is exhibited outside $\mathfrak{u}(1) \oplus \mathfrak{su}(2) \oplus \mathfrak{su}(3)$; the conditional matter results fail if the incidence identity, the target-blind response, the equivariant lift, the anomaly scan, the tensor-character enumeration or the refinement persistence receipt fails its control. The categorical route applies only inside its combined strictification, trivial-holonomy, localization and refinement and fibre receipts.

On the executable model the branch is tested by computing the complete accessible current of a specified law, with every admitted control and apparatus service in the menu. A complete accessible response that is not faithful, closed and twelve-dimensional with inner carrier action excludes that law's realization of A1 and A2, as the seventeen-dimensional exchange response excludes its grammar. A physical attachment that forced the carrier rechartings to be spatial rotations of an interacting relativistic field while they remain internal would be constrained by the Coleman–Mandula theorem. The particle paper [21] tests whether localized defects of the same transport carry holonomy valued in this structure.

A Local action, Cartan reduction and fermion current

The appendices carry the local covariant action on the declared matter grammar and the finite current constructions built on it.

A.1 A local gauge-covariant action on the declared matter grammar

Fix an oriented flat Lorentzian chart with metric $\eta = \mathrm{diag}(1,-1,-1,-1)$, its two-component Spin convention, and smooth compactly supported variations. These geometric and Spin data are inputs. Take the declared algebra $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ on its fundamental, dual and trivial carriers. The exterior grammar supplies

$$Q : (3,2)_{1/6}, \quad u^c : (\bar{3},1)_{-2/3}, \quad d^c : (\bar{3},1)_{1/3}, \quad L : (1,2)_{-1/2}, \quad e^c : (1,1)_1.$$

The Higgs choice $H : (1,2)_{1/2}$, family multiplicity, minimal first-derivative kinetic law, scalar polynomial truncation and Yukawa intertwiners are declared. No coupling, mass or Yukawa entry is selected.

Use anti-Hermitian connections, with coupling normalization absorbed into their coordinates, and set

$$D_\mu = \partial_\mu + \rho(A_\mu), \qquad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]. \tag{1}$$

Fermion values and first derivatives generate a complex exterior algebra. Conjugation is the antilinear anti-automorphism exchanging fermions with independent conjugate generators and reversing products. Scalar and connection coefficients are even. Write $\bar{\sigma}^\mu = (1,-\sigma^1,-\sigma^2,-\sigma^3)$ and $\epsilon_{01} = 1$. With color index i, weak indices a,b, spin indices α,β and family indices r,s, define

$$T^u_{rs} = \epsilon_{\alpha\beta}\epsilon_{ab} Q^{ia\alpha}_r H^b (u^c)^\beta_{s,i}, \qquad T^d_{rs} = \epsilon_{\alpha\beta} Q^{ia\alpha}_r \overline{H}_a (d^c)^\beta_{s,i}, \qquad T^e_{rs} = \epsilon_{\alpha\beta} L^{a\alpha}_r \overline{H}_a (e^c)^\beta_s. \tag{2}$$

The superscript c denotes charge conjugation. These are all-left-Weyl expressions, not four-component barred expressions with unchanged charges.

Theorem A.1 (Conditional local action composition)**.** *Under the stated representation, Spin and kinetic assumptions, the density*

$$\begin{aligned}\mathcal{L} = {} & \frac{1}{2}\sum_{s=3,2,1} \kappa_s \,\mathrm{Tr}(F^s_{\mu\nu} F_s^{\mu\nu}) + \sum_{r,\psi} \frac{i}{2}\left[\psi^\dagger_r \bar{\sigma}^\mu D_\mu \psi_r - (D_\mu \psi_r)^\dagger \bar{\sigma}^\mu \psi_r\right] \\ & + (D_\mu H)^\dagger D^\mu H - m^2 H^\dagger H - \lambda (H^\dagger H)^2 - \sum_{r,s,\chi=u,d,e} \left(Y^\chi_{rs} T^\chi_{rs} + \overline{Y^\chi_{rs}} (T^\chi_{rs})^\dagger \right)\end{aligned} \tag{3}$$

is Grassmann even, Hermitian and invariant under arbitrary infinitesimal local gauge transformations. The real m^2, λ, positive κ_s and complex Y^χ_{rs} remain free coefficients. The construction composes varying-field gauge, scalar and fermion kinetic terms and all three Yukawa channels.

Proof. For an anti-Hermitian local parameter θ, set $\delta\psi = \rho(\theta)\psi$, $\delta H = \rho_H(\theta) H$ and $\delta A_\mu = -\partial_\mu \theta + [\theta, A_\mu]$. The product rule gives

$$\delta(D_\mu \psi) = \rho(\theta) D_\mu \psi, \qquad \delta F_{\mu\nu} = [\theta, F_{\mu\nu}].$$

The first identity cancels the two opposite $(\partial_\mu \theta)\psi$ terms. The second uses symmetry of the second parameter jet and cancels the first-derivative terms against the connection commutators. No

equation of motion is used. Cyclic trace proves gauge-kinetic invariance; anti-Hermiticity and commutation of internal generators with $\bar{\sigma}^\mu$ prove kinetic invariance. The trace forms are ad-invariant, with the free normalizations permitted by the invariant-form classification.

The color fundamental–dual pairing and weak determinant tensor are invariant. The hypercharge sums are $1/6 + 1/2 - 2/3 = 0$, $1/6 - 1/2 + 1/3 = 0$ and $-1/2 - 1/2 + 1 = 0$. The supplied Spin epsilon contracts anticommuting Weyl components. Each T^{χ}_{rs} and its adjoint are gauge neutral, separately for every family pair. Every term has even Grassmann degree and the displayed conjugate combinations are Hermitian. The symmetrized kinetic expression differs from $i\psi^\dagger\bar{\sigma}^\mu D_\mu\psi$ by a total derivative. □

Proposition A.2 (Matter current of the same local action)**.** *For an anti-Hermitian basis element T_a, define $\delta\mathcal{L}_{\text{matter}} = j^\mu_a \delta A^a_\mu$ with matter jets fixed. Then*

$$j^\mu_a = i\sum_{r,\psi} \psi^\dagger_r \bar{\sigma}^\mu \rho_\psi(T_a)\psi_r + (\rho_H(T_a)H)^\dagger D^\mu H + (D^\mu H)^\dagger \rho_H(T_a)H. \tag{4}$$

Its covariant divergence vanishes on the matter Euler–Lagrange equations, using the dual adjoint connection on the current index.

Proof. Differentiate the two covariant kinetic terms. Anti-Hermiticity doubles the fermion contribution in the symmetrized expression; the potential and Yukawa tensors have no connection dependence. For a compactly supported gauge variation, the matter equations remove matter variations from the first variation of the action. Substituting $\delta A_\mu = -\partial_\mu\theta + [\theta, A_\mu]$ and integrating by parts gives the dual covariant-divergence identity. □

Exact local checks and scope. The executable fixture retains three families, 900 independent Grassmann value, derivative and conjugate generators, and all 27 complex Yukawa slots. Connection and scalar jets are exact Gaussian-rational test values, not physical coupling constants or measurements. Independent arithmetic checks reconstruct the density, retain first and symmetric second jets of the gauge parameter, and reject omitted inhomogeneous connection terms, omitted curvature commutators, wrong dual color action and mixed Yukawa charges. Grassmann anticommutation is checked separately from neutrality. Lean proves the noncommutative first- and second-jet identities, the cyclic-trace Ward identity and the spin-sign control. The full action and on-shell current argument are analytic, not full Lean formalizations.

The flat chart, Spin interpretation, kinetic choice, Higgs/Yukawa content, relative normalizations, family multiplicity and local connection remain declared. Identifying the connection with one reconstructed from source response histories is separate. There is no source-produced Standard Model action or current, broken-vacuum selection, gauge fixing or ghost construction, quantum anomaly cancellation from a fermionic measure, renormalization, continuum quantum-field existence theorem or numerical particle prediction. Global gauge quotient and physical line-spectrum attachments remain separate. The local action admits a consistent classical charged-scalar slice, with both Cartan gauge equations and their variational current retained. Its finite source-address execution uses a declared logarithmic plaquette action and model clock, as follows.

A.2 A consistent local Cartan–scalar reduction and current response

Work on the flat Spin chart and local classical action of Theorem A.1. All kinetic coefficients are positive constants; the mass, quartic and Yukawa coefficients remain declared. Use the standard

weak Cartan generator $i\sigma_3/2$ and central generator i. Writing the two real Cartan connection coordinates as $a = (a_2, a_1)$ gives

$$q = (1/2, 1/2)^T, \qquad K = \mathrm{diag}(\kappa_2, 2\kappa_1), \qquad d = q^T K^{-1} q, \qquad w = K^{-1} q/d.$$

Here the gauge density is $-\frac{1}{4} f^T K f$: the fundamental traces are $\mathrm{Tr}((i\sigma_3/2)^2) = -1/2$ and $\mathrm{Tr}(i^2) = -1$. The executable weak basis uses $i\sigma_3$, in which $q = (1, 1/2)^T$ and $K = \mathrm{diag}(4\kappa_2, 2\kappa_1)$; d agrees.

Theorem A.3 (Local classical solution embedding)**.** *Set all fermions and their independent conjugates, color connections and off-diagonal weak connections to zero. Let $H = (\varphi, 0)$ and $a = wA$. Every smooth solution, including the temporal gauge equation, of*

$$\mathcal{L}_{\mathrm{red}} = -\frac{F_{\mu\nu} F^{\mu\nu}}{4d} + |(\partial_\mu + iA_\mu)\varphi|^2 - m^2|\varphi|^2 - \lambda|\varphi|^4 \tag{5}$$

lifts to a solution of the full local classical equations, with ansatz-compatible initial and boundary data. All complex Yukawa entries remain unrestricted.

Proof. The identities $q^T w = 1$, $Kw = q/d$ and $w^T K w = 1/d$ give the density and both Cartan equations. In the current convention $\delta\mathcal{L}_{\mathrm{matter}} = j^\mu \delta A_\mu$, these equations are $q_a[d^{-1}\partial_\nu F^{\nu\mu} + j^\mu] = 0$. If $q^T n = 0$, the decomposition $a = wA + nB$ has no scalar coupling to B and no kinetic cross term, since $w^T K n = 0$; $B = 0$ solves its free equation. The Higgs and its covariant derivative lie in the upper weak line, so both off-diagonal weak currents vanish. The gauge covariant divergence of a Cartan curvature is Cartan. Color curvature and sources vanish. The diagonal connection preserves the lower Higgs line, whose potential derivative vanishes at zero. Every fermion equation contains a fermion factor; every Yukawa contribution to a Higgs equation contains two. Thus all discarded variations vanish without setting any Yukawa coefficient to zero. The temporal Cartan equation is included, so Gauss's law is retained. □

This is a local real-connection statement. Generic kinetic coefficients give a direction incommensurate with the full charge lattice. No compact subgroup, global quotient, large-gauge identification, electromagnetic vacuum, perturbative stability or quantum truncation follows. Retaining a real unwrapped connection is essential: its exponential alone need not determine the embedded connections in every representation.

A compatible spatial discretization. The same reduction holds for a declared semidiscrete gauge–scalar action with covariant link differences, invariant electric metric, and the local logarithmic plaquette energy. For the weak holonomy use its principal matrix logarithm on the smooth identity chart. On the Cartan slice, $\log U_f = i\sigma_3 w_{\mathrm{w}}(\mathrm{d}A)_f/2$ when $|w_{\mathrm{w}}(\mathrm{d}A)_f| < 2\pi$, where w_{w} is the weak component of w. The hypercharge curl is real and unwrapped. Consequently the plaquette energy restricts to the quadratic Maxwell curl with coefficient $1/d$. The first variation of $\mathrm{Tr}(\log U)^2$ is $2\,\mathrm{Tr}(\log U\, U^{-1}\delta U)$; discarded weak variations are off-diagonal and have zero trace pairing. Scalar link currents also have zero off-diagonal component. Electric transport along the Cartan line is geodesic for the invariant metric. These facts give the same omitted equations and weighted Cartan square. This statement requires the logarithm chart throughout the history. A Wilson cosine plaquette does not supply this quadratic reduction for arbitrary w.

On the prepared golden $q = 5$ tensor family use dimensionless positions x/L, $L = 2/\sqrt{\phi + 2}$. The ordered one-dimensional nodes are $0, 2\phi - 3, 4\phi - 6, \phi - 1, 3\phi - 4, 1$. The 4^3 interior nodes have lumped masses v_i; an internal edge has conductance c_e equal to its transverse dual area divided by its gap. The fixed zero scalar boundary contributes the usual nonnegative onsite stiffness b_i.

Declare $\kappa_1 = \kappa_2 = m^2 = 1$, $\lambda = 1/4$, so $d = 3/8$. For each internal plaquette set its Maxwell weight γ_f to $1/d$ times normal dual length divided by area, and set $\epsilon_e = c_e/d$. These are action quadratures, not causal counting weights.

With $\pi_i = v_i D_0 \psi_i$ and canonical real link momentum Π_e, the Hamiltonian is

$$\begin{aligned}\mathcal{H} = &\sum_i \frac{|\pi_i|^2}{v_i} + \sum_e \frac{\Pi_e^2}{2\epsilon_e} + \sum_{e=i\to j} c_e |e^{ia_e}\psi_j - \psi_i|^2 \\ &+ \sum_i [(v_i + b_i)|\psi_i|^2 + v_i|\psi_i|^4/4] + \tfrac{1}{2}\sum_f \gamma_f (\mathrm{d}a)_f^2. \end{aligned} \tag{6}$$

The complex scalar symplectic form is $2\,\mathrm{Re}(\mathrm{d}\overline{\pi}_i \wedge \mathrm{d}\psi_i)$. Differentiating this same energy gives

$$J_{i\to j} = -2c_e\,\mathrm{Im}(\overline{\psi}_i e^{ia_e}\psi_j), \qquad \rho_i = 2\,\mathrm{Im}(\overline{\psi}_i \pi_i), \qquad G_i = \rho_i + (\mathrm{div}\,\Pi)_i.$$

Here divergence is outward; $\dot{\rho} + \mathrm{div}\, J = 0$ and $\dot{\Pi} = J - \mathrm{d}^T \gamma \mathrm{d}a$ imply $\dot{G} = 0$.

Proposition A.4 (Local split evolution preserves Gauss constraints)**.** *Scalar kinetic drift, electric kinetic drift and every individual edge, onsite and plaquette potential kick preserve each G_i in ideal arithmetic. Their symmetric composition is symplectic and preserves the initial Gauss constraints. Disjoint local subflows commute.*

Proof. Scalar drift adds a real multiple of π_i to ψ_i, leaving ρ_i unchanged; electric drift changes only a. An edge kick changes the two charges by $-hJ_e, +hJ_e$ and Π_e by hJ_e. Onsite kicks add a real multiple of ψ_i to π_i. A plaquette kick adds a curl transpose to Π, whose divergence vanishes. Each map is its Hamiltonian subflow; disjoint subflows have disjoint dynamical variables. The potential kicks also commute because their coordinates are fixed during the kicks. □

Finite current and feedback control. The executable packet uses 64 scalar sites, 144 internal links and 108 plaquettes, with two steps of declared model duration 1/200. It compares the real preparation $\psi_i = 1/5$, $\pi_i = a_e = \Pi_e = 0$ with a local phase intervention $(3 + 4i)/5$ and a genuine gauge copy, which also changes the incident unwrapped connections. The intervention leaves charge and Gauss constraints unchanged but changes the edge current. At the selected edge $c_e = 1/4$ exactly, so $J_e = 2/125$ and its first electric half-kick is 1/25000. The gauge copy preserves invariant readouts. Complete operation records bind every dynamic read to its version's actual writer; decoded checkpoints contain scalar intensity, charge, edge current and electric feedback. Independent high-precision replay checks these binary64 records, including a nonempty off-influence control. This numerical agreement is separate from the exact local identities and gives no continuous-history error enclosure. The prepared addresses, local execution law, action, boundary and model tick remain supplied. No causal-count clock attachment, inherited fine-grid detector error, selected physical current or full Standard Model quantum evolution is asserted.

A.3 A continuous charged-current and separated-detector bound

The same finite Hamiltonian admits a useful continuous-time comparison without increasing the charged execution. Work with its exact golden weights and exact initial scalar magnitude $s_0 = 1/5$, zero scalar and electric momenta, and zero initial connections. Compare the baseline with the phase intervention $(3 + 4i)/5$ at site 21. The distinct detector site 37 is joined to it by the oriented edge of conductance $c = 1/4$. The reference gauge copy is the exact gauge transform of the baseline. Times are the declared dimensionless action parameter.

Theorem A.5 (Resolved nonlinear charged readout)**.** *The exact classical solutions exist on* $0 \le t \le T = 1/100$ *within the principal-log chart. On the selected edge their intervention current satisfies*

$$J(t) \ge 0.015110268, \qquad \Pi(T) > 0.0001543.$$

Writing $I_i = |\psi_i|^2$*, the spatially distinct detector obeys*

$$I_{37}^{\text{intervention}}(t) - I_{37}^{\text{baseline}}(t) \le -\frac{9}{20}t^2 \qquad (0 < t \le T).$$

Thus the response magnitude is at least 0.000045 *at* T *and at least* 0.00001125 *throughout* $[1/200, 1/100]$*. For the nine initial/completed decoded intensity checkpoints across all three preparations, the exact continuous-reference error is less than* 0.000006 *per checkpoint.*

Proof. Set $S = 1/4$, $A_* = 10^{-5}$, $B = 88$, and $D = 13/250$. On the box $|\psi_i| \le S$, $|a_e| \le A_*$, differentiation of the Hamiltonian gives

$$|\ddot{\psi}_i| \le \frac{2S\sum_{e\ni i} c_e + (v_i + b_i + v_i S^2/2)S}{v_i} < B,$$
$$|\ddot{a}_e| \le \frac{2c_e S^2 + 4A_*\sum_{f\ni e}\gamma_f}{\epsilon_e} < D.$$

Exact rational enclosures of the golden weights verify every inequality and $\sum_{f\ni e}\gamma_f < 78$. Since the initial velocities vanish, $|\psi_i(t) - \psi_i(0)| \le Bt^2/2$ and $|a_e(t)| \le Dt^2/2$. Both bounds lie strictly inside the box through T; a first-exit argument and local ODE continuation prove existence there. Also $|(4/3)(\mathrm{d}a)_f| \le (16/3)A_* < 1 < 2\pi$, preserving the chart. Gauge covariance transfers the invariant bounds to the ideal gauge copy.

The current product estimate gives

$$|J(t) - 2/125| \le C_2 t^2 + C_4 t^4, \quad C_2 = \tfrac{1}{4}(2s_0 B + s_0^2 D) = 8.80052, \quad C_4 = B^2/8 = 968.$$

The magnetic force is at most $2D(78)t^2$. Integrating $\dot{\Pi} = J - \mathrm{d}^T\gamma\mathrm{d}a$ therefore bounds $|\Pi(t) - (2/125)t|$ by $(C_2 + 156D)t^3/3 + C_4 t^5/5$, yielding the stated current and feedback.

For the detector, let $u = \sum_{e\ni 37} c_e/v_{37}$; its boundary stiffness vanishes. The cubic map $z \mapsto |z|^2 z$ is $3S^2$-Lipschitz on the complex disk. The phase estimate $|e^{ia} - 1| \le |a|$ consequently gives

$$|\ddot{\psi}_{37}(t) - \ddot{\psi}_{37}(0)| \le Et^2/2, \quad E = u(2B + s_0 D) + (1 + 3S^2/2)B < 30759.$$

The quadratic Taylor remainder is at most $r = 30759t^4/24$. Put $v = v_{37}$ and

$$a_0 = -51/250, \qquad a_1 = a_0 - 1/(50v) + i/(25v).$$

These are the exact initial accelerations of the two preparations. Each intensity differs from $|s_0 + a_j t^2/2|^2$ by at most $2(s_0 + Bt^2/2)r + r^2$. Subtraction gives a quadratic coefficient $s_0\,\mathrm{Re}(a_1 - a_0)$ and quartic coefficient $(|a_1|^2 - |a_0|^2)/4$. Exact rational bounds on v show that their sum with both remainders is at most $-(9/20)t^2$ for $0 < t \le T$.

Finally, decode ψ_{37} from the actual versioned reads at each initial or completed checkpoint. Interpret every stored binary64 component as its exact dyadic value. Its intensity discrepancy from the appropriate quadratic Taylor expression, plus the preceding analytic remainder, is an exact bound against the continuous IVP. The independent consumer checks all nine values, each below 6×10^{-6}, and excludes the intermediate split kick. It replays the parent custody and operations; copied public frames cannot replace the read events. □

This is a nonlinear classical current and readout result on the supplied family. The checkpoint bound does not enclose every state component or every intermediate observer operation. The exact reference preparation is distinct from its binary64 encoding. Quantum evolution, full fermion generation, a physical clock and native selection of the action or population are not supplied by this estimate.

A.4 Fermionic hypercharge transport on prepared source addresses

Fix the same 64 prepared golden addresses and 144 oriented nearest-neighbour links as the scalar execution. Give each address two Spin components of each of the five registered matter multiplets. Their internal multiplicities and integer charges are

$$(d_s) = (6, 3, 3, 2, 1), \qquad (q_s) = 6(Y_s) = (1, -4, 2, -3, 6).$$

This finite model supplies its spatial hopping action, canonical anticommutation relations, state preparation and model step. The source addresses do not select those inputs. The colour and weak indices are retained as multiplicities of an abelian transport law; the model does not execute the complete nonabelian or Yukawa dynamics.

Let $c_{s,a,v,\alpha}$ be the annihilation operators on the finite fermion Fock space, where $a = 1, \ldots, d_s$ and $\alpha = 1, 2$. They satisfy $\{c_i, c_j^\dagger\} = \delta_{ij}$ and $\{c_i, c_j\} = 0$. For an oriented link $e = (u, v)$ in direction j, a unit complex link U_e and a supplied real hopping weight κ_e, put

$$K_{s,e} = i\sigma_j U_e^{q_s}, \tag{7}$$

$$\widehat{H}_e = \kappa_e \sum_{s,a} (c_{s,a,u}^\dagger K_{s,e} c_{s,a,v} + c_{s,a,v}^\dagger K_{s,e}^\dagger c_{s,a,u}). \tag{8}$$

The Pauli matrices act on Spin, not on the internal multiplicity index. The sign and link orientation in this definition fix the current convention. If $U_e \mapsto g_u U_e \overline{g}_v$ and $c_{s,a,w} \mapsto g_w^{q_s} c_{s,a,w}$, the quadratic expression is unchanged. The link is held fixed during this matter update.

Occupy one normalized spatial–Spin orbital ψ_s for every internal index a. Distinct internal indices and multiplets are orthogonal, so their exterior product is a normalized fifteen-fermion Slater state. Its one-body density is the orthogonal projector

$$P = \bigoplus_s I_{d_s} \otimes |\psi_s\rangle\langle\psi_s|.$$

The complex orbital coefficients describe this fermionic state. They are not commuting replacements for the field operators. In particular $0 \le P \le 1$ and $\operatorname{Tr} P = 15$.

For one occupied orbital write $z = (\psi_{s,u}, \psi_{s,v})$ and

$$h_{s,e} = \kappa_e \begin{pmatrix} 0 & K_{s,e} \\ K_{s,e}^\dagger & 0 \end{pmatrix}.$$

For step $\delta > 0$, take the implicit midpoint equation

$$i\frac{z' - z}{\delta} = h_{s,e}\bar{z}, \qquad \bar{z} = \frac{z' + z}{2}. \tag{9}$$

Since $K_{s,e}$ is unitary, $h_{s,e}^2 = \kappa_e^2 I$. The update is the unitary Cayley transform $(I + i\delta h_{s,e}/2)^{-1}(I - i\delta h_{s,e}/2)$. It preserves the projector property and particle number, with $P \mapsto VPV^\dagger$, and is exact over Gaussian rationals when the link, step, weight and preparation lie in that field.

Proposition A.6 (Variational current and mean Gauss preservation)**.** *Define the mean site charge and midpoint current by*

$$\rho_w = \sum_s d_s q_s \|\psi_{s,w}\|^2, \tag{10}$$

$$J_e = -2\kappa_e \sum_s d_s q_s \, \mathrm{Im}(\bar{\psi}^\dagger_{s,u} K_{s,e} \bar{\psi}_{s,v}). \tag{11}$$

Then J_e is the derivative of the quadratic midpoint functional of (8) under $U_e \mapsto e^{i\theta} U_e$, with the midpoint orbitals fixed. These averages need not be normalized occupied orbitals; only the step endpoints represent the normalized fifteen-fermion Slater states. The charge changes satisfy

$$\rho'_u - \rho_u = -\delta J_e, \qquad \rho'_v - \rho_v = \delta J_e.$$

Let E_e be a classical link momentum and update $E'_e = E_e - \delta J_e$. With div *positive at the initial endpoint of each oriented link, every site residual $G_w = (\mathrm{div}\, E)_w - \rho_w$ is unchanged. This remains true for any finite composition of these edge factors.*

Proof. For the finite canonical anticommutation relations, $[c_i^\dagger c_j, c_k^\dagger c_l] = \delta_{jk} c_i^\dagger c_l - \delta_{li} c_k^\dagger c_j$; the degree-four terms cancel with their fermionic signs. Thus the quadratic Hamiltonian acts through its one-body matrix. Differentiating its link phase at fixed state gives the displayed J_e. The identity

$$\|x'\|^2 - \|x\|^2 = 2\,\mathrm{Re}\left(\frac{x'+x}{2}\right)^\dagger (x'-x)$$

and (9) give opposite endpoint transfers. The electric update cancels those changes in G_u, G_v; all other site charges and incident momenta are unchanged. Induction proves the finite composition statement. Gauge covariance follows by conjugating the one-body matrices and their Cayley transforms. □

Write $U_e = e^{i\theta_e}$. The same first-order action is

$$S_e = \int \left(i \sum_s d_s \psi_s^\dagger \dot{\psi}_s + E_e \dot{\theta}_e - \langle \widehat{H}_e \rangle \right) dt.$$

Its variations give the orbital equation, $\dot{E}_e = -\partial_{\theta_e} \langle \widehat{H}_e \rangle$ and $\dot{\theta}_e = 0$, since this factor has no E_e dependence in its Hamiltonian. Thus frozen links and electric feedback belong to one declared action factor. The midpoint scheme discretizes these equations. Exact discrete conservation does not identify its step with the continuous Hamiltonian exponential or prove a trajectory-error estimate. The electric momentum is driven by a fermionic expectation: this is a semiclassical update, not a quantized gauge field coupled to fermion Fock space. Electric drift, plaquette and Yukawa factors are absent.

A sharp quantum-constraint boundary. At every address choose $\psi_s = (3/40, 1/10)$ before the local phase intervention. Each orbital has squared site norm $1/64$ and unit global norm. Since $\sum_s d_s q_s = 0$, the initial mean charge vanishes at every site, and $E = 0$ solves the mean Gauss constraint. Nevertheless the local charge operator has variance

$$\mathrm{Var}(\widehat{Q}_w) = \sum_s d_s q_s^2 \frac{1}{64}\left(1 - \frac{1}{64}\right) = \frac{945}{512} > 0. \tag{12}$$

Each occupied internal orbital contributes an independent Bernoulli site occupation; this proves the formula. A site phase preserves it. Consequently the product preparation with sharp classical electric data is not annihilated by the quantum Gauss operator. In ordinary hypercharge units the variance is divided by 36. Vanishing mean residuals cannot certify a physical gauge-constrained quantum state.

Execution and interpretation. The executable construction in `code/sm_fermion_current/` retains the local orbital state, incident link ports, current readback and electric feedback as versioned software records. Its independent verifier rebuilds the charge dictionary, source addresses, Cayley updates, variational currents and consumed writer/value versions. Baseline, local phase intervention and gauge-related execution are separate controls. The generation's linear, cubic and mixed anomaly sums are representation identities; a finite canonical-anticommutation implementation does not construct a chiral continuum fermion measure.

This packet has one declared generation and supplied quantum preparation. It proves neither native source selection, all-matter Standard Model evolution, a controlled spatial or temporal continuum limit, laboratory current, a physical clock, nor compatibility with a sourced curved family. The preceding scalar reduction and its continuous-readout enclosure retain their own action and preparation; their error bounds do not transfer to this fermionic transport.

A.5 A flux-dressed fermion state satisfying abelian quantum Gauss

The mean-field obstruction has a precise remedy within a separately supplied quantum link model. It changes both the gauge-field Hilbert space and the preparation. Choose one oriented link (u, v) of the same golden carrier. Its electric space is $\ell^2(\mathbb{Z})$, with

$$\widehat{E}|n\rangle = n|n\rangle, \qquad \widehat{U}|n\rangle = |n+1\rangle, \qquad [\widehat{E}, \widehat{U}] = \widehat{U}.$$

The ambient flux space is infinite. A finite reachable support is computed from the preparation; no cyclic wrap or truncated shift replaces this unitary operator.

Index the fifteen internal channels by a, retaining their charges q_a. In each channel occupy the normalized left Spin orbital $\chi = (3/5, 4/5)$ and leave its orthogonal orbital empty. The active right orbital is $-i\sigma_j\chi$, so the Pauli link maps it back to χ. The two active modes have annihilators l_a, r_a. Fix the canonical fermion order $(l_1, r_1, l_2, r_2, \ldots)$ and set

$$\widehat{h}_a = \kappa_a \left(l_a^\dagger \widehat{U}^{q_a} r_a + r_a^\dagger \widehat{U}^{-q_a} l_a \right). \tag{13}$$

Each channel retains exactly one fermion. Distinct channel Hamiltonians commute: their even fermion bilinears have disjoint mode support and the flux shifts commute. Initially all fifteen fermions are at u, the right modes are empty, and the link is in $|0\rangle$. Since $\sum_a q_a = 0$, this preparation is locally neutral.

Proposition A.7 (Exact finite-support abelian quantum Gauss state)**.** *Let $\widehat{Q}_u = \sum_a q_a l_a^\dagger l_a$ and $\widehat{Q}_v = \sum_a q_a r_a^\dagger r_a$. Every $\widehat{h}_a$ commutes with*

$$\widehat{G}_u = \widehat{E} - \widehat{Q}_u, \qquad \widehat{G}_v = -\widehat{E} - \widehat{Q}_v.$$

Its Cayley transform preserves their joint kernel. Starting from the neutral preparation, the product of these transforms is the normalized state

$$|\Psi\rangle = \sum_{x \in \{0,1\}^{15}} \left[\prod_a A_a^{1-x_a} (-iB_a)^{x_a} \right] |x\rangle_F \otimes \left| -\sum_a q_a x_a \right\rangle_E, \tag{14}$$

where $x_a = 1$ means that channel a occupies its right mode, and

$$\eta_a = \delta\kappa_a/2, \qquad A_a = \frac{1-\eta_a^2}{1+\eta_a^2}, \qquad B_a = \frac{2\eta_a}{1+\eta_a^2}.$$

Every branch, and hence the complete state, satisfies both quantum Gauss constraints. All other vertices retain zero charge and incident flux.

Proof. In $l_a^\dagger \widehat{U}^{q_a} r_a$, both left charge and electric flux increase by q_a, while right charge decreases by q_a. The adjoint reverses all three changes. This proves the two commutators on the invariant finite-support core of the electric operator. The Hamiltonians are bounded; their Cayley transforms preserve the joint Gauss sectors and the electric domain. On the one-fermion channel sector, $\widehat{h}_a^2 = \kappa_a^2 I$, which gives the displayed two coefficients. The interleaved fermion convention fixes the even-hop signs. Channel commutation gives the product expansion. Its orthogonal Fock branches have squared norms summing to $\prod_a (A_a^2 + B_a^2) = 1$. On branch x, $Q_v = \sum_a q_a x_a$, $Q_u = -Q_v$ and $E = -Q_v$. Both Gauss residuals therefore vanish individually; no cancellation of opposite residuals in an expectation is used. □

The span of these 2^{15} dressed branches is invariant under every $\widehat{h}_a$: each admissible hop changes the occupation and flux by the same signed charge. Thus the finite electric support is exact for the coupled evolution of this sector, although the standalone shift $\widehat{U}$ does not preserve it. Channel commutation also gives the continuous solution of the supplied Hamiltonian $\sum_a \widehat{h}_a$: replace A_a, B_a in (14) by $\cos(\kappa_a t), \sin(\kappa_a t)$. This is an exact finite-sector quantum history, not a time-step approximation.

For $\delta = 1/2$, choose $\kappa_a = 1$ except on the e^c channel, where $\kappa_a = 2$. These species weights are declared; they do not establish a universal physical propagation speed. The coefficients are $(A, B) = (15/17, 8/17)$ on the first fourteen channels and $(3/5, 4/5)$ on the last. All 2^{15} Fock branches occur, with electric support $\{-18, \ldots, 18\}$. The link-phase current operator is

$$\widehat{J}_a = i q_a \kappa_a \left(l_a^\dagger \widehat{U}^{q_a} r_a - r_a^\dagger \widehat{U}^{-q_a} l_a \right), \qquad \dot{\widehat{E}} = -\sum_a \widehat{J}_a.$$

Exact evaluation in (14) gives

$$\langle \widehat{E} \rangle = -\frac{18144}{7225}, \qquad \sum_a \langle \widehat{J}_a \rangle = \frac{47232}{7225}.$$

The sum of the factor-midpoint currents is instead $36288/7225$; multiplied by $-\delta$, it equals the electric change. A final-state current and an integrated discrete transfer are different readouts.

For the continuous solution with the same species weights,

$$\langle \widehat{E} \rangle_t = 6(\sin^2 t - \sin^2 2t), \qquad \langle \widehat{J} \rangle_t = -6 \sin 2t + 12 \sin 4t.$$

The electric derivative equals minus the current. At the supplied model time $t = \pi/6$ they give -3 and $3\sqrt{3}$, respectively, with electric variance $45/2$. The flux fluctuations are correlated with charge, so quantum Gauss holds throughout this history. The time parameter, action and preparation are inputs; this solution calibrates no physical clock.

The executable producer and independent verifier in `code/sm_fermion_current/quantum_link.py` and `code/sm_fermion_current/verify_quantum_link.py` bind the channel and Spin maps, CAR signs, complete branch population, flux distribution and current matrix elements. This is a supplied finite quantum state and action factor on one prepared link. It is not the uniform semiclassical Slater preparation, a measured quantum history, full gauge-field evolution, nonabelian Gauss constraints or Yukawa evolution, or a chiral continuum regulator. Native state and operation selection, source-causal attachment and laboratory realization remain additional requirements.

B A coupled fermion current in a declared hybrid gauge action

The fermionic current can be coupled to electric drift and magnetic plaquette forces on the same prepared carrier. The construction below declares a regulated Hamiltonian and a finite splitting word. Its gauge fields are classical, while its matter state belongs to finite fermion Fock space. This hybrid interpretation differs from quantizing the gauge field in the preceding one-link construction.

Take the direct-product gauge cover $SU(3) \times SU(2) \times U(1)$, with the five registered representations and integer hypercharges $q_s = 6Y_s$ in each of three declared families. The compact abelian link is $U_e = e^{i\theta_e}$ with momentum E_e. For fixed $M, \lambda > 0$, choose

$$T_\lambda(E) = \frac{2E}{\lambda} \arctan \frac{\lambda E}{2M} - \frac{2M}{\lambda^2} \log \left(1 + \frac{\lambda^2 E^2}{4M^2}\right), \tag{15}$$

$$W_f = \prod_{e \in \partial f} U_e^{\epsilon_{fe}}, \qquad V_f = \beta_f (1 - \operatorname{Re} W_f). \tag{16}$$

Here ϵ_{fe} is the oriented face boundary. The reduced Hamiltonian is

$$\mathcal{H}_{\mathrm{red}} = \sum_e T_\lambda(E_e) + \sum_f V_f + \sum_e \langle \Psi, \widehat{H}_{\mathrm{CAR},e}(U_e)\Psi \rangle,$$

where the hopping operators and Spin matrices are those defined above. In temporal gauge, its first-order action contains $\sum_e E_e \dot{\theta}_e + i\langle \Psi, \dot{\Psi} \rangle - \mathcal{H}_{\mathrm{red}}$. The link-phase variation of the same matter Hamiltonian defines its current.

This is a specified kinetic law. In particular,

$$T'_\lambda(E) = \frac{2}{\lambda} \arctan \frac{\lambda E}{2M}, \qquad T''_\lambda(E) = \frac{1}{M(1 + (\lambda E/(2M))^2)} > 0.$$

Integrating $x - \arctan x = \int_0^x t^2/(1+t^2)\, dt$ gives

$$0 \le \frac{E^2}{2M} - T_\lambda(E) \le \frac{\lambda^2 E^4}{48 M^3}.$$

The quadratic electric energy is therefore a controlled small-field comparison of functions. It is not the exact kinetic energy of this model, and the displayed bound is not a trajectory-error estimate. The parameter λ remains fixed when describing this action; varying it changes the law. The separate charge-one abelian Wilson term is specified on the direct-product cover. No descent of that term to the global central quotient is assumed.

Proposition B.1 (Invariant expectation-level matter sector)**.** *Extend the reduced Hamiltonian by classical nonabelian links and momenta, quadratic nonabelian electric energies, Wilson magnetic energies, the covariant Higgs kinetic and spatial terms, and a potential with vanishing first derivative at zero Higgs field. Couple the finite CAR matter through its registered representation matrices and all twenty-seven complex Yukawa coefficients of three families. Interpret every matter force as its Fock-state expectation in the canonical classical-field/quantum-matter action.*

Set the nonabelian links to identity, their electric momenta to zero, and the Higgs field and conjugate momentum to zero. In each species of one family take normal covariance $P_s = I_{d_s} \otimes |\psi_s\rangle\langle\psi_s|$, with no covariance between distinct species or families. Let the other two families be Fock vacuum, and take zero anomalous covariance $\langle c_i c_j \rangle = 0$. These conditions define an invariant hybrid subsystem with Hamiltonian $\mathcal{H}_{\mathrm{red}}$.

Proof. Every nonabelian matter charge or link-current variation inserts a traceless internal generator T_s. The remaining Spin and spatial factor is scalar in that internal index, so the expectation factors through $\operatorname{tr} T_s = 0$. This checks the omitted nonabelian matter forces, including each color and weak generator. The nonabelian electric velocities vanish at zero momenta, and their magnetic first variations vanish at identity plaquette holonomy. Scalar kinetic, spatial and gauge-current variations vanish at zero Higgs field and momentum.

In all-left-Weyl notation, the Yukawa terms have the gauge-neutral forms QHu^c, $Q\overline{H}d^c$ and $L\overline{H}e^c$, with the required Spin, color and weak contractions and their adjoints. Their Higgs first variation is a linear combination of anomalous expectations $\langle cc\rangle$ and $\langle c^\dagger c^\dagger\rangle$. Both vanish by fermion-number selection, for every value of every Yukawa coefficient. This checks the Higgs force rather than only the value of the Yukawa energy at $H = 0$.

The surviving hopping Hamiltonian is number conserving, family diagonal and scalar on each internal multiplicity space. It preserves the stated covariance form, the vanishing anomalous covariance and the two vacuum families. The field equations hence preserve every imposed condition. No stability of the zero-Higgs equilibrium is needed for this invariance statement. □

The expectation hypothesis is essential. It does not identify independent Grassmann components with one commuting orbital, and it does not annihilate the nonabelian Gauss operators. For example, in the initial uniform preparation $p = 1/64$, the weak generator $\sigma_z/2$ on the occupied quark and lepton doublets has local charge variance

$$\operatorname{Var}(\widehat{Q}_T(v)) = \sum_s \operatorname{tr}(T_s^2)p(1-p) = \frac{63}{2048} > 0.$$

The absent nonabelian mean source therefore cannot certify a full gauge-constrained quantum state with sharp zero nonabelian electric data.

Proposition B.2 (Coupled transport and conserved readout)**.** *On a finite oriented lattice with closed face boundaries, compose the following factors of the declared hybrid action:*

1. *a fermionic implicit-midpoint hopping factor, with its variational current and the same classical electric kick;*

2. *an electric factor of duration λ,*

$$U_e' = \frac{1 + i\lambda E_e/(2M)}{1 - i\lambda E_e/(2M)} U_e, \qquad E_e' = E_e;$$

3. *a magnetic factor of duration h on face f,*

$$E_e' = E_e - h\beta_f \epsilon_{fe} \operatorname{Im} W_f \quad (e \in \partial f), \qquad U_e' = U_e.$$

For Gaussian-rational initial data and rational parameters these maps are exact over Gaussian rationals. Each preserves the expectation Gauss residual at every vertex. Their recorded current and charge transfers satisfy exact discrete continuity under finite composition, and the construction is covariant under residual time-independent vertex gauge transformations.

Proof. The midpoint matter identity proved above gives opposite endpoint charge transfers and precisely the electric kick that cancels them in Gauss. The midpoint equations are applied to the orbital, link-angle and momentum variables of that same factor action; its angle is constant during the factor because its Hamiltonian has no electric-momentum dependence.

The electric factor is the exact flow of T_λ for the stated duration: exponentiating $i\lambda T'_\lambda(E) = 2i \arctan(\lambda E/(2M))$ gives the rational phase. It leaves charge and electric divergence unchanged. For the magnetic factor, $\partial_{\theta_e} V_f = \beta_f \epsilon_{fe} \operatorname{Im} W_f$. Its momentum kick is the oriented face boundary, whose vertex divergence vanishes. These facts prove Gauss preservation for every factor and hence for their composition. The unit-modulus rational phase, Hermitian Cayley matter transform and rational Wilson product also prove the asserted arithmetic closure. Abelian gauge transformations conjugate the matter updates, leave E and W_f invariant and commute with the electric drift. □

This factor word is a declared discrete transport. It is not the exact exponential of the summed Hamiltonian; exact Gauss conservation does not imply exact total-energy conservation or a continuous-trajectory error bound.

Finite execution and returned current. Use the same sixty-four prepared addresses, 144 oriented links and 108 plaquettes, with $\lambda = h = 1/2$ and $M = \beta = \kappa = 1$. A full matter sweep is followed by all electric drifts and all magnetic kicks. One subsequent matter factor reads the evolved link and momentum on edge 14. That final factor is a readout in a partial second word, not a second completed full-Hamiltonian time step.

Across baseline, local intervention, gauge copy and a disabled-drift control, the retained prefixes contain 2,444 parent events. Each of the four continuations has 255 events: 144 electric factors, 108 magnetic factors, one returning matter factor and two public readouts. The control leaves the link unchanged instead of applying electric drift; it is a counterfactual operation word, not another execution of the full declared action. In the intervention and gauge copy the return current lies in $[-0.005355160745, -0.005355160744]$, and the final edge momentum lies in $[0.002253305815, 0.002253305816]$. The difference from the disabled-drift return current obeys the exact outward enclosure

$$0 < \frac{242747271}{2^{40}} \le J_{\text{return}} - J_{\text{disabled}} \le \frac{30343409}{2^{37}}.$$

Thus the electric record changes the link that the later fermionic current actually consumes. Gauge-related executions agree on the invariant readouts.

The package `code/sm_fermion_current/` contains the producer `coupled.py` and independent verifier `verify_coupled.py`. They reconstruct the closed face boundaries, exact factor updates, current, Gauss residuals and consumed writer/value versions. The observer-like software patches retain local orbital state, incident ports, public records and electric feedback; declared parent labels cannot substitute for actual consumption. Local magnetic energies are exact rationals, with their large sum bounded by outward dyadic intervals.

The action, regulator, family population, prepared addresses, routing and time labels are supplied. The invariant reduction does not execute an excited Higgs or nonabelian sector, select a chiral continuum regulator, identify a laboratory current or physical clock, or establish compatibility with a sourced curved geometry. Those statements require their own common-action and physical realization maps.

B.1 Spin transport and the information needed to select exchange statistics

A one-particle spin representation and a many-particle exchange rule are different data. Likewise, a current obtained from a supplied fermionic action need not distinguish that action from a bosonic alternative on the preparations actually used. The following finite statements separate these two questions without assuming a relativistic field reconstruction.

Proposition B.3 (A spin lift does not select a two-particle exchange sector)**.** *Let $\mathcal{H}$ be a finite-dimensional complex Hilbert space with $\dim \mathcal{H} \geq 2$, and let $D : G \to U(\mathcal{H})$ be a supplied unitary representation. Suppose a central element $z \in G$ acts as $D(z) = -I$. On $\mathcal{H}\otimes\mathcal{H}$, let $\tau(v\otimes w) = w\otimes v$. Both eigenspaces*

$$\mathcal{H}_+ = \ker(\tau - I) = \operatorname{Sym}^2 \mathcal{H}, \qquad \mathcal{H}_- = \ker(\tau + I) = \bigwedge^2 \mathcal{H}$$

are nonzero invariant subspaces of $D \otimes D$. The central element acts as $+I$ on both. For every nonzero $v \in \mathcal{H}$, the vector $v \otimes v$ is a nonzero member of $\mathcal{H}_+$, whereas its antisymmetric projection is zero. Thus the one-particle spin action, including its central minus sign, does not by itself determine the exchange sign or exclude two particles in the same one-particle mode.

Proof. For any linear operator A on $\mathcal{H}$,

$$\tau(A \otimes A) = (A \otimes A)\tau,$$

as is checked on simple tensors, which span the tensor product. Hence the orthogonal projections $\Pi_\pm = (I \pm \tau)/2$ commute with $D(g) \otimes D(g)$ for every g. The restrictions are therefore unitary representations. Since $(-I) \otimes (-I) = I$, they have the stated central action. If $v \neq 0$, then $\|v \otimes v\| = \|v\|^2 > 0$ and $\tau(v \otimes v) = v \otimes v$, so $\Pi_-(v \otimes v) = 0$. For orthonormal u, v, the unit vector $(u \otimes v - v \otimes u)/\sqrt{2}$ belongs to $\mathcal{H}_-$. Both sectors therefore exist for the same supplied spin representation, but they disagree on the doubled-mode state. □

This statement applies, in particular, to a supplied two-dimensional unitary realization of a binary rotation group whose central element is $-I$. A nonsplit double cover or a unique spin structure does not remove the displayed choice of two-particle sector. The statement does not construct a relativistic bosonic spinor field satisfying locality and a positive-energy spectrum condition, and it is not a counterexample to a relativistic spin–statistics theorem. It establishes only the insufficiency of the finite one-particle representation data just specified.

Proposition B.4 (Conserved single-occupancy channels cannot select statistics)**.** *Let*

$$\mathcal{K} = \bigoplus_{\lambda \in \Lambda} \mathcal{K}_\lambda$$

be a finite orthogonal decomposition into one-particle channel spaces. In both the bosonic and fermionic Fock spaces over $\mathcal{K}$, retain the sector with exactly one particle in every channel. Use the same initial normalized channel vectors ψ_λ, the same classical registers, and the same ordered finite sequence of the following operations:

1. *evolution by a self-adjoint channel-preserving quadratic Hamiltonian, or second quantization of a specified channel-preserving one-particle unitary;*
2. *readout of expectations of channel-preserving one-body operators;*
3. *classical register updates whose values are determined by those readouts and the retained classical registers.*

The two statistics give identical one-body densities, all such readouts, and every resulting classical register value. In particular, an expectation current, mean Gauss residual, and classical feedback history formed from these data cannot determine the exchange statistics.

Proof. Order the finite channel set once. On the algebraic tensor product define

$$J_{\pm}\Big(\bigotimes_{\lambda} v_{\lambda}\Big) = \prod_{\lambda} a_{\pm}^{\dagger}(v_{\lambda})\,|0\rangle,$$

where $+$ denotes the bosonic and $-$ the fermionic creation operators, and the product follows the fixed order. Orthogonality of the channel spaces and the canonical relations show that the inner product of two such vectors is $\prod_{\lambda}\langle v_{\lambda}, w_{\lambda}\rangle$ in both cases. The maps therefore extend to unitary identifications between $\bigotimes_{\lambda}\mathcal{K}_{\lambda}$ and the respective fixed-channel-occupation sectors.

For a one-body operator B_{λ} supported in channel λ, its number-conserving second quantization obeys

$$J_{\pm}^{*}\, d\Gamma_{\pm}(B_{\lambda})\, J_{\pm} = I \otimes \cdots \otimes B_{\lambda} \otimes \cdots \otimes I.$$

In the fermionic calculation, the signs from moving annihilation and creation through the preceding occupied channels cancel. In the bosonic calculation there is no sign, and occupation one gives the same matrix coefficient. Consequently the restriction of a channel-diagonal quadratic Hamiltonian $d\Gamma_{\pm}(\bigoplus_{\lambda} h_{\lambda})$ is the same sum of tensor-slot Hamiltonians. The restrictions of second-quantized channel-preserving unitaries are likewise the same tensor products. Starting from the same tensor-product vector therefore gives the same channel vectors after every such operation. Their common one-body density is

$$P = \bigoplus_{\lambda} |\psi_{\lambda}\rangle\langle\psi_{\lambda}|, \qquad \langle d\Gamma_{\pm}(B)\rangle = \operatorname{Tr}(PB)$$

for channel-diagonal B. Equal classical inputs select equal subsequent operations and equal readouts give equal classical updates. Induction on the operation sequence proves the assertion, including adaptive mean-field feedback. A channel declared to remain empty contributes its vacuum factor and zero one-body density in both constructions. □

For the finite hypercharge-current construction, take $\lambda = (s, a)$ with $a = 1, \ldots, d_s$ and $\mathcal{K}_{\lambda} = \mathbb{C}^{64} \otimes \mathbb{C}^{2}$, the prepared site and Spin coordinates. There is one occupied orbital in each of the fifteen channels. Its abelian hopping and phase interventions act within each channel. Replacing the supplied CAR Fock state by the bosonic number state with one particle in each of the same channels preserves

$$P = \bigoplus_{s} I_{d_s} \otimes |\psi_s\rangle\langle\psi_s|, \qquad 0 \le P \le I, \qquad \operatorname{Tr} P = 15.$$

The projector property in this preparation is therefore not itself a statistics discriminator. The multiplicity index is an orthogonal one-particle label; repeating the same spatial–Spin orbital across different multiplicity indices does not prepare two particles in one complete one-particle mode.

The local Cayley updates use the same one-particle matrices in the two constructions. This comparison second-quantizes each one-particle Cayley unitary; it does not replace it by the Cayley transform of the full many-body Hamiltonian. The midpoint orbitals, and hence the midpoint variational functional and its current, also agree algebraically. Those midpoint averages are not normalized intermediate Fock states. Therefore the charge transfers, electric kicks, and expectation Gauss residuals coincide. The subsequent electric drifts and Wilson plaquette kicks use only the common classical registers, so they preserve this equality before the next matter factor reads the evolved link.

The retained site-charge variance supplies no distinction either. A single particle in channel (s, a) has site probability $p_{s,v} = \|\psi_{s,v}\|^{2}$, and its site-number projector squares to itself for both statistics. The product over channels gives

$$\operatorname{Var}(Q_v) = \sum_{s} d_s q_s^{2} p_{s,v}(1 - p_{s,v})$$

in both states. The nonzero variance excludes an operator Gauss constraint with the specified classical electric numbers. Its equality does not turn that constraint into a consequence of mean Gauss.

These identical histories concern the executed abelian, channel-preserving restriction. They do not assert equivalence of the full nonabelian or Yukawa theories, of channel-mixing evolutions, or of preparations with two particles in one conserved channel. Nor do they change the validity of the CAR current calculation under its stated assumptions. They show why that current calculation alone does not select those assumptions. An exchange-sensitive comparison must specify additional preparation and readout operations outside this restriction, together with their source realization. A finite comparison of supplied alternatives is distinct from deriving physical spin–statistics from observer consistency.

Proposition B.5 (A two-particle port discriminator)**.** *Let L, R be orthonormal modes of one declared channel, let $a = 3/5$, $b = 4/5$, and supply the one-particle unitary*

$$U = \begin{pmatrix} a & -ib \\ -ib & a \end{pmatrix}.$$

Prepare $\Psi_\pm = (L \otimes R \pm R \otimes L)/\sqrt{2}$ and apply $U \otimes U$. For the coincidence effect

$$F = |LR\rangle\langle LR| + |RL\rangle\langle RL|,$$

the antisymmetric and symmetric preparations have probabilities

$$p_- = 1, \qquad p_+ = (a^2 - b^2)^2 = \frac{49}{625}.$$

A distinguishable-particle preparation $L \otimes R$ gives $p_{\mathrm{dist}} = a^4 + b^4 = 337/625$. All three preparations have the same summed one-particle density I before and after the operation, so this contrast requires a two-particle readout.

Proof. The matrix is unitary because $a^2 + b^2 = 1$, and its determinant is one. Direct expansion gives

$$(U \otimes U)\Psi_- = \Psi_-, \qquad (U \otimes U)\Psi_+ = (a^2 - b^2)\Psi_+ - i\sqrt{2}ab\,(LL + RR).$$

The three vectors Ψ_+, LL, RR are orthonormal, so the bosonic coincidence probability is $(a^2 - b^2)^2$, with probability $2a^2b^2 = 288/625$ at each bunched port. The antisymmetric output has no bunched component. For the distinguishable product input the coincidence amplitudes are a^2 and $-b^2$ in orthogonal labelled outcomes; their squared moduli add to $337/625$. Its bunching probability at each port is $a^2b^2 = 144/625$. Finally the two orthogonal occupied orbitals give summed one-particle density I, and $UIU^\dagger = I$. □

The port labels denote a supplied invariant channel basis. They do not assert that the two physical ports carry identical raw spin-coordinate vectors: a spin-dependent hopping matrix can use different local basis vectors. The rational coefficients agree with a finite hopping factor, but this discriminator supplies its own two-particle preparation and coincidence effect. It inherits no electric flux dressing, operator Gauss constraint, clock identification or laboratory instrument from a one-particle-per-channel history.

Indistinguishability is essential. If the particles carry normalized unobserved internal vectors α, β, unchanged by the port unitary, write $\eta = |\langle \alpha, \beta \rangle|^2$. Expanding the normalized symmetrized or antisymmetrized input and tracing over those vectors gives

$$p_{\mathrm{fermion}} = \frac{337 + 288\eta}{625}, \qquad p_{\mathrm{boson}} = \frac{337 - 288\eta}{625}.$$

The cross term is the product $\langle\alpha,\beta\rangle\langle\beta,\alpha\rangle$; its sign is the exchange sign. Orthogonal hidden states give $\eta = 0$ and erase the contrast. The ideal values above require $\eta = 1$.

There is also a conditional stability bound. Let ρ be either ideal output density and $\widetilde{\rho}$ an actual normalized output density with $\|\widetilde{\rho} - \rho\|_1/2 \le \epsilon$. If an actual effect $0 \le \widetilde{F} \le I$ satisfies $\|\widetilde{F} - F\| \le \delta$, then

$$\left|\mathrm{Tr}(\widetilde{\rho}\widetilde{F}) - \mathrm{Tr}(\rho F)\right| \le \epsilon + \delta.$$

Indeed, split the difference into $\mathrm{Tr}((\widetilde{\rho} - \rho)F)$ and $\mathrm{Tr}(\widetilde{\rho}(\widetilde{F} - F))$. The positive and negative parts of the trace-zero density difference each have trace at most ϵ, so the first term has magnitude at most ϵ; the second is bounded by δ. A preparation trace-distance bound transfers unchanged through an exactly common unitary; imperfect dynamics instead require a bound on the actual output. For a common error budget $\epsilon + \delta < 288/625$, the two ideal-model probability intervals are disjoint because their separation is $576/625$. These are supplied error bounds, not a calibration or a bound on finite sampling error. An observed test additionally needs authenticated preparation, indistinguishability, operation and readout evidence.

B.2 Conditional exclusion from a stable generator and a scalar deformation law

One can select exclusion without assuming it, provided the quantization class and the relation between occupation and dynamics are specified. The following theorem permits a real scalar deformation parameter rather than declaring either canonical commutation or anticommutation relations. Its phase-generator and stability hypotheses are additional physical inputs; a finite spin representation alone supplies neither.

Theorem B.6 (Stability selects exclusion within a scalar deformation class)**.** *Let $\mathcal{K}$ be a finite-dimensional complex one-particle Hilbert space. On a positive Hilbert space $\mathcal{F}$, let $v \mapsto a^\dagger(v)$ be complex-linear, with $a(v)$ its adjoint on a common dense invariant domain $\mathcal{D}$. Assume $\mathcal{D}$ is invariant under these operators and a self-adjoint Hamiltonian H, and is contained in their domains. Supply:*

1. *a common real number q such that, for every unit v,*

$$a(v)a^\dagger(v) - q\,a^\dagger(v)a(v) = I$$

   on $\mathcal{D}$;

2. *the exact occupation/phase law, for every unit v,*

$$N_v = a^\dagger(v)a(v), \qquad [N_v, a^\dagger(v)] = a^\dagger(v);$$

3. *a normalized $\Omega \in \mathcal{D}$ with $a(v)\Omega = 0$ for all v and $H\Omega = E_0\Omega$;*

4. *a unit mode v_- and $\kappa > 0$ such that*

$$[H, a^\dagger(v_-)] = -\kappa a^\dagger(v_-)$$

   on $\mathcal{D}$, and a finite uniform lower bound for H.

Then $q = -1$, and on $\mathcal{D}$,

$$\{a(v), a^\dagger(w)\} = \langle v, w\rangle I, \qquad \{a^\dagger(v), a^\dagger(w)\} = 0, \qquad \{a(v), a(w)\} = 0.$$

Here the inner product is conjugate-linear in its first argument. In particular $(a^\dagger(v))^2 = 0$ for every mode v. The operators extend boundedly to $\mathcal{F}$, where the same relations hold. The reference vacuum Ω is not assumed to be a ground state.

Proof. Fix a unit mode, abbreviate its operators to $a, a^\dagger, N$, and work throughout on the common invariant domain. The deformation law gives

$$Na^\dagger = a^\dagger + q\, a^\dagger N.$$

Comparison with the occupation/phase law yields $(q-1)a^\dagger N = 0$. The vector $|1\rangle = a^\dagger\Omega$ has norm one and $N|1\rangle = |1\rangle$. If $q \neq 1$, it follows that $a^\dagger|1\rangle = 0$. Taking its squared norm using the deformation law gives $0 = 1 + q$. Thus q is either 1 or -1.

If $q = 1$, induction using $aa^\dagger = I + a^\dagger a$ gives

$$|n\rangle = (a^\dagger)^n\Omega, \qquad \| \, |n\rangle\|^2 = n!, \qquad N|n\rangle = n|n\rangle$$

for every integer $n \geq 0$. Apply this to v_-. The Hamiltonian commutator and the vacuum eigenvalue imply $H|n\rangle = (E_0 - n\kappa)|n\rangle$. All these vectors are nonzero and belong to $\mathcal{D}$, contradicting the lower bound. This is an arbitrary-n argument, not a finite occupation scan. Hence $q = -1$.

For any unit mode, the phase identity then gives $a^\dagger N = 0$, whereas the deformation law gives $Na^\dagger = a^\dagger$. Consequently

$$0 = a^\dagger N a^\dagger = (a^\dagger)^2.$$

Linearity and scaling extend this square-zero identity to every v, including $v = 0$. Expanding it at $v + w$ proves creation anticommutation, and taking adjoints on the dense invariant domain proves annihilation anticommutation. The diagonal relation is $\{a(v), a^\dagger(v)\} = \|v\|^2 I$ for every v. Polarization at $v + w$ and $v + iw$ gives the mixed relation. Finally, for $\psi \in \mathcal{D}$,

$$\|a(v)\psi\|^2 + \|a^\dagger(v)\psi\|^2 = \|v\|^2\|\psi\|^2.$$

Both operators therefore have bounded extensions, and density extends their identities to all of $\mathcal{F}$. □

Corollary B.7 (Finite-dimensional alternative to the stability hypothesis)**.** *Retain the theorem's common scalar deformation law, linear mode fields, exact occupation/phase law and normalized annihilation vacuum, with $\mathcal{K} \neq \{0\}$. If $\mathcal{F}$ itself is nonzero and finite-dimensional, the same exclusion and CAR conclusions hold without assuming a Hamiltonian or a negative ladder.*

Proof. A dense linear domain in finite dimension is the whole space. The vacuum argument gives $q \in \{1, -1\}$. If $q = 1$, the trace of $aa^\dagger - a^\dagger a = I$ is both zero and $\dim \mathcal{F} > 0$, a contradiction. The square-zero and polarization arguments for $q = -1$ require no Hamiltonian. □

This alternative requires finite dimension of the Hilbert space carrying the full creation and annihilation algebra. A finite one-particle mode space alone is insufficient. A finite observer response space does not by itself identify that operator representation with physical matter. Applying the corollary to observer finite capacity requires precisely this additional identification, as well as the stated deformation and phase laws.

The negative-mode hypothesis has a concrete one-particle instance in the supplied finite matter hopping action. For a unitary spin-link matrix K and $\kappa > 0$, its edge matrix is

$$h_e = \kappa \begin{pmatrix} 0 & K \\ K^\dagger & 0 \end{pmatrix}, \qquad h_e^2 = \kappa^2 I.$$

For any unit spin vector χ, the vector $v_- = (-K\chi, \chi)/\sqrt{2}$ is normalized and satisfies $h_e v_- = -\kappa v_-$. The quantum dynamical law $[H, a^\dagger(v)] = a^\dagger(h_e v)$ would therefore supply the required ladder. This

law is a hypothesis linking the one-particle action to its quantization. A one-particle matrix, a measured unitary step, or the existence of a spin lift does not establish that law.

There is a finite nonvacuity witness. In ordinary fermionic Fock space over the edge modes, take the empty vacuum and $H = d\Gamma(h_e)$ with its normal ordering fixed at that vacuum. The negative-energy modes can each be occupied once, and the ground energy is the sum of all negative one-particle eigenvalues. Filling those modes gives the ground state; particles in positive modes and holes in filled negative modes both have excitation energy κ. For the two-spin edge with fifteen internal channels of the supplied matter census, there are thirty negative modes. Its ground energy is -30κ. With

$$(d_s) = (6, 3, 3, 2, 1), \qquad (q_s) = (1, -4, 2, -3, 6), \qquad \sum_s d_s q_s = 0,$$

the filled ground state has total hypercharge zero. Subtracting its constant energy gives a positive excitation Hamiltonian and preserves the field commutator. The reference empty vacuum and the filled ground state are different vectors.

For comparison, canonical bosonic Fock space on the same fixed edge action admits occupation n in one negative mode of every internal channel. These states have total hypercharge zero and energy $-15n\kappa$. Thus even the globally neutral sector has no uniform lower bound. Adding a chemical potential proportional to total hypercharge cannot repair this sequence. This neutrality statement does not impose local quantum Gauss constraints or nonabelian singlet constraints, and is not a claim about a gauge-invariant physical Hilbert space.

The hypotheses exclude several concrete alternatives. At fixed total particle number the bosonic Hamiltonian is bounded below, so stability across all occupation sectors is essential. Replacing the one-particle law by $h_e + cI$ with $c \geq \kappa$ stabilizes its full bosonic Fock Hamiltonian: its second-quantized change is cN_{total}, not a constant vacuum-energy subtraction. Positive number interactions can also bound the energy, while changing the negative ladder. Neither change is ruled out by the theorem. A positive spectral replacement such as $|h_e|$ likewise changes the supplied dynamics.

Even the phase law and stability together do not force exclusion. On a three-dimensional Hilbert space with basis $|0\rangle, |1\rangle, |2\rangle$, set

$$a^\dagger|0\rangle = |1\rangle, \qquad a^\dagger|1\rangle = \sqrt{2}|2\rangle, \qquad a^\dagger|2\rangle = 0.$$

Then $N = a^\dagger a = \operatorname{diag}(0, 1, 2)$ satisfies $[N, a^\dagger] = a^\dagger$, and $H = -\kappa N$ is bounded below, but $(a^\dagger)^2|0\rangle \neq 0$. This model violates the scalar deformation hypothesis: $aa^\dagger = \operatorname{diag}(1, 2, 0)$, whose middle entry requires $q = 1$ while the top entry then contradicts $aa^\dagger - qN = I$. A hard occupation cutoff or a more general exchange algebra is therefore a genuine alternative outside the stated class.

The theorem is a conditional Pauli-exclusion derivation within a specified quantization class. It is insensitive to spin, so it cannot attach half-integer spin to fermionic statistics or exclude a different quantization class for a different field. The common scalar deformation law, linear field structure, exact occupation/phase law, unrestricted ladder domain and stable physical generator remain supplied. The finite filled-sea construction supplies no continuum vacuum or relativistic locality theorem. Deriving these inputs and selecting the physical matter action from observer consistency remain separate requirements.

B.3 What finite source operations must supply for exclusion

Finite-dimensional response, unitary covariance and a quadratic occupation generator do not alone imply the scalar deformation law used in the stability theorem. A normalized operational interpretation can replace that law, but requires an exact identification between a field operator and an observer operation. The following constructions distinguish these requirements.

Proposition B.8 (Globally covariant finite occupation cutoff)**.** *Let $\mathcal{K} = \mathbb{C}^m$ with $m \geq 1$, and*

$$\mathcal{F}_{\leq 2} = \mathbb{C} \oplus \mathcal{K} \oplus \operatorname{Sym}^2 \mathcal{K}.$$

This positive Hilbert space has dimension $(m+1)(m+2)/2$; the two-mode case has dimension six and the four-mode spin edge has dimension fifteen. Compress the usual bosonic creation and annihilation operators to this total-occupation cutoff, writing $C(v) = P_{\leq 2} b^\dagger(v) P_{\leq 2}$ and $A(v) = C(v)^\dagger$. Then C is linear in v, the normalized empty vacuum is annihilated by all $A(v)$, and for every unit v,

$$N_v = C(v)A(v), \qquad [N_v, C(v)] = C(v).$$

For every genuine unitary U on $\mathcal{K}$,

$$\Gamma_{\leq 2}(U) C(v) \Gamma_{\leq 2}(U)^\dagger = C(Uv).$$

For any Hermitian one-particle h, its restricted second quantization $H = d\Gamma(h)$ obeys $[H, C(v)] = C(hv)$ and is bounded below. Nevertheless, $C(v)^2\Omega \neq 0$ for every nonzero v, and no scalar q satisfies $A(v)C(v) - qC(v)A(v) = I$ even for one fixed unit mode.

Proof. Annihilation lowers total occupation, so $C(v)A(v)$ is the restriction of the ordinary bosonic number operator $d\Gamma(|v\rangle\langle v|)$. It preserves total-occupation sectors and commutes with the cutoff projection. The ordinary one-particle commutator identities therefore survive compression. At total occupation two the creation operator vanishes, so both sides of the claimed raising commutators vanish there as well. The cutoff also commutes with every $\Gamma(U)$ and with $d\Gamma(h)$, proving full unitary covariance and the dynamical identity. The finite spectrum gives

$$H \geq 2 \min\{0, \lambda_{\min}(h)\} I.$$

The normalized one-mode vectors $|v\rangle$ and $|2v\rangle$ have number eigenvalues one and two, while $\|C(v)^2\Omega\|^2 = 2$. On $|v\rangle$ the proposed scalar law requires $2 - q = 1$, hence $q = 1$. On $|2v\rangle$ it instead requires $-2q = 1$. More generally its exact commutator defect is

$$A(v)C(v) - C(v)A(v) = I - (N_v + I)P_2,$$

where P_2 projects onto total occupation two. This state-dependent boundary term cannot be replaced by a scalar deformation parameter. □

For a supplied hopping matrix with eigenvalues $\pm\kappa$, the cutoff Hamiltonian has minimum -2κ and admits double occupation of its negative mode. Thus its covariance, exact quadratic phase law and stability coexist with failure of exclusion. The construction is a counterexample to this specified finite implication. It is not a model of the complete observer axioms: in particular a projective one-particle transport does not automatically lift to a projective transport on all occupation sectors, since $\Gamma(e^{i\theta} I) = e^{i\theta N_{\text{total}}}$ need not be scalar. No source production of the cutoff field operators is asserted.

Proposition B.9 (Normalized operations and an exact occupation generator)**.** *On a finite-dimensional positive Hilbert space, let C be a creation operator, $A = C^\dagger$, and $N = CA = A^\dagger A$. Assume*

$$0 \leq N \leq I, \qquad [N, C] = C.$$

Then $C^2 = 0$. Consequently, for a complex-linear family $v \mapsto C(v)$ satisfying these conditions for every unit v, one has $\{C(v), C(w)\} = 0$ for all v, w. No scalar deformation law is required.

Proof. Choose an orthonormal eigenbasis of N, with eigenvalues $\lambda_i \in [0,1]$. The commutator identity gives

$$(\lambda_i - \lambda_j)C_{ij} = C_{ij}.$$

Every nonzero matrix entry therefore maps a zero-eigenvalue vector to a one-eigenvalue vector. A second application vanishes, so $C^2 = 0$. Scaling extends this conclusion from unit vectors to all modes, and expanding $C(v+w)^2 = 0$ gives the anticommutation identity. □

Under the same phase law the converse holds: if $C^2 = 0$, then $CC^\dagger C - C^2C^\dagger = C$ gives $CC^\dagger C = C$. Thus C is a partial isometry and a contraction. Within this phase-law class, contraction is equivalent to exclusion. Identifying an unscaled field operator with a normalized observer operation therefore carries the full selection content; it is not a harmless change of units.

This conclusion alone does not construct ordinary many-fermion Fock space. On $\mathbb{C}\Omega \oplus \mathcal{K}$, the operators $C(v) = |v\rangle\langle\Omega|$ satisfy the contraction and phase hypotheses for every unit v, but allow at most one particle in total. For $\dim \mathcal{K} \geq 2$, their mixed anticommutator is

$$\{C(v)^\dagger, C(w)\} = \langle v, w\rangle |\Omega\rangle\langle\Omega| + |w\rangle\langle v|,$$

which is not generally $\langle v, w\rangle I$. Additional structure is required to recover the full canonical field algebra.

For a linear family on m orthonormal modes, the exclusion test reduces to a finite set of exact operator identities. It suffices to check the contraction and unit phase conditions on every e_i and every $(3e_i + 4e_j)/5$ with $i < j$. The proposition gives $C(e_i)^2 = 0$; expanding the square at the pair vector then gives $\{C(e_i), C(e_j)\} = 0$. Linearity proves exclusion for every mode. To obtain full mixed CAR by the paired-instrument route, it suffices to check its completeness identity at the basis vectors, these real pairs, and the imaginary pairs $(3e_i + 4ie_j)/5$. The two pair identities force the symmetric and antisymmetric cross terms to vanish. These are exact matrix identities, not conclusions from finitely sampled outcome frequencies.

Basis-mode checks alone are insufficient. Set $\sigma_+ = |1\rangle\langle 0|$. Two commuting hard-core bit raisers $C_1 = \sigma_+ \otimes I$, $C_2 = I \otimes \sigma_+$ each satisfy the contraction and phase conditions, but $C = (3C_1 + 4C_2)/5$ has $C^2|00\rangle = (24/25)|11\rangle \neq 0$ and fails the unit phase law. Inserting the fermionic sign on the second raiser changes no occupation-basis transition probability. Coherent superposition tests, or source-derived identities that imply them, must therefore carry the relative-sign information.

The bound $N \leq I$ follows if the unscaled removal operator A is itself a Kraus operator of a trace-nonincreasing observer operation. Equivalently, A is a contraction; its adjoint C is a contraction too. This is an operational sufficient condition for exclusion only after the same operator's quadratic form $A^\dagger A$ is identified with the exact unit-step occupation generator. A normalized probability effect and a particle-number operator are not interchangeable merely because both are positive.

A stronger operational interface supplies the mixed relation as well. If a complete two-outcome instrument has exactly the adjoint-paired Kraus operators $A(v), C(v)$ for each unit v, trace preservation gives

$$C(v)A(v) + A(v)C(v) = I.$$

With the exact phase law and linearity, the preceding proposition and complex polarization recover all canonical anticommutation relations. The adjoint pairing, exhaustive two-outcome list and normalization for every input state are essential hypotheses; they are not consequences of probability normalization alone.

The cutoff example demonstrates the distinction. For a unit mode its operator $N_v + A(v)C(v)$ has eigenvalues at most three. Consequently

$$E_{\text{remove}} = \tfrac{1}{4}N_v, \qquad E_{\text{add}} = \tfrac{1}{4}A(v)C(v), \qquad E_{\text{idle}} = I - \tfrac{1}{4}\big(N_v + A(v)C(v)\big)$$

are positive and sum to identity. Kraus operators $A(v)/2$, $C(v)/2$ and $\sqrt{E_{\mathrm{idle}}}$ define a normalized instrument which retains the possibility of double occupation. Its removal effect is $N_v/4$, not the occupation generator N_v. In particular the phase commutator with that effect has coefficient $1/4$, not one. The extra idle outcome and the operator normalization cannot be suppressed in transferring an instrument to the exclusion theorem.

Projective readout has a different algebraic role. A Lüders outcome with Kraus operator $P = P^\dagger = P^2$ has $[PP^\dagger, P] = 0$, so a nonzero projector cannot satisfy the unit-step creation law. For the binary coordinate effects, an alternative pair $|1\rangle\langle 0|$, $|0\rangle\langle 1|$ is adjoint-paired, complete and satisfies the single-mode exclusion laws, but swaps the outcome-certain states rather than preserving them. The declared phase instrument and its swap-twisted comparison therefore illustrate different operations, despite carrying the same outcome effects. Their matrix-level semantics do not select a native particle-creation instrument. The distinction is explicit in `Lean/EventAlgebra/LuedersPhaseInstrument.lean` and `Lean/EventAlgebra/SourcePhaseInstrumentOutcomeBridge.lean`.

There is a separate obstruction in the finite history-count interface of `Lean/QFT/SourceHistoryGNSDynamics.lean`. Its eight three-bit histories carry the repair count

$$Q(x_0, x_1, x_2) = \mathbf{1}_{x_0 \neq x_1} + \mathbf{1}_{x_1 \neq x_2}.$$

The eigenspaces of its diagonal operator have dimensions two, four and two at eigenvalues zero, one and two respectively. No operator A on this full eight-dimensional carrier can obey both $Q = A^\dagger A$ and $[Q, A^\dagger] = A^\dagger$. Indeed the adjoint commutator sends A from the eigenvalue-one space into the eigenvalue-zero space. On the former space $A^\dagger A = Q$ would make A an isometry, which cannot map four dimensions into two. This excludes identifying that particular repair count with one quadratic mode occupation; it excludes neither multiple mode generators nor different source observables or larger carriers.

Counted transitions can supply a weaker positive statement. If a record basis has count labels n_x and an operator T has nonzero entry T_{yx} only when $n_y = n_x + 1$, then $[\mathrm{diag}(n_x), T] = T$ exactly, for arbitrary transition amplitudes. This follows entry by entry. It does not identify $\mathrm{diag}(n_x)$ with $TT^\dagger$: that additional equality constrains transition magnitudes, ranks and degeneracies. The native history packet fixes its diagonal counts and empirical state. Its extension to the complete matrix algebra does not authenticate these occupation-changing operations or their quantum instrument semantics.

Finite covariant response therefore does not force the scalar deformation law, while normalized operations can imply exclusion under an exact quadratic occupation/phase identification. The stated history-count interface supplies neither that identification nor an exhaustive adjoint-paired quantum instrument. These finite results leave open an observer-native construction of the required operations, a different quantization mechanism, and the physical spin–statistics attachment.

Credited prior work

The results of this paper build on the following established work. Each entry names an established result as it is used here and cites its sources.

- A5 representation theory, compact Lie classification, covering groups, exterior unification, and signed graph operators [127, 99, 92, 157].
- Ad-invariant quadratic kinetic forms on reductive Lie algebras and gauge kinetic terms [152].
- Algebraic renormalization and BV/BRST cohomology [144, 95, 94, 124].

- Analytic S-matrix and resonance-pole theory [112].
- Complex-pole renormalization, Nielsen identities, FJ tadpole schemes, and finite-order electroweak calculations [147, 140, 116, 113, 107].
- DHR/DR/Tannaka reconstruction, compact-group representation theory, and Standard Model anomaly algebra [109, 110, 111, 148, 99, 96, 120].
- Finite fermion Fock space and lattice gauge current identities [98, 130].
- Finite lattice connections, Wilson-loop holonomy, vertex recharting, algebraic two-path character phases (an Aharonov–Bohm analogy only), and characters of cyclic groups [155, 91, 145].
- Finite lattice gauge theory and the Standard Model action [155, 138].
- Finite spectral theory, Kraus instruments and truncated symmetric Fock spaces [122, 132, 104, 98].
- Finite-dimensional Hamiltonian ODE enclosures and nonlinear detector Taylor bounds [139, 123].
- Finite-group equivariant current algebras, compact Lie algebras, normed approximation, and exact algebraic representation certificates [115, 129, 97].
- Finite-group heat kernels, character projectors, and lattice-gauge toy models [108, 145, 131].
- Ginsparg-Wilson fermions and determinant-line formulations [118, 136].
- Hamiltonian lattice gauge theory and constrained quantization [130, 124].
- Icosahedral Galois conjugacy and classical topological degree [102, 137].
- Icosahedral graph theory, finite group actions, equiangular tight frames, and universal optimality [100, 134, 101].
- Invariant quadratic forms on compact semisimple Lie algebras and finite Gibbs product laws [129, 93].
- Osterwalder-Schrader reconstruction [141, 142].
- Product gauge algebras and simple-GUT X/Y generators [117, 146].
- Scalar q oscillators, canonical Fock quantization and the finite trace obstruction [119, 153, 98].
- Standard Model fermion bilinears with charge-diagonal structure [143, 133].
- SU(5)-style exterior matter embeddings and Standard Model anomaly algebra [117, 92, 96, 120].
- Super-Tannakian reconstruction, exterior-algebra matter embeddings, binary icosahedral covers, and anomaly algebra [106, 92, 103, 96].
- Symmetric and exterior Fock representations [114, 98].
- Tannaka duality, representation equivalence, and commuting-diagram methods [105, 128, 110].
- Textbook electroweak Lagrangian assembly of gauge kinetic, Higgs potential, gauge mass, and Yukawa sectors [150, 143, 156].
- two-particle bunching and antibunching [125, 135].
- Weinberg and Wilczek-Zee dimension-six baryon-number-violating operator classification and the SMEFT operator basis [151, 154, 121].
- Z6 center-character descent bookkeeping for Standard Model global gauge-group forms [149, 126].

References

[1] L. Bombelli, J. Lee, D. Meyer, and R. D. Sorkin, *Space-time as a causal set*, Phys. Rev. Lett. **59**, 521–524 (1987). https://doi.org/10.1103/PhysRevLett.59.521.

[2] S. Surya, *The causal set approach to quantum gravity*, Living Rev. Relativ. **22**, 5 (2019). https://doi.org/10.1007/s41114-019-0023-1.

[3] L. Bombelli and D. A. Meyer, *The origin of Lorentzian geometry*, Phys. Lett. A **141**, 226–228 (1989). https://doi.org/10.1016/0375-9601(89)90474-X.

[4] G. Brightwell and R. Gregory, *Structure of random discrete spacetime*, Phys. Rev. Lett. **66**, 260–263 (1991). https://doi.org/10.1103/PhysRevLett.66.260.

[5] S. Major, D. Rideout, and S. Surya, *On recovering continuum topology from a causal set*, J. Math. Phys. **48**, 032501 (2007). https://doi.org/10.1063/1.2435599.

[6] S. Major, D. Rideout, and S. Surya, *Stable homology as an indicator of manifoldlikeness in causal set theory*, Class. Quantum Grav. **26**, 175008 (2009). https://doi.org/10.1088/0264-9381/26/17/175008.

[7] L. Glaser and S. Surya, *Towards a definition of locality in a manifoldlike causal set*, Phys. Rev. D **88**, 124026 (2013). https://doi.org/10.1103/PhysRevD.88.124026.

[8] J. Myrheim, *Statistical geometry*, CERN preprint TH-2538 (1978).

[9] D. A. Meyer, *The Dimension of Causal Sets*, PhD thesis, Massachusetts Institute of Technology (1988).

[10] D. D. Reid, *Manifold dimension of a causal set: Tests in conformally flat spacetimes*, Phys. Rev. D **67**, 024034 (2003). https://doi.org/10.1103/PhysRevD.67.024034.

[11] D. M. T. Benincasa and F. Dowker, *The scalar curvature of a causal set*, Phys. Rev. Lett. **104**, 181301 (2010). https://doi.org/10.1103/PhysRevLett.104.181301.

[12] F. Dowker and L. Glaser, *Causal set d'Alembertians for various dimensions*, Class. Quantum Grav. **30**, 195016 (2013). https://doi.org/10.1088/0264-9381/30/19/195016.

[13] L. Bombelli, J. Henson, and R. D. Sorkin, *Discreteness without symmetry breaking: A theorem*, Mod. Phys. Lett. A **24**, 2579–2587 (2009). https://doi.org/10.1142/S0217732309031958.

[14] S. W. Hawking, A. R. King, and P. J. McCarthy, *A new topology for curved space–time which incorporates the causal, differential, and conformal structures*, J. Math. Phys. **17**, 174–181 (1976). https://doi.org/10.1063/1.522874.

[15] D. B. Malament, *The class of continuous timelike curves determines the topology of spacetime*, J. Math. Phys. **18**, 1399–1404 (1977). https://doi.org/10.1063/1.523436.

[16] B. Müller, J. Kim, D. Matscheko, and J. Hill, *Observation-Determined Normal Forms: Stability, Obstructions, and Refinement in Constraint and Rewrite Systems*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/extra/observable_normal_forms.pdf.

[17] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, M. A. Visser, D. Matscheko, and B. de La Fournière, *Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency*, 2026. Available at GitHub PDF.

[18] B. Müller, M. Poneder, A. Osika, K. Xue, P. Nguyen, D. Matscheko, and D. A. Friedman, *Standard Model Gauge Structure under Complete-Response and Observer-Transport Premises*, 2026. Available at GitHub PDF.

[19] B. Müller, K. Xue, J. Kim, K. A. Anirudha, D. Matscheko, and J. Hill, *Reality as a Consensus Protocol: The Fixed-Point Computation That Implements Physics*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/reality_as_consensus_protocol.pdf.

[20] B. Müller, A. Osika, K. Xue, B. Cassie, B. de La Fournière, and D. A. Friedman, *Twelve-Port Icosahedral Carriers and Native Seam Repair: The Carrier Layer of Observer-Patch Holography*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/screen_microphysics_and_observer_synchronization.pdf.

[21] B. Müller, M. Poneder, A. Osika, K. Xue, and D. A. Friedman, *Particles as Localized Topological Defects of Native Overlap Transport: Finite Theorems, a Native Decision Test and Mass Obstructions*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/deriving_the_particle_zoo_from_observer_consistency.pdf.

[22] B. Müller and D. Matscheko, *Observer-Patch Holography and the Dark Sector: Modular Charge, the Anomalous Collar Source, and the Deep Galaxy Law*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/cosmology/oph_dark_matter_paper.pdf.

[23] B. Müller and J. Hill, *A Conditional Identification of the Yang–Mills Gap with a Finite Repair Gap: Dobrushin Repair Floors and a $\mathbb{Z}_2$ Test of the Transfer Receipt*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/extra/yang_mills_gap_clay_problem.pdf.

[24] B. Müller, *Observer-Patch Holography as a String-Vacuum Selector: Observers, Clocks, Edge Strings, and the Bouchard-Donagi Witness*, 2026. Archived manuscript. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/archive/observer_patch_holography_as_string_vacuum_selector.pdf.

[25] B. Müller, M. Poneder, A. Osika, K. Xue, B. Cassie, P. Nguyen, J. Kim, D. Matscheko, D. A. Friedman, J. Hill, W. T. Glynn, M. A. Visser, K. A. Anirudha, and B. de La Fournière, *Observers Who Read Their Universe into Existence: Finite Consensus, an Executable Observer, and Standard Physics*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/flagship/from_observer_consensus_to_standard_physics.pdf.

[26] B. Müller, M. Poneder, J. Kim, and D. A. Friedman, *An Executable Observer for Observer Patch Holography: Specification, Executed Components and Ordered Joins*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/paper/executable_observer_construction.pdf.

[27] B. Müller, *The de Sitter Time-Advance Sign from a Finite Screen with Fixed Capacity*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography/blob/r2055/extra/de_sitter_time_advance_sign_from_fixed_screen_capacity.pdf.

[28] B. Müller, *OPH-FPE: finite simulator and receipt engine for Observer-Patch Holography physics experiments*, 2026. Source available at https://github.com/muellerberndt/oph-physics-sim. Large-run source revision: https://github.com/muellerberndt/oph-physics-sim/tree/ce17921eb7504106fef1ba445e1349b5367aa676. Exact-federation archive: https://github.com/FloatingPragma/observer-patch-holography/tree/r2055/evidence/exact_federation_L6_canonical_20260909. Closure-loop archive: https://github.com/FloatingPragma/observer-patch-holography/tree/r2055/evidence/closure_loop. Active refinement-readout revision: https://github.com/muellerberndt/oph-physics-sim/tree/036b12b608a9ca4c19d69f5e951b3b83a7a51cd8. Finite local-domain revision and data: https://github.com/muellerberndt/oph-physics-sim/tree/550fe77dc67dc8bef2dd8927bcf72ec98cfa3506, https://github.com/muellerberndt/oph-physics-sim/tree/550fe77dc67dc8bef2dd8927bcf72ec98cfa3506/data/local_domain. Exact repository-local archive and checker: https://github.com/FloatingPragma/observer-patch-holography/tree/r2055/evidence/local_domain. Causal-poset evidence package: https://github.com/FloatingPragma/observer-patch-holography/tree/r2055/evidence/source_net_causal_poset. Closure-loop evidence package: https://github.com/FloatingPragma/observer-patch-holography/tree/r2055/evidence/closure_loop. Evidence index: https://github.com/FloatingPragma/observer-patch-holography/tree/r2055/evidence.

[29] Pragma Research, *Observer-Patch Holography: machine-checked proofs and executable certificates*, 2026. Available at https://github.com/FloatingPragma/observer-patch-holography.

[30] Pragma Research, *Frozen Prediction Register and Primitive Twelve-Port Prediction Receipt*, 2026. Immutable source receipt: https://github.com/FloatingPragma/observer-patch-holography/blob/66176656dc1143f9ec50ba1a6e409c403545857f/code/a5_fingerprint/runtime/spin_six_primitive_port_prediction_receipt.json. Append-only custody packet: https://github.com/FloatingPragma/observer-patch-holography/tree/bb73f80d459113ccc82d5a5b5219d1dda78e9a79/evidence/custody/falsification/frozen_targets/fz11_2026-07-31.

[31] DESI Collaboration, "DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints," 2025, arXiv:2503.14738. https://arxiv.org/abs/2503.14738. Official chains and checksum manifest: https://data.desi.lbl.gov/public/papers/y3/bao-cosmo-params/README.html.

[32] M. Filzinger et al., "Improved Limits on the Coupling of Ultralight Bosonic Dark Matter to Photons from Optical Atomic Clock Comparisons," *Physical Review Letters* **130**, 253001 (2023). https://doi.org/10.1103/PhysRevLett.130.253001.

[33] F. Lelli, S. S. McGaugh, and J. M. Schombert, "SPARC: Mass Models for 175 Disk Galaxies with Spitzer Photometry and Accurate Rotation Curves," *Astronomical Journal* **152**, 157 (2016), arXiv:1606.09251. https://arxiv.org/abs/1606.09251. CDS/VizieR catalogue: https://cdsarc.cds.unistra.fr/ftp/J/AJ/152/157/.

[34] S. S. McGaugh, F. Lelli, and J. M. Schombert, "Radial Acceleration Relation in Rotationally Supported Galaxies," *Physical Review Letters* **117**, 201101 (2016), arXiv:1609.05917. https://arxiv.org/abs/1609.05917.

[35] L. Onsager, "Reciprocal Relations in Irreversible Processes. I," *Physical Review* **37**, 405–426 (1931), https://doi.org/10.1103/PhysRev.37.405.

[36] M. S. Green, "Markoff Random Processes and the Statistical Mechanics of Time-Dependent Phenomena. II," *Journal of Chemical Physics* **22**, 398–413 (1954), https://doi.org/10.1063/1.1740082; R. Kubo, "Statistical-Mechanical Theory of Irreversible Processes. I," *Journal of the Physical Society of Japan* **12**, 570–586 (1957), https://doi.org/10.1143/JPSJ.12.570.

[37] I. Csiszár, "I-Divergence Geometry of Probability Distributions and Minimization Problems," *The Annals of Probability* **3**, 146–158 (1975), https://doi.org/10.1214/aop/1176996454.

[38] J. E. Marsden and M. West, "Discrete Mechanics and Variational Integrators," *Acta Numerica* **10**, 357–514 (2001), https://doi.org/10.1017/S096249290100006X.

[39] I. Esteban, M. C. Gonzalez-Garcia, M. Maltoni, I. Martinez-Soler, J. P. Pinheiro, and T. Schwetz, *NuFit-6.0: Updated global analysis of three-flavor neutrino oscillations*, JHEP **12** (2024) 216, arXiv:2410.05380, with the NuFIT 6.1 (2025) profile-table release at https://www.nu-fit.org/?q=node/309.

[40] M. H. A. Newman, "On theories with a combinatorial definition of 'equivalence'," *Ann. of Math. (2)* **43** (1942), 223–243.

[41] E. H. Lieb and D. W. Robinson, "The finite group velocity of quantum spin systems," *Commun. Math. Phys.* **28** (1972), 251–257.

[42] J. J. Bisognano and E. H. Wichmann, "On the duality condition for a Hermitian scalar field," *J. Math. Phys.* **16** (1975), 985–1007.

[43] J. J. Bisognano and E. H. Wichmann, "On the duality condition for quantum fields," *J. Math. Phys.* **17** (1976), 303–321.

[44] R. Brunetti, D. Guido, and R. Longo, "Modular Structure and Duality in Conformal Quantum Field Theory," *Commun. Math. Phys.* **156** (1993), 201–219, arXiv:funct-an/9302008.

[45] H.-W. Wiesbrock, "Half-Sided modular inclusions of von-Neumann-Algebras," *Commun. Math. Phys.* **157** (1993), 83–92.

[46] H. Araki and L. Zsidó, "Extension of the structure theorem of Borchers and its application to half-sided modular inclusions," *Rev. Math. Phys.* **17** (2005), 491–543, arXiv:math/0412061.

[47] D. Guido, R. Longo, and H.-W. Wiesbrock, "Extensions of Conformal Nets and Superselection Structures," *Commun. Math. Phys.* **192** (1998), 217–244, arXiv:hep-th/9703129.

[48] H.-W. Wiesbrock, "Modular Intersections of von-Neumann-Algebras in Quantum Field Theory," *Commun. Math. Phys.* **193** (1998), 269–285.

[49] W. G. Unruh, "Notes on black-hole evaporation," *Phys. Rev. D* **14** (1976), 870–892.

[50] T. Jacobson, "Thermodynamics of spacetime: The Einstein equation of state," *Phys. Rev. Lett.* **75** (1995), 1260–1263, doi:10.1103/PhysRevLett.75.1260, arXiv:gr-qc/9504004.

[51] T. Jacobson, "Entanglement equilibrium and the Einstein equation," *Phys. Rev. Lett.* **116** (2016), 201101, doi:10.1103/PhysRevLett.116.201101, arXiv:1505.04753.

[52] G. W. Gibbons and S. W. Hawking, "Cosmological event horizons, thermodynamics, and particle creation," *Phys. Rev. D* **15** (1977), 2738–2751, doi:10.1103/PhysRevD.15.2738.

[53] Y. Chen, D. Stanford, H. Tang, and Z. Yang, "Negative shocks versus static patch holography," arXiv:2607.14042 [hep-th] (2026). https://arxiv.org/abs/2607.14042

[54] Planck Collaboration, "Planck 2018 results. VI. Cosmological parameters," *Astron. Astrophys.* **641** (2020), A6, doi:10.1051/0004-6361/201833910, arXiv:1807.06209.

[55] R. Bousso, Z. Fisher, J. Koeller, S. Leichenauer, and A. C. Wall, "Proof of the quantum null energy condition," *Phys. Rev. D* **93** (2016), 024017, arXiv:1509.02542.

[56] R. Bousso, Z. Fisher, S. Leichenauer, and A. C. Wall, "Quantum focusing conjecture," *Phys. Rev. D* **93** (2016), 064044, arXiv:1506.02669.

[57] S. Balakrishnan, T. Faulkner, Z. U. Khandker, and H. Wang, "A general proof of the quantum null energy condition," *J. High Energy Phys.* **2019** (2019), 20, arXiv:1706.09432.

[58] E. H. Lieb and M. B. Ruskai, "Proof of the strong subadditivity of quantum-mechanical entropy," *J. Math. Phys.* **14** (1973), 1938–1941.

[59] E. H. Wichmann, "Density matrices arising from incomplete measurements," *J. Math. Phys.* **4** (1963), 884–896.

[60] F. Hiai, M. Ohya, and M. Tsukada, "Sufficiency, KMS condition and relative entropy in von Neumann algebras," *Pacific J. Math.* **96** (1981), 99–109.

[61] M. Takesaki, "Conditional expectations in von Neumann algebras," *J. Funct. Anal.* **9** (1972), 306–321.

[62] D. Petz, "Sufficient subalgebras and the relative entropy of states of a von Neumann algebra," *Commun. Math. Phys.* **105** (1986), 123–131.

[63] D. Petz, "Sufficiency of channels over von Neumann algebras," *Quart. J. Math.* **39** (1988), 97–108.

[64] O. Fawzi and R. Renner, "Quantum conditional mutual information and approximate Markov chains," *Commun. Math. Phys.* **340** (2015), 575–611, arXiv:1410.0664.

[65] P. Hayden, R. Jozsa, D. Petz, and A. Winter, "Structure of states which satisfy strong subadditivity of quantum entropy with equality," *Commun. Math. Phys.* **246** (2004), 359–374.

[66] S. Doplicher and J. E. Roberts, "A new duality theory for compact groups," *Invent. Math.* **98** (1989), 157–218.

[67] S. Doplicher and J. E. Roberts, "Why there is a field algebra with a compact gauge group describing the superselection structure in particle physics," *Commun. Math. Phys.* **131** (1990), 51–107.

[68] T. Tannaka, "Über den Dualitätssatz der nichtkommutativen topologischen Gruppen," *Tohoku Math. J.* **45** (1938), 1–12.

[69] M. G. Krein, "A principle of duality for a bicompact group and a square block algebra," *Dokl. Akad. Nauk SSSR* **69** (1949), 725–728.

[70] H. Georgi and S. L. Glashow, "Unity of all elementary-particle forces," *Phys. Rev. Lett.* **32** (1974), 438–441.

[71] S. L. Glashow, J. Iliopoulos, and L. Maiani, "Weak interactions with lepton-hadron symmetry," *Phys. Rev. D* **2** (1970), 1285–1292.

[72] E. Witten, "An SU(2) anomaly," *Phys. Lett. B* **117** (1982), 324–328.

[73] S. Dimopoulos, S. Raby, and F. Wilczek, "Supersymmetry and the scale of unification," *Phys. Rev. D* **24** (1981), 1681–1683.

[74] U. Amaldi, W. de Boer, and H. Fürstenau, "Comparison of grand unified theories with electroweak and strong coupling constants measured at LEP," *Phys. Lett. B* **260** (1991), 447–455.

[75] D. J. Gross and W. Taylor, "Two-dimensional QCD is a string theory," *Nucl. Phys. B* **400** (1993), 181–208, arXiv:hep-th/9301068.

[76] F. Peter and H. Weyl, "Die Vollständigkeit der primitiven Darstellungen einer geschlossenen kontinuierlichen Gruppe," *Math. Ann.* **97** (1927), 737–755.

[77] A. W. Knapp, *Lie Groups Beyond an Introduction*, second edition, Progress in Mathematics **140**, Birkhäuser, 2002.

[78] A. Bullivant, M. Calçada, Z. Kádár, P. Martin, and J. Faria Martins, "Topological phases from higher gauge symmetry in 3+1D," *Phys. Rev. B* **95** (2017), 155118, arXiv:1606.06639.

[79] S. Chandrasekharan and U.-J. Wiese, "Quantum link models: A discrete approach to gauge theories," *Nucl. Phys. B* **492** (1997), 455–471, arXiv:hep-lat/9609042.

[80] W. Donnelly and A. C. Wall, "Entanglement entropy of electromagnetic edge modes," *Phys. Rev. Lett.* **114** (2015), 111603, arXiv:1412.1895.

[81] F. Pastawski, B. Yoshida, D. Harlow, and J. Preskill, "Holographic quantum error-correcting codes: Toy models for the bulk/boundary correspondence," *JHEP* **06** (2015), 149, arXiv:1503.06237.

[82] M. A. Levin and X.-G. Wen, "String-net condensation: A physical mechanism for topological phases," *Phys. Rev. B* **71** (2005), 045110, arXiv:cond-mat/0404617.

[83] D. Laghi, G. Carullo, J. Veitch, and W. Del Pozzo, "Quantum black hole spectroscopy: probing the quantum nature of the black hole area using LIGO-Virgo ringdown detections," *Class. Quantum Grav.* **38** (2021), 095005, arXiv:2011.03816.

[84] Clay Mathematics Institute, "Yang–Mills & the Mass Gap." Available at https://www.claymath.org/millennium/yang-mills-the-maths-gap/.

[85] A. Jaffe and E. Witten, "Quantum Yang–Mills Theory," official Clay Mathematics Institute problem description. Available at https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf.

[86] L. Wu, "Poincaré and transportation inequalities for Gibbs measures under the Dobrushin uniqueness condition," *Annals of Probability* **34** (2006), 1960–1989, arXiv:math/0611635.

[87] K. Osterwalder and R. Schrader, "Axioms for Euclidean Green's functions," *Communications in Mathematical Physics* **31** (1973), 83–112.

[88] K. Osterwalder and R. Schrader, "Axioms for Euclidean Green's functions II," *Communications in Mathematical Physics* **42** (1975), 281–305.

[89] Particle Data Group, "Review of Particle Physics," *Phys. Rev. D* **110** (2024), 030001. Available at https://pdg.lbl.gov/2024/download/db2024.pdf.

[90] Particle Data Group, "Electroweak Model and Constraints on New Physics," Available at https://pdg.lbl.gov/2024/reviews/rpp2024-rev-standard-model.pdf.

[91] Aharonov, Y., and Bohm, D. (1959). "Significance of electromagnetic potentials in the quantum theory." *Physical Review* **115**, 485–491. https://doi.org/10.1103/PhysRev.115.485

[92] Baez, J., and Huerta, J. (2010). "The algebra of grand unified theories." *Bulletin of the American Mathematical Society* **47**, 483–552. https://doi.org/10.1090/S0273-0979-10-01294-2

[93] Barndorff-Nielsen, O. (2014). *Information and Exponential Families: In Statistical Theory*. Wiley (Wiley Series in Probability and Statistics). https://doi.org/10.1002/9781118857281

[94] Barnich, G., Brandt, F., and Henneaux, M. (2000). "Local BRST cohomology in gauge theories." *Physics Reports* **338**, 439–569. https://doi.org/10.1016/S0370-1573(00)00049-1

[95] Becchi, C., Rouet, A., and Stora, R. (1976). "Renormalization of gauge theories." *Annals of Physics* **98**, 287–321. https://doi.org/10.1016/0003-4916(76)90156-1

[96] Bouchiat, C., Iliopoulos, J., and Meyer, P. (1972). "An anomaly-free version of Weinberg's model." *Physics Letters B* **38**, 519–523. https://doi.org/10.1016/0370-2693(72)90532-1

[97] Boyd, S., and Vandenberghe, L. (2004). *Convex Optimization*. Cambridge University Press. https://doi.org/10.1017/CBO9780511804441

[98] Bratteli, O., and Robinson, D. W. (1997). *Operator Algebras and Quantum Statistical Mechanics 2*. Springer. https://doi.org/10.1007/978-3-662-03444-6

[99] Bröcker, T., and tom Dieck, T. (1985). *Representations of Compact Lie Groups*. Springer. https://doi.org/10.1007/978-3-662-12918-0

[100] Brouwer, A. E., Cohen, A. M., and Neumaier, A. (1989). *Distance-Regular Graphs*. Springer, Berlin. https://doi.org/10.1007/978-3-642-74341-2

[101] Cohn, H., and Kumar, A. (2007). "Universally optimal distribution of points on spheres." *Journal of the American Mathematical Society* **20**, 99–148. https://doi.org/10.1090/S0894-0347-06-00546-7

[102] Coxeter, H. S. M. (1973). *Regular Polytopes*. Dover. ISBN 0-486-61480-8.

[103] Coxeter, H. S. M., and Moser, W. O. J. (1980). *Generators and Relations for Discrete Groups*. Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-662-21943-0

[104] Davies, E. B., and Lewis, J. T. (1970). "An operational approach to quantum probability." *Communications in Mathematical Physics* **17**, 239–260. https://doi.org/10.1007/BF01647093

[105] Deligne, P., and Milne, J. S. (1982). "Tannakian Categories." In *Hodge Cycles, Motives, and Shimura Varieties*, 101–228. Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-540-38955-2_4

[106] Deligne, P. (2002). "Catégories tensorielles." *Moscow Mathematical Journal* **2**, 227–248. https://doi.org/10.17323/1609-4514-2002-2-2-227-248

[107] Denner, A. (1993). "Techniques for the calculation of electroweak radiative corrections at the one-loop level and results for W-physics at LEP 200." *Fortschritte der Physik* **41**, 307–420. https://doi.org/10.1002/prop.2190410402

[108] Diaconis, P. (1988). *Group Representations in Probability and Statistics*. Institute of Mathematical Statistics (Lecture Notes–Monograph Series). https://doi.org/10.1214/lnms/1215467407

[109] Doplicher, S., Haag, R., and Roberts, J. E. (1971). "Local observables and particle statistics I." *Communications in Mathematical Physics* **23**, 199–230. https://doi.org/10.1007/BF01877742

[110] Doplicher, S., and Roberts, J. E. (1989). "A new duality theory for compact groups." *Inventiones Mathematicae* **98**, 157–218. https://doi.org/10.1007/BF01388849

[111] Doplicher, S., and Roberts, J. E. (1990). "Why there is a field algebra with a compact gauge group describing the superselection structure in particle physics." *Communications in Mathematical Physics* **131**, 51–107. https://doi.org/10.1007/BF02097680

[112] Eden, R. J., Landshoff, P. V., Olive, D. I., and Polkinghorne, J. C. (1966). *The Analytic S-Matrix*. Cambridge University Press. ISBN 0-521-04869-9.

[113] Fleischer, J., and Jegerlehner, F. (1981). "Radiative corrections to Higgs-boson decays in the Weinberg-Salam model." *Physical Review D* **23**, 2001–2026. https://doi.org/10.1103/PhysRevD.23.2001

[114] Fock, V. (1932). "Konfigurationsraum und zweite Quantelung." *Zeitschrift für Physik* **75**, 622–647. https://doi.org/10.1007/BF01344458

[115] Fulton, W., and Harris, J. (1991). *Representation Theory: A First Course*. Springer, New York (Graduate Texts in Mathematics). https://doi.org/10.1007/978-1-4612-0979-9

[116] Gambino, P., and Grassi, P. A. (2000). "Nielsen identities of the SM and the definition of mass." *Physical Review D* **62**, 076002. https://doi.org/10.1103/PhysRevD.62.076002

[117] Georgi, H., and Glashow, S. L. (1974). "Unity of All Elementary-Particle Forces." *Physical Review Letters* **32**, 438–441. https://doi.org/10.1103/PhysRevLett.32.438

[118] Ginsparg, P. H., and Wilson, K. G. (1982). "A remnant of chiral symmetry on the lattice." *Physical Review D* **25**, 2649–2657. https://doi.org/10.1103/PhysRevD.25.2649

[119] Greenberg, O. W. (1991). "Particles with small violations of Fermi or Bose statistics." *Physical Review D* **43**, 4111–4120. https://doi.org/10.1103/PhysRevD.43.4111

[120] Gross, D. J., and Jackiw, R. (1972). "Effect of Anomalies on Quasi-Renormalizable Theories." *Physical Review D* **6**, 477–493. https://doi.org/10.1103/PhysRevD.6.477

[121] Grzadkowski, B., Iskrzyński, M., Misiak, M., and Rosiek, J. (2010). "Dimension-six terms in the Standard Model Lagrangian." *Journal of High Energy Physics* **2010**, 85. https://doi.org/10.1007/JHEP10(2010)085

[122] Halmos, P. R. (1958). *Finite-Dimensional Vector Spaces*. Van Nostrand. https://doi.org/10.1007/978-1-4612-6387-6

[123] Hartman, P. (2002). *Ordinary Differential Equations*. Society for Industrial and Applied Mathematics (Classics in Applied Mathematics). https://doi.org/10.1137/1.9780898719222

[124] Henneaux, M., and Teitelboim, C. (1992). *Quantization of Gauge Systems*. Princeton University Press. https://doi.org/10.1515/9780691213866

[125] Hong, C. K., Ou, Z. Y., and Mandel, L. (1987). "Measurement of subpicosecond time intervals between two photons by interference." *Physical Review Letters* **59**, 2044–2046. https://doi.org/10.1103/PhysRevLett.59.2044

[126] Hucks, J. (1991). "Global structure of the standard model, anomalies, and charge quantization." *Physical Review D* **43**, 2709–2717. https://doi.org/10.1103/PhysRevD.43.2709

[127] James, G., and Liebeck, M. (2001). *Representations and Characters of Groups*. Cambridge University Press. https://doi.org/10.1017/CBO9780511814532

[128] Joyal, A., and Street, R. (1991). "An introduction to Tannaka duality and quantum groups." In *Category Theory (Como, 1990)*, 413–492. Springer Berlin Heidelberg. https://doi.org/10.1007/BFb0084235

[129] Knapp, A. W. (1996). *Lie Groups Beyond an Introduction*. Birkhäuser, Boston (Progress in Mathematics). https://doi.org/10.1007/978-1-4757-2453-0

[130] Kogut, J., and Susskind, L. (1975). "Hamiltonian formulation of Wilson's lattice gauge theories." *Physical Review D* **11**, 395–408. https://doi.org/10.1103/PhysRevD.11.395

[131] Kogut, J. B. (1979). "An introduction to lattice gauge theory and spin systems." *Reviews of Modern Physics* **51**, 659–713. https://doi.org/10.1103/RevModPhys.51.659

[132] Kraus, K. (1983). *States, Effects, and Operations: Fundamental Notions of Quantum Theory*. Springer. https://doi.org/10.1007/3-540-12732-1

[133] Langacker, P. (2017). *The Standard Model and Beyond*. CRC Press. https://doi.org/10.1201/b22175

[134] Lemmens, P. W. H., and Seidel, J. J. (1973). "Equiangular lines." *Journal of Algebra* **24**, 494–512. https://doi.org/10.1016/0021-8693(73)90123-3

[135] Loudon, R. (1998). "Fermion and boson beam-splitter statistics." *Physical Review A* **58**, 4904–4909. https://doi.org/10.1103/PhysRevA.58.4904

[136] Lüscher, M. (1999). "Abelian chiral gauge theories on the lattice with exact gauge invariance." *Nuclear Physics B* **549**, 295–334. https://doi.org/10.1016/S0550-3213(99)00115-7

[137] Milnor, J. W. (1997). *Topology from the Differentiable Viewpoint*. Princeton University Press (Princeton Landmarks in Mathematics). ISBN 0-691-04833-9.

[138] Montvay, I., and Münster, G. (1994). *Quantum Fields on a Lattice*. Cambridge University Press. https://doi.org/10.1017/CBO9780511470783

[139] Moore, R. E., Kearfott, R. B., and Cloud, M. J. (2009). *Introduction to Interval Analysis*. SIAM. https://doi.org/10.1137/1.9780898717716

[140] Nielsen, N. K. (1975). "On the gauge dependence of spontaneous symmetry breaking in gauge theories." *Nuclear Physics B* **101**, 173–188. https://doi.org/10.1016/0550-3213(75)90301-6

[141] Osterwalder, K., and Schrader, R. (1973). "Axioms for Euclidean Green's functions." *Communications in Mathematical Physics* **31**, 83–112. https://doi.org/10.1007/BF01645738

[142] Osterwalder, K., and Schrader, R. (1975). "Axioms for Euclidean Green's functions II." *Communications in Mathematical Physics* **42**, 281–305. https://doi.org/10.1007/BF01608978

[143] Peskin, M. E., and Schroeder, D. V. (1995). *An Introduction to Quantum Field Theory*. Addison-Wesley. https://doi.org/10.1201/9780429503559

[144] Piguet, O., and Sorella, S. P. (1995). *Algebraic Renormalization: Perturbative Renormalization, Symmetries and Anomalies*. Springer. https://doi.org/10.1007/978-3-540-49192-7

[145] Serre, J. P. (1977). *Linear Representations of Finite Groups*. Springer. https://doi.org/10.1007/978-1-4684-9458-7

[146] Slansky, R. (1981). "Group theory for unified model building." *Physics Reports* **79**, 1–128. https://doi.org/10.1016/0370-1573(81)90092-2

[147] Stuart, R. G. (1991). "Gauge invariance, analyticity and physical observables at the Z0 resonance." *Physics Letters B* **262**, 113–119. https://doi.org/10.1016/0370-2693(91)90653-8

[148] Tannaka, T. (1939). "Über den Dualitätssatz der nichtkommutativen topologischen Gruppen." *Tohoku Mathematical Journal, First Series* **45**, 1–12. https://www.jstage.jst.go.jp/article/tmj1911/45/0/45_0_1/_article

[149] Tong, D. (2017). "Line operators in the Standard Model." *Journal of High Energy Physics* **2017**, 104. https://doi.org/10.1007/JHEP07(2017)104

[150] Weinberg, S. (1967). "A model of leptons." *Physical Review Letters* **19**, 1264–1266. https://doi.org/10.1103/PhysRevLett.19.1264

[151] Weinberg, S. (1979). "Baryon- and Lepton-Nonconserving Processes." *Physical Review Letters* **43**, 1566–1570. https://doi.org/10.1103/PhysRevLett.43.1566

[152] Weinberg, S. (1996). *The Quantum Theory of Fields, Volume II*. Cambridge University Press. https://doi.org/10.1017/CBO9781139644174

[153] Wielandt, H. (1949). "Über die Unbeschränktheit der Operatoren der Quantenmechanik." *Mathematische Annalen* **121**, 21. https://doi.org/10.1007/BF01329611

[154] Wilczek, F., and Zee, A. (1979). "Operator Analysis of Nucleon Decay." *Physical Review Letters* **43**, 1571–1573. https://doi.org/10.1103/PhysRevLett.43.1571

[155] Wilson, K. G. (1974). "Confinement of quarks." *Physical Review D* **10**, 2445–2459. https://doi.org/10.1103/PhysRevD.10.2445

[156] Yukawa, H. (1935). "On the interaction of elementary particles. I." *Proceedings of the Physico-Mathematical Society of Japan. 3rd Series* **17**, 48–57. https://doi.org/10.11429/ppmsj1919.17.0_48

[157] Zaslavsky, T. (1982). "Signed graphs." *Discrete Applied Mathematics* **4**, 47–74. https://doi.org/10.1016/0166-218X(82)90033-6

AI Assistance Disclosure

This research project used research-grade commercial models, including Anthropic's Fable and OpenAI's GPT-6 Astra, for research support, software development, editing, and synthesis. The authors are responsible for the paper's claims, methods, and final text.